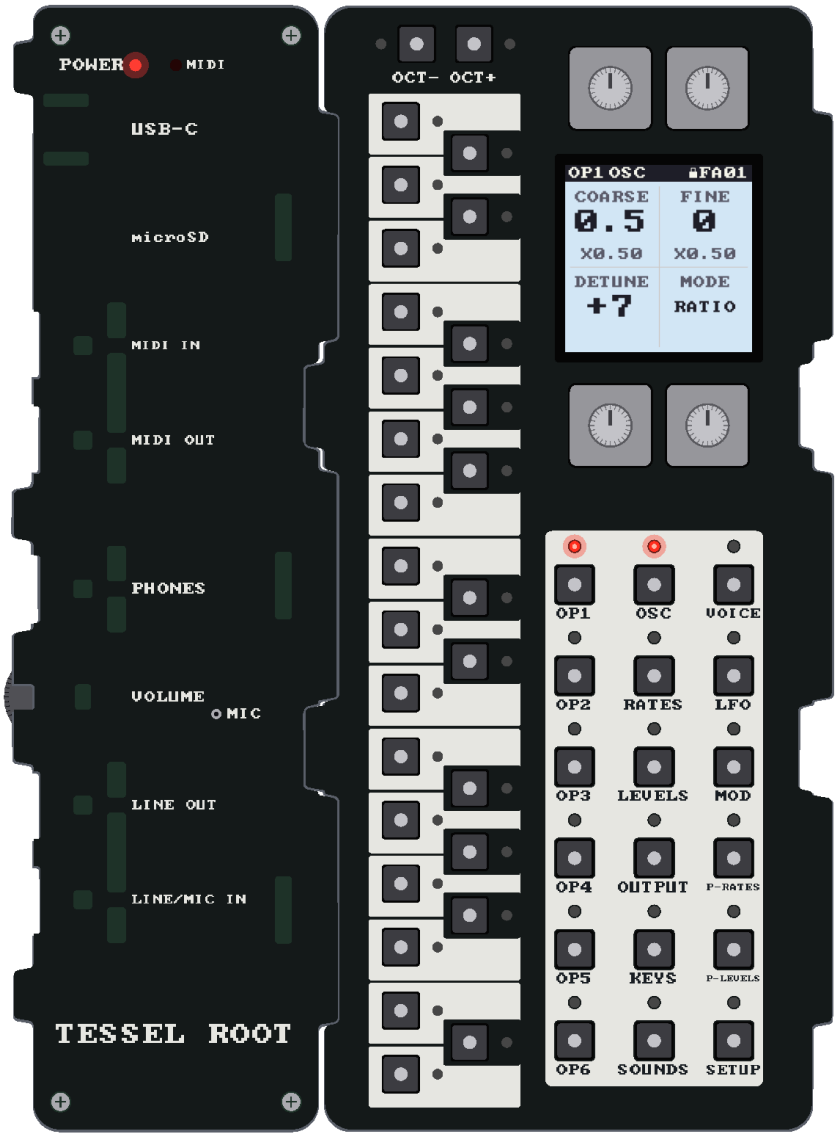
Making Sounds

The Hidden Art of Electronic Synthesis



UTOPIAN INSTRUMENTS

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What you need

Before you read a chapter, have these to hand.

> **A Root**, the end module: it gives a unit its power and its sound, and your headphones plug into it.

> **One unit**, flashed as the synth whose chapter you're reading. Every unit is the same box until you choose what it becomes, and you can change it as often as you like: flash it as Glass for chapter 3, then as Volt for chapter 4, and so on. The Quick Start guide in the box says how. Your sounds aren't lost when you do: they're kept kind by kind (most of them on the Root, section 1.8), so when you flash your unit back as Glass, your Glass sounds are there, just as you left them.

> **Headphones**, or speakers, with an ordinary 3.5 mm plug, into the Root's PHONES socket. Use headphones if you can: a lot of what this book asks you to listen for is small.

That's everything the book needs: the starter pack, a Root and one unit, and a pair of headphones. Every synth in it makes its sounds on its own.

Optional

> **A MIDI keyboard**: not essential, but a big help. Each unit's own keys are for trying sounds out: they play every note at the same strength, and they're two octaves wide. A MIDI keyboard plugged into the Root gives you more keys, pitch bend and mod wheels, a sustain pedal, and notes that get louder or brighter the harder you play, which several chapters ask you to try. Any keyboard with a MIDI OUT socket will do, however small (which cable, below).

> **A second unit, for effects.** Chorus, echo and reverb come from Haze, a module of its own (chapter 2, section 2.5, and chapter 10). With one unit you can read about them and hear them in this book's examples; to put them on your own sounds, join Haze on beside your synth.

> **A microSD card** in the Root, for keeping copies of your sounds and loading other people's (Glass's banks, chapter 3, section 3.13).

> **A computer**, for updating the firmware, and for sending sounds to a unit from editing programs.

> **Step**, the sequencer, which plays patterns on your synths, with accents and slides: the subject of the Music Production manual.

Which keyboard, and which cable

The Root's MIDI IN is a small 3.5 mm socket, like a headphone socket. Plenty of small keyboards have exactly that socket for their MIDI OUT, and a plain 3.5 mm stereo cable joins them. Bigger keyboards have the big round five-pin MIDI OUT socket instead, so you'll need an adapter: a "TRS MIDI Type A" one, which is the commonest kind.

A keyboard with only a USB socket

Look for a keyboard with a MIDI OUT socket. Most small ones get their power from their USB socket, and any phone charger or power bank will do for that: no computer needed.

A keyboard that only has USB can't plug straight into the Root: the Root's USB socket is for a computer (and for the power). Plug the keyboard into a computer, and the computer into the Root, and the computer passes the notes on. Or use a small "USB MIDI host" box, which turns a USB keyboard's notes into an ordinary MIDI OUT.

Before you start - why make your own sounds?

The sounds are half the music.

If you want to make music, whether that's techno, acid, film music, rock or anything else, the sounds you use matter as much as the notes you play. A tune on a piano and the same tune on a distorted bass synth aren't the same piece of music. Every unit in this range comes with factory sounds, and they're good: good enough to make whole tracks with, and good places to start. Everybody uses presets, and always has.

But presets only take you so far. One electric piano sound on one 1980s synthesiser turned up on so many hit records that you can still date a song by it. The music people remember, and keep coming back to, very often has a sound in it that nobody had heard before, because somebody went looking for it. Wendy Carlos made a bestseller of Bach in 1968 by building every sound from scratch on a big modular synthesiser. Kraftwerk spent years shaping the sounds that made them Kraftwerk. Vangelis's music for *Blade Runner* is as much the sound of one synthesiser, played and shaped by him, as it is the notes.

That's what this book is for. Not to make you a scientist, but to give you the handful of ideas every synthesiser is built from, so that when a sound is nearly right you know which control to reach for, and when you stumble on something wonderful you know how to find it again.

The best sounds were mistakes

Here's something worth knowing before you start: a lot of the most famous sounds in music were never meant to happen.

In 1980 Roland brought out the TR-808 drum machine. It was meant to sound like a real drummer, for musicians to practise and make demos with, and it didn't: its kick drum was a long electronic boom, its handclap was a burst of hiss, its cowbell sounded like nothing on earth. It sold badly, and Roland stopped making it after about three years. Then, second-hand and cheap, it fell into the hands of people who didn't care that it sounded wrong. They loved how it sounded. Marvin Gaye's "Sexual Healing" and Afrika Bambaataa's "Planet Rock", both in 1982, were built on it, and its boom has been under hip hop ever since.

Roland's TB-303 Bass Line, a little box meant to play bass lines for guitarists practising on their own, failed the same way and was gone by 1984. A few years later, in Chicago, a group called Phuture turned its knobs while it played, and "Acid Tracks" (1987) began acid house. Distorted guitar went the same way: in 1964 the Kinks' Dave Davies slashed a speaker cone with a razor blade to get the fuzzy sound on "You Really Got Me", and a fault became the sound of rock.

None of those people were following instructions. They were playing, and they kept what they liked. So use this book to learn the ideas, then go and break them. Turn things too far. Use controls for the "wrong" job. The only measure of success is whether you've found a sound you like.

How to use this book

Chapter 1 is how every unit works, and chapter 2 is what's in a sound, whatever makes it. Then there's a chapter for each synth, and each one stands on its own: if you've got one unit, read chapters 1 and 2 and then its chapter. Each unit chapter starts with a little history of its way of making sounds, who uses it and what for, and the famous instruments that made it. A Tessel unit isn't any of those instruments, but what it teaches will serve you well on all of them.

Instrument names in this book (such as TR-808 and TB-303, which belong to Roland Corporation, and DX7, which belongs to Yamaha Corporation) are trademarks of their owners. They're mentioned for their history. Tessel isn't connected with any of these companies or endorsed by them.

Chapter 1 - Getting around

Everything that's the same on every unit. Learn it once.

Every Tessel synth is the same box. The screen is in the same place, the four encoders are in the same place, the keys run down the same side, and the buttons that matter most, SOUNDS and SETUP, sit in the same spots. What changes from one unit to the next is what the buttons are called and what they do. So this chapter is about the box, and every chapter after it is about one thing you can turn the box into.

If you've just built your first unit and flashed it, you already know where the headphones go and how to turn it up. That's all you need. Have it in front of you as you read: everything here is quicker to learn by doing than by reading.

1.1 How this book tells you what to press

Most of this book is instructions of the "press this, then turn that" kind, and writing them out in full sentences gets tiring for everybody very quickly. So instead they're written as a little strip of coloured shapes, in the order you do them. Here's one:



That reads: press the button marked VCF, then turn encoder 2 (the box on the screen it changes is called RES) until it says 5.0. Every route in the book is built from the same few pieces:

■ VCF	A **button**: press it. White with a small dark square, like a button on the panel's white strip.
■ VECTOR VECTOR ENV	A button with a **page name** on it: press the button until the bar along the top of the screen says that name. Some buttons have two pages, and pressing again flips between them, so you might need one press or two. The name in the pill is the name you'll see at the top of the screen.
■ CHORUS *again*	Press a button **again** to switch something on or off rather than to change page.
■ *hold* OCT+	**Hold** a button down.
❷ RES	An **encoder**, by its number, and the name of the box on the screen it changes: turn it.
push ❶	**Push** an encoder, like a button, and let go.
hold ❹	**Hold** an encoder pushed in until the bar on the screen fills up.
`5.0`	The **value** to set, written exactly as the screen shows it.
C3	A **key**, by its note.

Apart from the routes, there are three kinds of box, and they always mean the same thing.

Try this

Something to do with your hands. These are the most important part of the book, and they're short.

Watch out

A trap, or something that surprises nearly everybody the first time.

Keep

Something worth remembering after you've closed the book.

1.2 A tour of the box

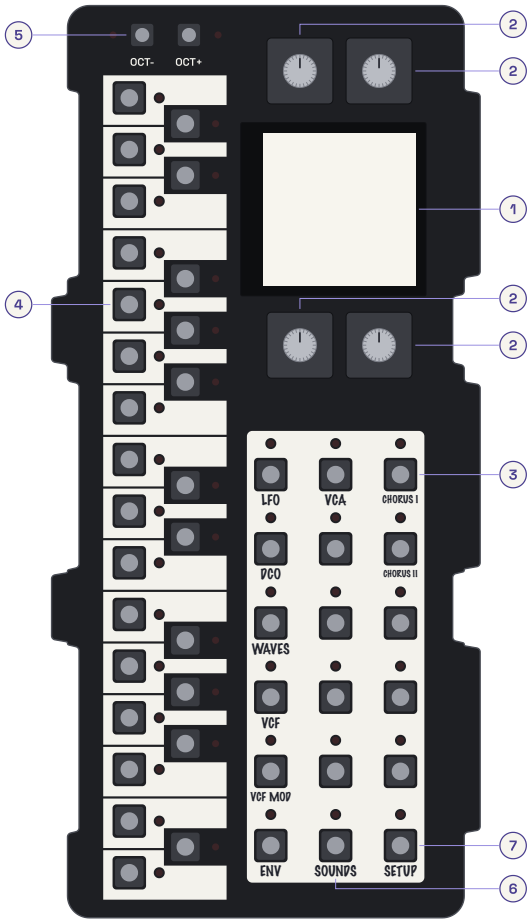


Figure 1.1 Every Tessel unit, from above (this one is Ember). 1: the screen. 2: the four encoders, two above the screen and two below. 3: one of the function buttons (one per page). 4: the keys, down the left. 5: OCT- and OCT+, above them. 6 and 7: SOUNDS and SETUP, the same on every unit.

Everything on the panel is numbered in the picture.

1 **The screen**, at the top right. It shows one page at a time, and every page is laid out the same way, which section 1.5 is about.

2 **The four encoders**, two above the screen and two below. They turn forever in either direction, with no end stop, and they push in like buttons as well.

3 **The function buttons**, the grid of buttons under the screen, three across and six down. Each one takes you to a page: on Ember they're called things like LFO, DCO and VCF, on Glass OP1 to OP6, and so on.

4 **The keys**, down the left-hand side: white keys along the edge, black keys beside them.

5 **OCT- and OCT+**, the two small buttons above the keys. They move the keys down or up; section 1.4 has the details.

Two of the function buttons are the same on every unit, and they're always in the same place:

6 **SOUNDS**, in the middle of the bottom row, where you choose a sound and keep your own (section 1.8).

7 **SETUP**, at the bottom right, for the settings that aren't part of any sound (section 1.9).

Where's the Root?

The pictures in this book show a unit on its own, to keep things clear, but a unit on its own does nothing at all: it has no battery, no socket and no speaker. It needs the Root, the end module, joined onto its left-hand side, the side with the keys. The Root gives it power and sound, and that's where your headphones plug in.

Your buttons, your labels

If your unit has the plain top panel, the words under the buttons are the ones you wrote on the white strip when you built it. This book uses the names each button has in the firmware, the same ones that appear in the top bar of the screen, so if the book says VCF and your strip says something else, the screen will settle it.

1.3 Two ways to hold it

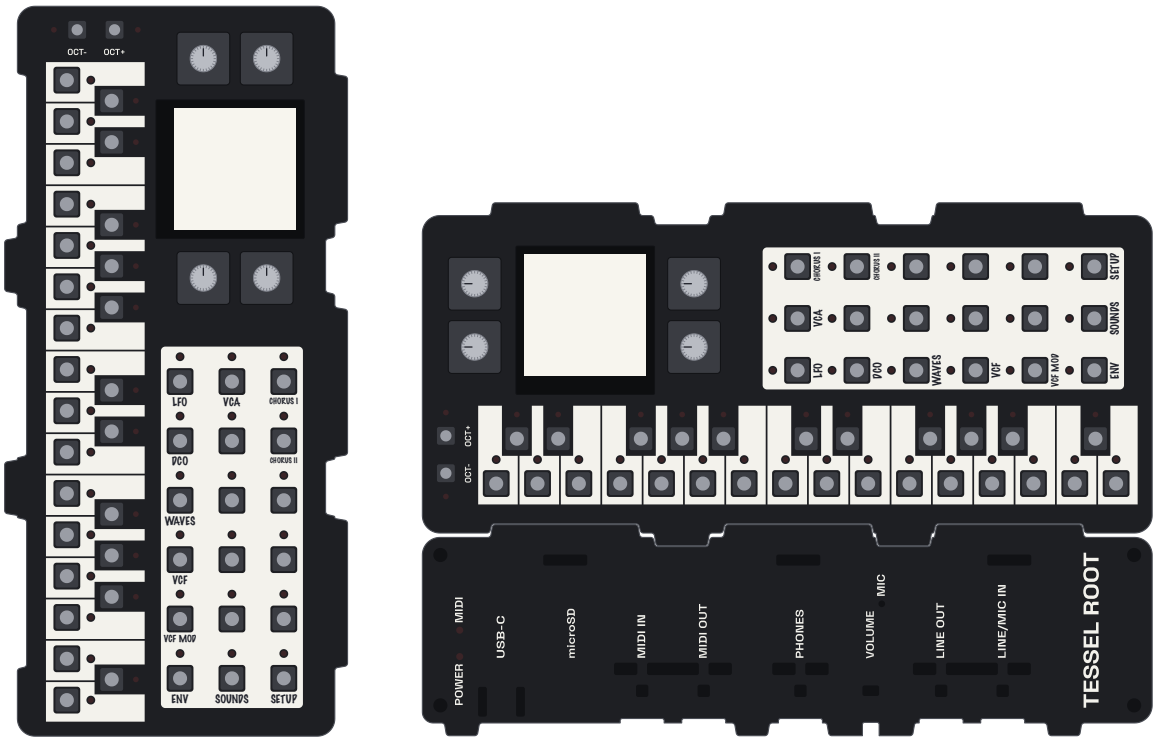


Figure 1.2 Upright, for setting a sound up; turned a quarter to the left, for playing it: the lowest note on the left, the white keys nearest you, like a piano, and the Root along the front.

Upright, the way the pictures in this book show it, is for setting a sound up: the screen's at the top and the encoders fall nicely under your fingers.

Turned a quarter of the way round to the left, it becomes a little piano: the lowest note on the left, the white keys nearest you. That's for playing a sound to hear what it's like. The screen doesn't turn round with it, so it'll be on its side, but you won't be reading it much while you play.

When you turn the unit round to play it, the Root comes round too, and lies along the front, between you and the keys. That's why the turned picture shows it: it's the one time you need to know where it'll be.

Once you've got a few units joined together, you'll mostly be playing them from a MIDI keyboard or from Step, the sequencer, and the keys on each unit are just for trying a sound out. That's what they're for.

1.4 The keys

There are twenty-seven keys: two octaves and a little bit, from C3 at the top (when the unit's upright) to D5 at the bottom. Middle C, C4, is the eighth white key down, just above halfway. They aren't touch-sensitive, so every note plays at the same loudness however hard you press. A MIDI keyboard plugged into the Root fixes that, and section 1.10 says how.

OCT- and OCT+ move the keys down or up:

- **Tap** one to move the keys an octave, up to three octaves either way. While the keys are moved, the light beside that button is on, so you can always see that you've moved them.
- **Hold** one, and the KEYBOARD page appears for as long as you hold it: encoder 1 moves the keys by octaves, encoder 2 by single semitones, and the screen shows the lowest and highest notes the keys now play.
- **Press both together** and the keys go back to where they started.

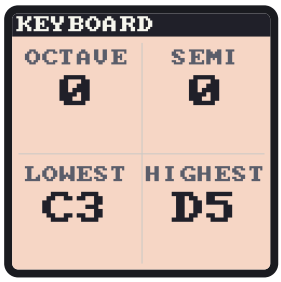


Figure 1.3 Holding OCT+ (or OCT-) brings up the KEYBOARD page: encoder 1 moves the keys by octaves, 2 by semitones, and it shows the lowest and highest notes they now play.

Moving the keys only changes what your fingers play. It isn't part of the sound, and it doesn't change notes coming from a MIDI keyboard. Strike, the drum unit, has no OCT buttons at all, because its keys are drums rather than notes.

1.5 Pages, and the four corners of the screen

Each function button has its own page. Press ■ VCF on Ember and the VCF page comes up, with the filter's settings on it. Press ■ ENV and you're on the envelope page instead. There's no menu to dig through, and nothing to scroll: one button, one page.

A few buttons have a second page, which you get to by pressing the same button again. Press it once more and you're back on the first. You'll meet these as they come up, and they're always written as a route with the page's name in it, like ■ VECTOR VECTOR ENV. Press until the top bar says VECTOR ENV.

Now the most useful thing in this whole chapter. Every page has up to four boxes, one in each corner of the screen, and each box belongs to an encoder. The encoders sit in the same pattern as the boxes: two above the screen, two below.

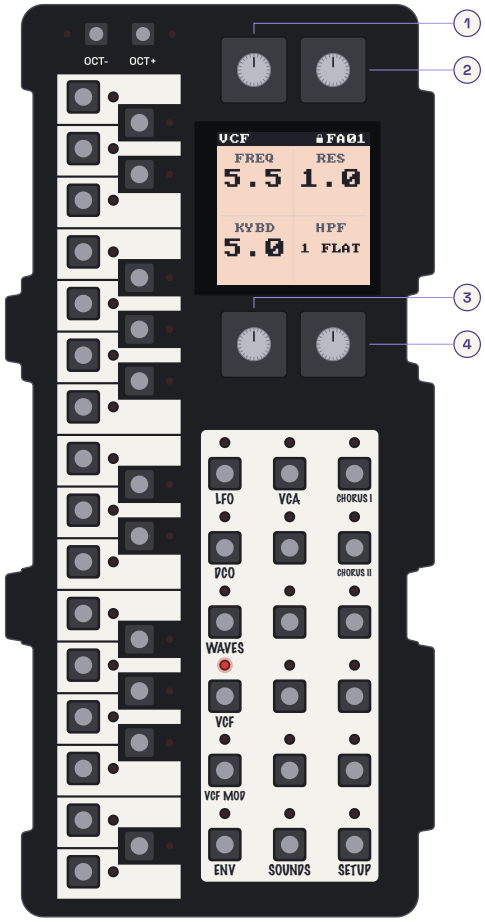


Figure 1.4 The encoders and the screen's four boxes go together: encoder 1 changes the top-left box, 2 the top-right, 3 the bottom-left and 4 the bottom-right. Here, on Ember's VCF page, encoder 2 is RES.

1 Encoder 1, top left, changes the top-left box.

2 Encoder 2, top right, changes the top-right box.

3 Encoder 3, bottom left, changes the bottom-left box.

4 Encoder 4, bottom right, changes the bottom-right box.

That's it. You never have to remember which encoder does what: look at the screen, find the box you want, and turn the encoder in that corner. When you turn one, the name in its box changes colour, so you can see which one you touched last.

Try this

Press any function button. Look at the four boxes, then turn each encoder in turn, a few clicks one way and back again, and watch which box changes. Do it on two or three different pages. After about a minute you'll stop thinking about it, which is the point.

Encoders do three things:

- **Turn.** Slowly, and each click moves the value by one step. Quickly, and it moves several steps a click, so you can cross a big range in one sweep and then slow down for the last bit.
- **Push.** On some boxes, pushing the encoder in does a second job: switching something on and off, or moving to the next letter of a name. The book says when.
- **Hold.** Anything that can't be undone, such as copying over one of your sounds or resetting the unit, needs the encoder held in. A bar appears along the bottom of the screen and fills up, and only when it's full does the thing happen.



Figure 1.5 Holding an encoder for something that can't be undone: the bar appears and fills. Let go before it's full and nothing happens.

Changed your mind?

Let go before the bar is full and nothing happens at all. That's what the bar is for.

1.6 The top bar

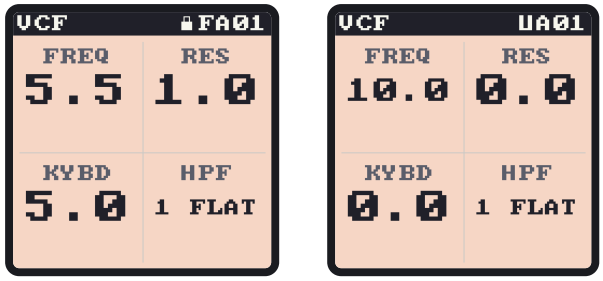


Figure 1.6 The top bar: the page's name on the left, where the sound lives on the right. Left, a factory sound (FA01, with the padlock); right, one of your own (UA01).

Along the top of every page there's a dark strip with two things in it.

On the left, the **name of the page**. It's the same name the button has, and it's the name in the route pills. If you're ever not sure where you are, look here.

On the right, **where the sound you're playing lives**, as a short code: FA01, UB07. The first letter is F for one of the factory sounds or U for one of your own; the second letter is which bank (A, B, C…); the number is which sound in that bank. FA01 is the first factory sound, UB07 is the seventh of your own in your second bank.

A small **padlock** next to the code means it's a factory sound: you can change it while you play, but nothing you do is ever kept in it. A small **star** in the padlock's place means the sound has changes that aren't kept anywhere yet. Section 1.8 explains both.

1.7 The lights

Every key and every button has a small red light beside it, and the lights are always telling you something.

- **Bright** beside a function button: that's the page you're on. Beside a key: that note is playing. Beside OCT- or OCT+: the keys are moved that way.
- **Dim** and **blinking** mean different things on different units: on Glass, a dim light means a part of the sound has been switched off, and a blinking one means you're editing a part that's switched off. Each unit's chapter says what they mean there.
- **Off**: nothing to say.

When you first turn a unit on, the screen shows its name and the firmware version for a moment. Press anything, or turn anything, and it goes away at once.

If the Root carries newer firmware for the unit than the unit has, the next screen says `NEW FIRMWARE`, with the old and new version numbers. Hold encoder 4 until the bar fills to update it: your sounds are kept. Or press or turn anything else to carry on as you were; it will ask again the next time the unit is

turned on.

1.8 Sounds: the factory's and your own

Every unit comes with its own collection of sounds, made at the factory, and room for plenty of your own. They live in **banks** of up to 32 sounds each: one or more factory banks (FAC A, FAC B…) and four banks of your own (USER A to USER D). The SOUNDS button is where you choose between them.

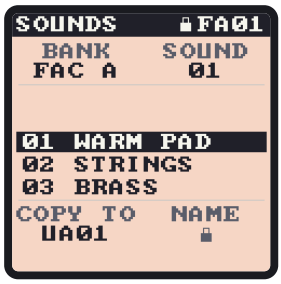


Figure 1.7 The SOUNDS page. Encoder 1 picks the bank, 2 the sound in it, 3 where COPY TO will copy it, 4 its name (in your own banks).

On the SOUNDS page:

1. **BANK**, encoder 1: which bank.
2. **SOUND**, encoder 2: which sound in it. The sound changes the moment you turn it, so you can play a few notes, turn, play again.
3. **COPY TO**, encoder 3: turn it to choose one of your own slots, then hold it in to copy the sound you're playing into that slot.
4. **NAME**, encoder 4: on one of your own sounds, turn it to change a letter of the name, and push it to move on to the next letter.

> **Try this**
>
> ■ SOUNDS › ❷ SOUND and go through the first factory bank, playing a few notes on each sound. Don't be careful about it. Find two or three you like and remember their codes from the top bar. We'll be coming back to them.

Changing a factory sound

Turn something on a factory sound and you'll hear it change. Go ahead: that's the quickest way to find out what a knob does. But the factory sound itself never changes. The padlock in the top bar turns into a small **star**, which means "changed, but not kept anywhere". That's deliberate: the factory sounds are the book's examples, and they're also what you can always go back to, exactly as they came.

So the undo is simple: choose the sound again on the SOUNDS page (turn SOUND away and back), and it's as it came. Your changes also go when you choose any other sound, so the first time you turn away from a changed sound the screen says `EDITS NOT KEPT` and stays where it is: copy it if you want to keep it, or turn again to move on. They do survive turning the unit off, so you can unsnap it and carry on later.

To keep what you've done, copy it into one of your own slots:



The copy takes your changes with it. The unit switches to the copy, the star disappears from the top bar, and the code changes to UA01. From now on, every change you make to it is kept.

> ### The habit
>
> Like what you've done to a factory sound? Copy it before you choose another one. And before you change one of your own sounds that you care about, copy it to a spare slot first: there's no undo button on your own sounds, so the copy is the undo.

Where your sounds are kept

Your own sounds aren't kept in the unit alone: they're kept on the Root, which every unit of the same kind shares. Two Glass units joined to one Root have the same banks of your sounds, and a change on one shows on the other straight away. Every kind of unit works this way, Taste and Grit with their samples too.

Join a unit to a different Root, a friend's or a new one, and if its sounds aren't the same as the ones that Root keeps, the unit asks before anything happens. Turn encoder 1 to choose one of three: **ROOT OVERWRITES GLASS** (the unit takes the Root's sounds), **GLASS OVERWRITES ROOT** (yours replace the Root's) or **DON'T OVERWRITE** (both stay as they are, and the unit uses its own until it's next turned on). Then hold encoder 4 for three seconds. Nothing is ever mixed together, and a brand-new Root, with nothing on it, simply takes your units' sounds without asking.

Every one of your own slots starts as a plain, simple sound called INIT SOUND (on Glass it's INIT VOICE, on Strike INIT KIT, on Haze INIT FX). It's the blank page you build a sound from scratch on.

It saves itself

There's no SAVE button. A unit saves your sounds, and which page and sound you were on, a moment after you change anything. You can unplug it whenever you like, even halfway through turning a knob, and when you plug it back in it'll be exactly where you left it.

1.9 SETUP

SETUP, at the bottom right, is where the settings that aren't part of any sound live. It has two parts, and pressing SETUP again (or pushing any encoder) moves on through their pages.

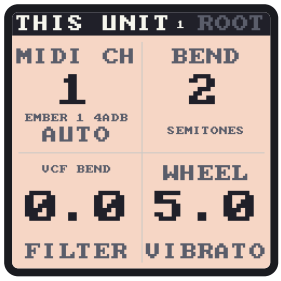


Figure 1.8 SETUP's first page, THIS UNIT: encoder 1 is the unit's MIDI channel, 2-4 its own settings.

THIS UNIT comes first: the settings that belong to this one unit. Encoder 1 is always the unit's MIDI channel (section 1.10), and the others are that unit's own settings, such as how far pitch bend goes, or (VELO) whether notes from a MIDI keyboard play louder when you hit them harder. Each unit's chapter lists its own. A unit with more settings than fit has a second THIS UNIT page; the top bar shows which page you're on as a small number.

The last THIS UNIT page is **RESET**, which puts things back the way they left the factory, and looks after the unit itself. Encoder 1 chooses what to do:

- **SETTINGS** puts back SETUP's settings, the MIDI channel and the keys' octave, and keeps your sounds.
- **EVERYTHING** does that and wipes your sounds as well, so every one of your slots goes back to INIT SOUND. Your sounds are kept on the Root (section 1.8), so this wipes them for every unit of that kind on the Root, as the screen warns.
- **REFLASH** turns the unit into another kind of unit: encoder 3, in the bottom-left box that shows what the unit is now, chooses which, from the ones the Root carries. This kind's sounds stay in the unit, for when you flash it back.
- **Another type's name**, such as TASTE, if the unit has been flashed as that type before: each type keeps its own sounds in the unit (see "What you need"), and this clears that type's, leaving the rest alone. The screen says how much room they take.
- **EVERY TYPE** wipes every type's sounds and settings: for when you pass a unit on to someone else.

- **TOTAL RESET** wipes everything: every kind's sounds and settings, and its MIDI channel. The unit becomes a blank unit, as it came out of the box, ready to be flashed as anything.
- **TEST** checks the unit is working: every unit carries the same test it was given in the factory. The unit restarts into the test, which shows the screen in five plain colours, plays a short note through the Root's headphones, then draws a picture of the panel. Press every button and turn every encoder both ways (the picture shows how many clicks are left) and push it, until nothing is left. At the end it says PASS or FAIL, and encoder 4 brings the unit back exactly as you left it: nothing is lost. If you ever need help with a unit, this is the first thing to try, and encoder 1 on the results shows a QR code that tells us what we need.

Then hold encoder 4 for three seconds. The firmware version is on this page too, in the bottom-left box.



Figure 1.9 The RESET page, the last of THIS UNIT's pages: choose SETTINGS or EVERYTHING, then hold encoder 4 for three seconds.



Figure 1.10 REFLASH on the RESET page: encoder 3 chooses what the unit becomes; hold encoder 4 for three seconds.

> ### EVERYTHING means everything
>
> There's no getting your sounds back after a reset to EVERYTHING, unless you've made a BACKUP to the Root's memory card first (ROOT, encoder 4, below). If you've got a card, do that first.

ROOT comes after THIS UNIT, and it's the settings of the Root, the end module, which has no screen or buttons of its own: which input it listens to (LINE, MIC, or its own BUILT-IN microphone) and how loud (GAIN), what its MIDI output sends, and jobs for its memory card. Every unit's SETUP can reach them, and

they're the same whichever unit you use. A second ROOT page, ROOT2, decides how MIDI from outside reaches the units, which is the next section.

- **MIDI OUT** sends what's playing to a computer or another synth: NOTES (Step's notes, each on the channel of the unit it plays), CLOCK (Step's tempo, with start and stop), BOTH, THRU (a copy of what comes in at MIDI IN), or ALL of them.
- **CARD**: hold encoder 4 to run a job on the Root's memory card. **BACKUP** copies every unit on the chain, every type each one has been, and the Root's own settings, into a new folder, BACKUP-1, BACKUP-2 and so on, never over an old one. **RESTORE** puts the newest backup back into the units that are on the chain, and says if any from the backup weren't there; each unit then starts up again with its things back. **FORMAT** empties the card. A big backup takes a while: the screen shows how far it's got.
- **INPUT** chooses which of the Root's inputs units record from: LINE (the LINE/MIC socket, at line level), MIC (the same socket, with more gain and power for a plug-in microphone) or BUILT-IN (the little microphone in the Root's top). **GAIN** sets how loud it comes in. Both are also on Taste's and Grit's recording pages, where you need them.

ROOT2 also has **SOUNDS**, for a Root that already has sounds on it, such as a second-hand one. Encoder 3 chooses **UNITS'** (each kind's sounds come from the leftmost unit of that kind on the chain, replacing the Root's) or **ROOT'S** (every unit takes the Root's); hold encoder 4 for three seconds. It settles every kind at once, without the question each unit would otherwise ask.

1.10 Playing from a MIDI keyboard

Plug a MIDI keyboard into the Root's **MIDI IN** socket ("What you need", at the start of the book, says which keyboards and which cable) and you can play any unit from it: with the pitch bend and mod wheels, a sustain pedal, and notes that can get louder the harder you play: most units have a VELO setting for that on SETUP, and on Glass each sound decides for itself.

Each unit has a name made of what it is and a number, such as EMBER 2 or GLASS 1. The number is its **MIDI channel**, and that's how a keyboard or a computer picks which unit it's talking to. Units are given a channel automatically, each its own, so you don't need to do anything; but you can choose one yourself on THIS UNIT, encoder 1, and a channel you've chosen is never changed.

On ROOT2, MIDI IN decides how the Root passes notes on: **CHANNELS**, where each unit plays only the notes on its own channel (the usual way, with a computer), or **FOLLOW**, where the Root sends your keyboard's notes to whichever unit you last touched. FOLLOW is the easy way to play several units from one keyboard: turn a knob on the unit you want and play.

1.11 When something seems wrong

- **There's a star in the top bar.** The sound has changes that aren't kept anywhere. Copy it to one of your own slots to keep them, or choose the sound again on the SOUNDS page to throw them away (section 1.8).
- **The screen says `COPY IT TO RECORD` or `COPY IT TO LOAD`.** On Taste and Grit, a new sample has to belong to one of your own sounds, so copy the factory sound to one of your slots first (section 1.8).
- **There's no sound.** Check the volume on the Root, and that the unit is joined on properly (its screen will be lit). On a few units, some sounds play what another unit is playing and make no noise by themselves; their chapters say so.
- **The keys play the wrong notes.** Look at OCT- and OCT+: if one is lit, the keys are moved. Press both together to put them back.
- **You're lost.** Look at the top bar. It always says which page you're on.
- **The screen says `FLASH FULL`.** The unit's memory is full, almost always of samples (Taste's or Grit's) or songs, perhaps from when it was another type. Nothing is ever thrown away to make room: save what you want to keep to the card, then clear what you don't need on SETUP's RESET page, which lists what's using the space.
- **The screen says `CARD NEEDED`.** The Root keeps about 40 seconds of your own samples, Taste's and Grit's together, and that's full. Put a memory card in the Root and there's room for as many as the card holds; or delete samples you no longer use (Taste's and Grit's SAMPLES pages).
- **The screen says `LOADING`.** On Taste and Grit: the sound you've chosen uses a sample this unit hasn't got yet, and it's being fetched from the Root (or its card). It takes a few seconds, more for a long sample; until it's here, the notes that need it are silent.
- **You've made a mess of a sound.** On a factory sound, choose it again on the SOUNDS page. On one of your own, copy the factory sound you started from again (section 1.8). That's what they're for.

That's chapter one, and it's the same on every unit, so you'll only ever need to learn it once: whichever unit you pick up next, it works the same way. Next, a short chapter about what a sound is actually made of, and then one chapter for each unit, each starting from a sound you like and ending with one you've made yourself.

Chapter 2 - What's in a sound

Six things your ears are listening for, all the time, without being asked.

Before any of the units, a chapter about sound itself. It's short, it's mostly listening, and everything in the rest of the book leans on it.

Here's the idea. When you hear a note, your ears are quietly answering six questions about it. How high is it? What colour is it: bright and buzzy, or dark and soft? How does it change, from the moment it starts to the moment it's gone? Does anything in it move while it lasts? Is it close and dry, or swirling, echoing, far off in a big room? And how loud is it?

Every synth in the range is a machine for changing the answers. They go about it in very different ways, which is what the unit chapters are for, but the questions are always the same six. Learn to hear them here and you'll know what you're reaching for on any unit.

> **Try this**
>
> Every idea in this chapter comes with a sound to try, Sound 2.1, Sound 2.2 and so on, and each one says how to make it on the unit in front of you. Do them as you read: some of the differences are small, so use headphones if you can.

2.1 Pitch

Pitch is how high or low a note is, and it's the one you already know best. Every key plays a different pitch, and that's most of what a tune is.

Sound 2.1 The same note, C, four times, an octave higher each time.

Make it on your unit

Any unit: play C3, then C4 and C5 (the white keys with a gap of seven between them), then press ■ OCT+ and play C5 again.

Those four notes are all called C, and they sound like the same note, only higher each time. That's an **octave**, and it's the oddest fact in music: count twelve steps up from any key, black and white keys together, and you arrive back at the same note, one floor up. Underneath, the sound is vibrating exactly twice as fast. Each octave doubles it again.

Sound 2.2 Five notes, each a semitone above the last: the smallest step on the keyboard.

Make it on your unit

Any unit: play C4 and the next four keys up, black and white, one at a time.

Strike: in your own kit, ■ EDIT, play LO TOM, and turn ❷ TUNE a step at a time (section 5.5).

Taste: on MY KEYS (section 8.8), play up and down: one note on every key. A kit key's pitch is ■ TUNE › ❶ COARSE (section 8.5).

Grit: ■ PITCH › ❶ PITCH moves every grain by semitones, without changing the speed (section 9.6).

The smallest step on the keyboard, from one key to the very next, black or white, is a **semitone**. Twelve of them make an octave.

Almost every unit can move its whole sound up or down without you changing the keys you press, in octaves and in semitones, and many of them can do it in much smaller steps too. Two sounds a hair apart in pitch, played together, rub against each other and make a slow wobble. That's not a mistake: it's one of the oldest ways of making a sound bigger and warmer, and you'll meet it as **detune**.

Find it on your unit: Glass, section 3.8 · Volt, section 4.4 · Strike, section 5.5 · Ember, section 6.4 · Taste, section 8.5 · Grit, section 9.6

2.2 Tone

Play the same note on a flute and on a trumpet. Same pitch, same loudness, and you'd still never mix them up. What's different is the **tone**, or the colour of the sound, and it's the thing synths are best at changing.

A note isn't one pitch. It's the pitch you hear, plus a stack of quieter ones above it, called **harmonics**, that your ear doesn't hear separately but blends into a colour. Few harmonics, and the sound is soft and pure. Lots of them, especially high ones, and it's bright and buzzy.

Sound 2.3 The four basic waves at the same pitch and loudness: sine, triangle, sawtooth, square. Then noise, which has no pitch at all.

Make it on your unit

Glass: INIT VOICE (UA01, section 3.2) is the sine. For something near the sawtooth: ■ VOICE › ❶ ALGO › 32, OP1 LEVEL 0, OP6 LEVEL 99, ❷ FEEDBK › 6 (section 3.7).

Volt: on ONE OSC (section 4.3), step ■ OSC 1 › ❸ WAVE through TRIANGLE, SAWTOOTH and SQUARE (Volt has no sine: TRIANGLE is the nearest). For noise, switch OSC 1 off and NOISE on with ■ MIXER › *push* ❶ and *push* ❹ (section 4.5).

Strike: the hats, cymbals and maracas are noise and metallic clang, with no pitch at all; the kick is close to a sine (section 5.2).

Ember: on SAW 1 (section 6.3), ■ WAVES: SAW on its own, PULSE on its own, then SUB alone for a square; NOISE for the hiss (section 6.4). Ember has no sine.

Axis: on CORNERS (section 7.2), each corner is one wave: move the point with ■ VECTOR › ❸ X and ❹ Y from the sine to the square and on (section 7.3). Waves 0-15 are the plain shapes.

Taste: a sampler plays recordings, not waves: any of the starter set's samples (section 8.4). Its RAIN is close to noise.

Grit: grains are made from samples too; very short ones, ■ GRAIN › ❶ SIZE under 10 MS, turn any sound into a buzz (section 9.4).

Those are the basic shapes most synths start from, all at the same pitch and loudness. The **sine** is the pure one: no harmonics at all, just the note. The **triangle** has a few, faintly. The **sawtooth** has every harmonic there is, and it's the brightest and buzziest, which is why it's the one you'll start from most. The **square** has every other harmonic, which gives it that hollow, woody sound. The last one is **noise**: every pitch at once, so no pitch at all. It's the hiss of wind, waves, cymbals and breath.

The filter

Most synths then hand the sound to a **filter**, which takes harmonics away. Close it a little and the top of the sound goes; close it further and the sound turns dark and muffled, like hearing music through a wall.

Sound 2.4 One sawtooth, four times, with the filter closing a little more each time.

Make it on your unit

Glass: it has no filter, and takes harmonics away by turning a modulator down. On SINE 1 set ■ OP2 › ■ OUTPUT › ❶ LEVEL › `80`, hold a note and turn LEVEL slowly down to 0 (section 3.6).

Volt: on ONE OSC, hold a low note and turn ■ FILTER › ❶ CUTOFF down from +5.0 (section 4.6).

Ember: on SAW 1, hold a chord and turn ■ VCF › ❶ FREQ down from 10 (section 6.5).

Axis: turn ■ FILTER › ❶ CUTOFF down from the top (section 7.5).

Taste: ■ FILTER › ❶ CUTOFF down from 10, on any key (section 8.8).

Grit: ■ FILTER › ❶ CUTOFF down from 10 with TYPE on LOW (section 9.8).

The filter has a second control, usually called **resonance**. It makes the filter ring at the point where it's cutting, so the harmonics right there stand out. A little gives the sound an edge. A lot makes it whistle.

Sound 2.5 A filter part closed, with no resonance, then some, then a lot: listen for the whistle.

Make it on your unit

Glass: it has no filter, so no resonance. Its nearest thing is a modulator at a high COARSE, which picks out a high harmonic (section 3.6).

Volt: on ONE OSC with CUTOFF at about 0, turn ■ FILTER › ❷ EMPHASIS up: from about 7 it whistles (section 4.6). In ACID, RESONANCE (section 4.12).

Ember: on SAW 1 with FREQ at about 4, turn ■ VCF › ❷ RES up (section 6.5).

Axis: with CUTOFF part way down, turn ■ FILTER › ❷ RES up (section 7.5).

Taste: ■ FILTER › ❷ RES up, with CUTOFF part way down (section 8.8).

Grit: ■ FILTER › ❷ RES up; TYPE on BAND rings like a tuned pipe (section 9.8).

Bright and dark

When people talk about a sound being bright, dark, buzzy, soft, hollow or thin, they're talking about harmonics: how many there are, and which. A filter is the commonest way of changing them, but it isn't the only one. Glass has no filter at all, and makes its harmonics a completely different way (its chapter explains how).

Find it on your unit: Glass, section 3.6 · Volt, section 4.6 · Strike, section 5.2 · Ember, section 6.5 · Axis, section 7.4 · Taste, section 8.10 · Grit, section 9.4

2.3 The shape of a note

Strike a piano key and the note is loudest straight away and then dies away, however long you keep your finger down. Draw a bow across a violin and the note creeps up from nothing. Play an organ and the note stays exactly as loud as long as you hold it, and stops the moment you let go. Hit a bell and it goes on ringing for ten seconds whatever you do.

That difference, how loud a note is at each moment between the key going down and silence coming back, is called the **envelope**, and it does more to tell your ear what kind of instrument it's hearing than almost anything else.

Sound 2.6 One note, four shapes: an organ, something like a piano, something like a bowed string, a pluck.

Make it on your unit

Glass: the four recipes in section 3.5: ORGAN, PLUCK, BELL and SWELL, on OP1 of SINE 1.

Volt: the four shapes on the LOUDNESS page in section 4.7.

Strike: drums are nearly all attack and decay: in your own kit, ■ EDIT, play KICK, and turn 3 DECAY (section 5.6).

Ember: the four shapes on the ENV page in section 6.6.

Axis: the AMP ENV's five points: times and levels (section 7.6).

Taste: ■ ENV's ATTACK, DECAY, SUSTAIN and RELEASE, on top of the sample's own shape (section 8.8).

Grit: ■ ENV's four, for each note's cloud (section 9.8).

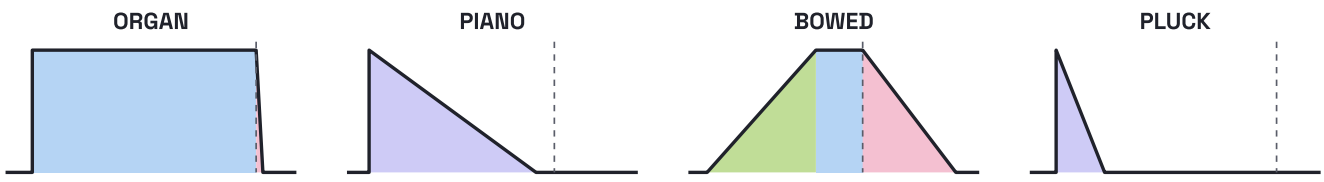


Figure 2.1 The four shapes from sound 2.6. The dashed line is where the key is let go.

Most synths draw the envelope with four controls, and you'll meet these four names on nearly every unit:

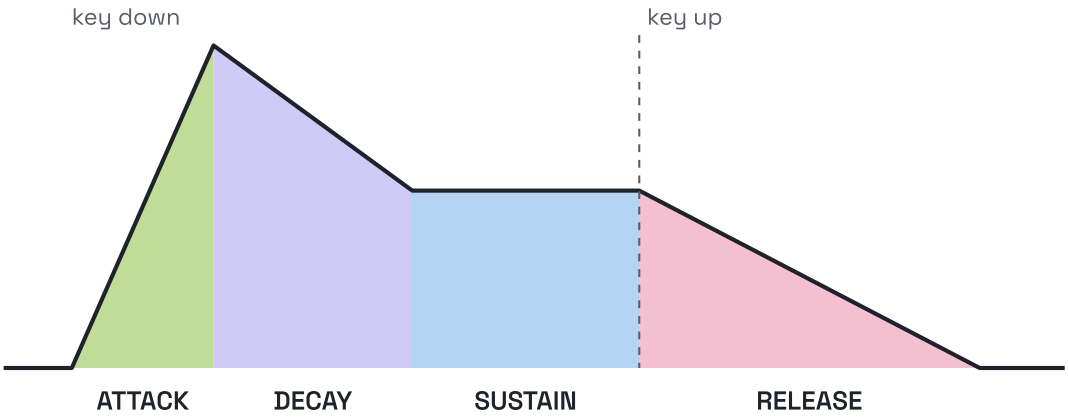


Figure 2.2 The shape of a note, as most synths draw it. Attack: how quickly it gets going. Decay: how quickly it falls back. Sustain: the level it holds while your finger stays down. Release: how long it takes to die away once you let go.

A **Attack**: how long the note takes to get going. Short for a piano or a drum; long for strings creeping in.

D **Decay**: how long it takes to fall back from the top.

S **Sustain**: the level it holds while your finger stays down. It's a level, not a time, because you choose the time, with your finger. All the way up is an organ; all the way down, and the note dies away whether you hold the key or not, like a piano.

R **Release**: how long it takes to die away once you let go.

Sound 2.7 A short tap, twice: first with no release, then with a long one.

Make it on your unit

Glass: on SINE 1, ■ OP1 › ■ RATES › 4 R4 › `40` and play short notes; then put R4 back to 99 (section 3.5).

Volt: the DECAY switch, ■ CONTROL › 3 DECAY: ON, notes die away after you let go; OFF, they stop (section 4.7).

Ember: on SAW 1, ■ ENV › 4 R up to 6 and play short notes (section 6.6).

Axis: on AMP ENV, point 4's TIME: how long it takes to die away after you let go (section 7.6).

Taste: on MY KEYS, ■ ENV › 4 RELEASE (section 8.8).

Grit: ■ ENV › 4 RELEASE (section 9.8).

Letting go early

Letting go of a key doesn't wait for anything. Whatever the note was doing at that moment, even if it was still creeping up, it heads straight for silence at the release speed. So a sound with a slow attack never reaches full loudness if you play it in short stabs. That isn't a fault: slow sounds want long, held notes.

Brightness over time

Here's the bit that makes synths sound alive. The envelope doesn't have to change only the loudness: it can open and close the filter too. A real plucked string is brightest the instant it's plucked and gets darker as it rings, and a filter that opens at the start of each note and closes again does the same.

Sound 2.8 A dark note, then the same note with the filter opened at the start of each note and closing again: bright, then darker, like a plucked string.

Make it on your unit

Glass: build E PIANO 1 (section 3.6): OP2's envelope dies away sooner than OP1's, so the brightness goes first.

Volt: PLUCK BASS in section 4.7: the filter contour opens the filter on every note and closes it again.

Ember: STAB in section 6.6: the envelope opens the filter on each note and closes it again.

Axis: FILTER's ENV up, and the FILTER ENV's points shape it (sections 7.5 and 7.6).

Taste: ■ FILTER › 3 ENV above 0: each note starts brighter (MY KEYS, section 8.8).

Find it on your unit: Glass, section 3.5 · Volt, section 4.7 · Strike, section 5.6 · Ember, section 6.6 · Axis, section 7.6 · Taste, section 8.8 · Grit, section 9.8

2.4 Movement

Everything so far does the same thing on every note: press a key, the shape runs, and if you hold the note it sits there being exactly what it is. Real instruments don't. A held violin note shivers. A singer's long note starts straight and then starts to move.

The usual way to put movement into a sound is an **LFO**, a low frequency oscillator. It's an oscillator like the ones that make the sound, except it runs so slowly you can't hear it as a pitch: a few wobbles a second rather than a few hundred. Because you can't hear it, it isn't used to make sound. It's used to move something else.

Point it at the pitch and you get **vibrato**. Many LFOs can wait a moment before they start, the way a singer holds a long note steady and only then lets it go.

Sound 2.9 A held note: plain, with vibrato, and with vibrato that waits a moment before it starts.

Make it on your unit

Glass: ■ MOD › ❶ VIBRATO › `20`, then ■ LFO › ❷ DELAY › `40` for the late start (section 3.9).

Volt: oscillator 3 as a slow modulator, with OSC MOD on (section 4.8). Volt has no delayed vibrato.

Ember: ■ DCO › ❷ LFO with the LFO's DELAY up, for the late start (section 6.7).

Axis: MOD slot: LFO 1 to PITCH, with LFO 1's DELAY up (section 7.7).

Taste: ■ LFO › ❷ PITCH, as on VOX PAD (section 8.8).

Grit: a gentle vibrato is ■ PITCH › ❸ RANDOM at a few cents instead: every grain slightly out of tune (section 9.6).

Point it at the filter and the tone goes round and round: brighter, darker, brighter.

Sound 2.10 A filter slowly opening and closing on its own, over and over.

Make it on your unit

Glass: on SINE 1 with OP2 LEVEL 70: ■ OP2 › ■ OUTPUT › ❸ AMP MOD › `3` and ■ MOD › ❸ TREMOLO › `60`. The brightness wobbles, the way a filter's would (section 3.9).

Volt: oscillator 3 as a slow modulator, with FILT MOD on (section 4.8).

Ember: ■ VCF MOD › ❸ LFO with RES up a little (section 6.7).

Axis: MOD slot: LFO 1 to CUTOFF (section 7.7).

Taste: ■ LFO › ❸ FILTER (section 8.8).

Grit: ■ LFO › ❹ FILTER, as on MIST MOVING (section 9.8).

Find it on your unit: Glass, section 3.9 · Volt, section 4.8 · Strike, section 5.8 · Ember, section 6.7 · Axis, section 7.7 · Grit, section 9.5

2.5 Effects

Everything so far happens inside the note. Effects happen to the sound after it's made: they take what comes out and do something to all of it, every note at once. A guitarist's pedals are effects, and so is the sound of the room a band plays in.

There are dozens of them, but four do most of the work, and you'll hear them on nearly every record.

Chorus copies the sound, wobbles the copy's pitch very slightly and mixes the two together, the way a choir is never quite in tune with itself. It makes a sound wider and warmer, and it's half the secret of many famous synth sounds.

Sound 2.11 A chord, plain and then with chorus.

Make it on your unit

If you have Haze as well: press its ninth white key, CHORUS, and play: the key sends every synth in. To make your own, MY CHORUS (chapter 10, section 10.9). With one unit on its own, just listen for now (section 2.5).

Ember: press ■ CHORUS I: Ember has a chorus of its own (section 6.8).

Axis: press ■ CHORUS, then again to switch it on (section 7.9).

A **phaser** is a close cousin: it mixes the sound with a copy that's been shifted slightly, so some harmonics cancel out, and then sweeps where the cancelling happens. You hear it as a slow whoosh, like a jet going over.

Sound 2.12 A chord, plain and then through a phaser: the sweeping, whooshing one.

Make it on your unit

If you have Haze as well: its eleventh white key, PHASER, and hold a chord; MOD's MODE PHASER (chapter 10, section 10.9).

Echo plays the sound back a moment later, then again, quieter each time, like shouting across a valley. Short echoes thicken a sound; long ones make rhythms of their own.

Sound 2.13 A short note, plain and then with echo: the note comes back, quieter each time.

Make it on your unit

If you have Haze as well: its fifth white key, TAPE ECHO, and play short notes; MY ECHO (chapter 10, section 10.8).

Reverb is the room: the thousands of tiny echoes off the walls, floor and ceiling that blur together into a tail. A little puts the sound in a room; a lot puts it in a cathedral.

Sound 2.14 A short chord, plain and then with reverb: the same chord in a big room.

Make it on your unit

If you have Haze as well: its second white key, BIG HALL, and play short notes; MY PLATE and MY HALL (chapter 10, section 10.7).

> ### With one unit, these four have to wait
>
> If you've got one unit, the one in the starter pack, you can't make most of these four sounds yourself yet: in this range, effects come from a second module, Haze. (Ember and Axis each have a chorus of their own: sections 6.8 and 7.9.) That's fine. Read what each one does, and when you hear them on a record you'll know what's being done: they're on nearly all of them. Every other sound in this chapter you can make on the unit you've got.

In this range, effects are mostly Haze's job (chapter 10). Join it on and its echo, reverb and chorus are shared by every synth on the chain, each sending in as much as you like. A few units have an effect of their own built in, such as the chorus on Ember and Axis, because it's part of how those sounds were always made.

Find it on your unit: Glass, section 3.11 · Volt, section 4.13 · Strike, section 5.10 · Ember, section 6.8 · Axis, section 7.9 · Taste, section 8.13 · Haze, section 10.7 · Haze, section 10.8 · Haze, section 10.9 · Haze, section 10.11 · Haze, section 10.12 · Haze, section 10.13

2.6 Loudness

Loudness is the sixth thing, and the simplest: every sound has a level, and every unit has somewhere to set it. Only two things about it are worth knowing now. One: louder almost always sounds better at first, so when you compare two versions of a sound, make them equally loud before you decide. Two: loudness can follow how hard you play, if you play from a MIDI keyboard that notices (section 1.10), and that does a lot to make a sound feel like an instrument.

Find it on your unit: Glass, section 3.4 · Volt, section 4.9 · Strike, section 5.9 · Ember, section 6.9 · Axis, section 7.10

2.7 Where next

That's the whole list: pitch, tone, shape, movement, effects and loudness. Every knob on every unit changes one of those, and every unit's chapter will tell you which.

> ### What you have at the end of this chapter
>
> Six questions to ask of any sound you hear, and a name for each part of the answer. At the end of each section above is where your unit's chapter deals with it. Go to that chapter next.

Chapter 3 - Glass

Six sine waves, and everything they can be made to do to each other.

Everything Glass does is made of sine waves. Not "based on" or "a bit like", but literally: there is nothing else inside it. It has six sine wave generators, and every one of its 128 factory sounds is those same six waves arranged in some particular way.

This chapter starts the way chapter 2 said every unit chapter would: with a sound you like, and no theory at all. Then it switches five of the six off, looks very carefully at the one that's left, and builds back up from there. By the end you'll have made an electric piano and a bell from nothing, and you'll be able to open up any of the factory sounds and see how it works.

FM synthesis: where it came from, and who uses it

FM synthesis was discovered by accident, which fits the introduction nicely. In the late 1960s John Chowning, a composer at Stanford University, was experimenting with vibrato on a computer. He made it faster and faster, and at some point it stopped sounding like vibrato and turned into a new tone colour: bright, metallic, bell-like, and controlled by just a couple of numbers. Section 3.6 explains exactly what he heard. Stanford patented it and licensed it to Yamaha.

In 1983 Yamaha brought out the DX7, and it became one of the best-selling synthesisers ever made. Its sounds were nothing like the warm, buzzy analogue synths before it: glassy electric pianos, bells, plucked basses, brass with a bite. They're all over 1980s pop, and Brian Eno, among many others, made a great deal of music with it. FM went into home computers and games consoles too: the sound chips in early PC sound cards, and in Sega's Mega Drive, were FM, so a lot of 1980s and 1990s game music is FM. And a small rack-mounted FM synth from the same family, the TX81Z, gave 1990s house music one of its favourite bass sounds.

FM is good at the sounds other methods find hard: anything that rings, chimes, clangs or plucks, and basses with a hard edge. It's also famous for being hard to program, which is why so many people only ever used the presets. You won't be one of them.

Today you'll find FM on synthesisers such as Elektron's Digitone and Korg's opsix, and in software such as Dexed, which is free. Glass isn't any of those, but it has every setting of the classic six-operator kind, so everything this chapter teaches works on all of them, and it can load the forty years of free sounds made for the DX7 (section 3.13).

3.1 Find a sound you like

Glass comes with four factory banks, FAC A to FAC D: 128 sounds in all, made by people who knew this kind of synth extremely well. The only job in this section is to find out which of them you like.

> **Try this**
>
> ■ SOUNDS › 2 SOUND and go through all four banks, changing bank with 1 BANK. Not carefully: turn, play two or three notes, turn again. It takes about ten minutes, and it's probably the most useful ten minutes you'll spend on Glass all year.
>
> Write down the codes of the five you like best (FA11, FC07 and so on, from the top bar). We'll be coming back to them.

Now one more thing to try before any explanation. Changes to a factory sound aren't kept (chapter 1, section 1.8), and you'll want to keep this one, so copy a sound into your second bank, USER B. In this chapter USER B is your scrapbook, and USER A is where the sounds you build will go.

■ SOUNDS › 1 BANK › FAC A › 2 SOUND › FA11 then 3 COPY TO › UB01 › *hold* 3

FA11 is an electric piano. You're now playing the copy, UB01, and everything on every page will do what you ask.

> **Try this**
>
> ■ VOICE, hold a note down, and while it's still sounding turn 1 ALGO slowly, one click at a time. A diagram appears and changes as you turn, and so does the sound.
>
> Some will sound better than the original. Plenty will sound worse. A few will come out thin, or buzzy, or so quiet you'll wonder whether it's still on. All of that is normal, none of it has broken anything, and by the end you'll have a feel for what this encoder does long before anyone explains it. Push *push* 1 to put the diagram away.

Ignore the diagram for now. We're not going to talk about what it means for a few sections yet, and there's a good reason: it makes far more sense once you've heard what it does than the other way round. If you've made a mess of UB01, copy FA11 over it again. That's what copying is for.

3.2 Six sine waves

Each of the six sine wave generators is called an **operator**. It's an odd word for something that just makes a tone, but it's what they've been called since the early 1980s and it's what everyone who works with these instruments calls them, so it's worth getting used to. On Glass they're OP1 to OP6, the six buttons down the left of the grid.

We could take a factory sound apart to see how it works, and that's a perfectly good way to learn, but it's much easier to hear what one thing does when there's only one thing there. So we're going to start from a blank sound instead. Every one of your own slots starts as one: **INIT VOICE**.

■ SOUNDS › ❶ BANK › USER A › ❷ SOUND › UA01

Play a note. It should be plain, soft and a little dull, not far off the sound of blowing across the top of a bottle. It isn't exciting, and it isn't meant to be. What you're listening to is a single operator, OP1, making a pure sine wave, with the other five turned all the way down to nothing.

> ### If it sounds like something else entirely
>
> Then UA01 has been used before, by you or by someone else. Any slot you haven't touched is INIT VOICE, and the SOUND box lists them by name, so turn ❷ SOUND until you find one that says INIT VOICE, and use that slot wherever this chapter says UA01.

The operator you can hear is doing the simplest job an operator can do: making a sound and sending it to the output. An operator doing that job is called a **carrier**.

The other five haven't gone anywhere. They're still sitting there, switched on, set to zero. In section 3.6 we're going to turn one of them up, and it won't get louder: it'll get brighter. That single surprising fact is what this whole instrument is built on.

The sounds you'll build

This chapter builds five sounds of your own, in the first five slots of USER A. Each starts as a copy of SINE 1, the first one, so you're never far from a sound that works:

Slot	Sound	Sections	What it shows
UA01	SINE 1	3.4	one operator, one sine wave: the place to start
UA02	PLUCK 1	3.5	an envelope that falls away to nothing
UA03	SWELL 1	3.5	an envelope that creeps in
UA04	E PIANO 1	3.6	a modulator: brightness from a sine you can't hear
UA05	THREE OPS	3.6 and 3.7	two modulators side by side, then stacked

Section 3.4 says how to copy a sound into another slot; after that, each section says which slot to use.

3.3 How Glass is laid out

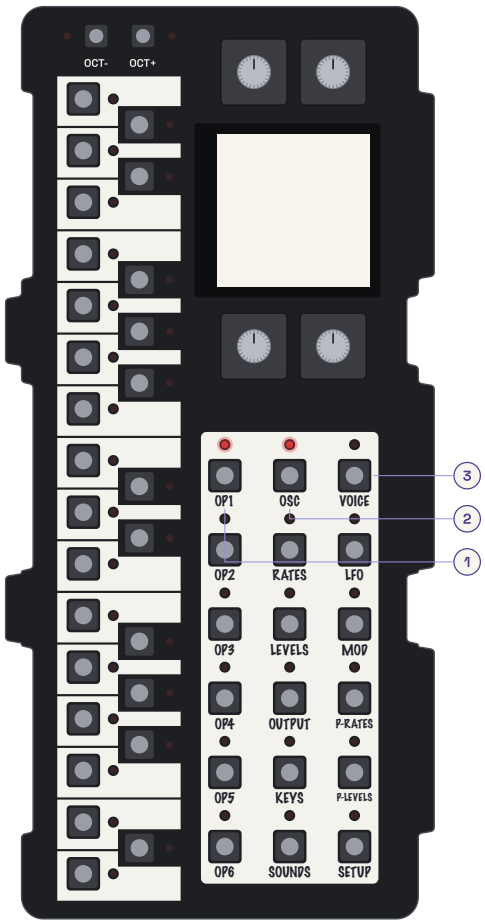


Figure 3.1 Glass's buttons. 1: OP1 to OP6 choose an operator. 2: the operator's pages, then SOUNDS. 3: the pages for the whole sound, then SETUP.

Glass's eighteen buttons are three columns of six, and each column has one job.

1. **OP1 to OP6** choose which operator you're working on. They don't change the page: press OP3 while you're on the RATES page and you're on OP3's RATES.

2. **The operator's pages**: OSC, RATES, LEVELS, OUTPUT and KEYS. Each operator has its own set, so these show whichever operator you chose. Then SOUNDS.

3. **The pages for the whole sound**: VOICE, LFO, MOD, P-RATES and P-LEVELS, which belong to the sound rather than to any one operator. Then SETUP.

On an operator's page the top bar says which operator as well as which page, so `OP1 OSC` is OP1's OSC page. On the other pages it just says the page. One button, one page, every time: no page remembers anything that you have to undo before following an instruction.

The lights tell you the same thing. The operator you've chosen is lit, and so is the page you're on.

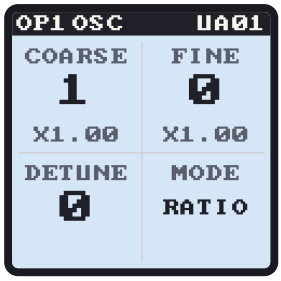


Figure 3.2 OP1's OSC page on INIT VOICE. Under COARSE and FINE, what they make together: X1.00, the operator running at exactly the note you played.

Listening to one operator at a time

Hold an OP button for half a second and that operator goes silent; hold it again to bring it back. Its light goes dim so you can see which ones are muted (and blinks if the muted one is the operator you've chosen), and the diagram on the VOICE page fades it out.

This is the most useful button on Glass for learning, because the next few sections argue that some operators are heard directly and some are only heard through others. You don't have to take that on trust. Mute them one at a time and listen: silence one and the sound disappears; silence another and the tone changes without the sound going away. The first kind is a carrier. The second is a modulator, which is section 3.6.

> #### Mutes are kept with your own sounds
>
> On a factory sound a mute is a change like any other: the star appears in the top bar, and it's forgotten as soon as you choose another sound. On one of your own, the mute is part of the sound and saved with it. So if one of your sounds seems to have lost something, look down the OP buttons for a dim light.

Looking inside the factory sounds

Every page works on a factory sound too, so you can open any of them and see exactly how it's made, which, by the end of this chapter, you'll be able to read. Turn something and you'll hear what it does; the star in the top bar says it's changed, and choosing the sound again puts it back.

3.4 One note, louder and quieter

Time to change something. Make sure you're on UA01, your INIT VOICE, and press ■ OP1 then ■ OUTPUT.

■ OP1 › ■ OUTPUT › ❶ LEVEL

> **LEVEL** (encoder 1, 0-99) how much this operator puts out. Hold a note down and turn it all the way to 0. Silence. Bring it back to 99 and the sound returns.

So LEVEL is the volume of this operator, which might seem a slightly odd thing to have, given there's already a volume control on the Root.

> ### Worth remembering
>
> It isn't really volume. It only looks like volume today, because OP1 is a carrier and there's nothing else switched on for it to work with.
>
> Turn the very same encoder on an operator that isn't a carrier and it changes how bright the sound is instead: same page, same box, entirely different result. Which one you get depends on what job that operator has been given. That's section 3.6.

The other three boxes on OUTPUT come later: KEY VEL and RATE SC in section 3.10, AMP MOD in section 3.9.

The tuning page

■ OP1 › ■ OSC

> **COARSE** (encoder 1, 0.5 and 1-31) what pitch this operator runs at, compared with the key you pressed. On a carrier all it can do is move the note up in jumps, which is mildly interesting at best.

> **FINE** (encoder 2, 0-99) the same, in smaller steps. Under both boxes the screen shows what COARSE and FINE make together: X1.00 is exactly the note you played, X2.00 twice it.

> **DETUNE** (encoder 3, -7 to +7) a very small amount either side of in tune.

> **MODE** (encoder 4, RATIO or FIXED) section 3.6 has this one.

The three tuning controls become some of the most important on Glass in section 3.6, once the operator you're retuning is one you can't directly hear. Put anything you moved back where it was before you go on: COARSE 1, FINE 0, DETUNE 0, LEVEL 99.

Keeping it

You've made a sound. It isn't a very good sound, but it's yours, and the next section builds on top of it, so give it a name. On the SOUNDS page, turn **4 NAME** to change the letter under the cursor, and push it to move on to the next letter. Call it **SINE 1**. There's no saving to do: it already has.

Then make two copies of it to work in, because the next section turns them into two other sounds:

■ SOUNDS › **3 COPY TO** › **UA02** › *hold* **3** and again to UA03

Copying moves you onto the copy, so go back to UA01 afterwards with **2 SOUND**.

> ### What you have so far
>
> A sound called SINE 1 in UA01: one operator, one sine wave, nothing else running at all. UA02 and UA03 hold copies. It's the plainest noise Glass can make, and it's also the only sensible place to start from. You'll find yourself coming back to it every time you want to build something new.

3.5 The shape of a note

Play SINE 1. Hold a note. Let go.

Whatever else is wrong with that sound, the biggest problem is that it arrives at full volume the instant you touch the key and vanishes the instant you let go. Chapter 2 (section 2.3) said what's missing: the **envelope**, the shape of the note. Most synths give you four controls for it, attack, decay, sustain and release. Glass does something more flexible, and it's worth understanding on its own terms rather than squashing it into the usual four.

Four corners

Think of it as drawing a shape. You place four corners, and you say how quickly the sound travels between them.

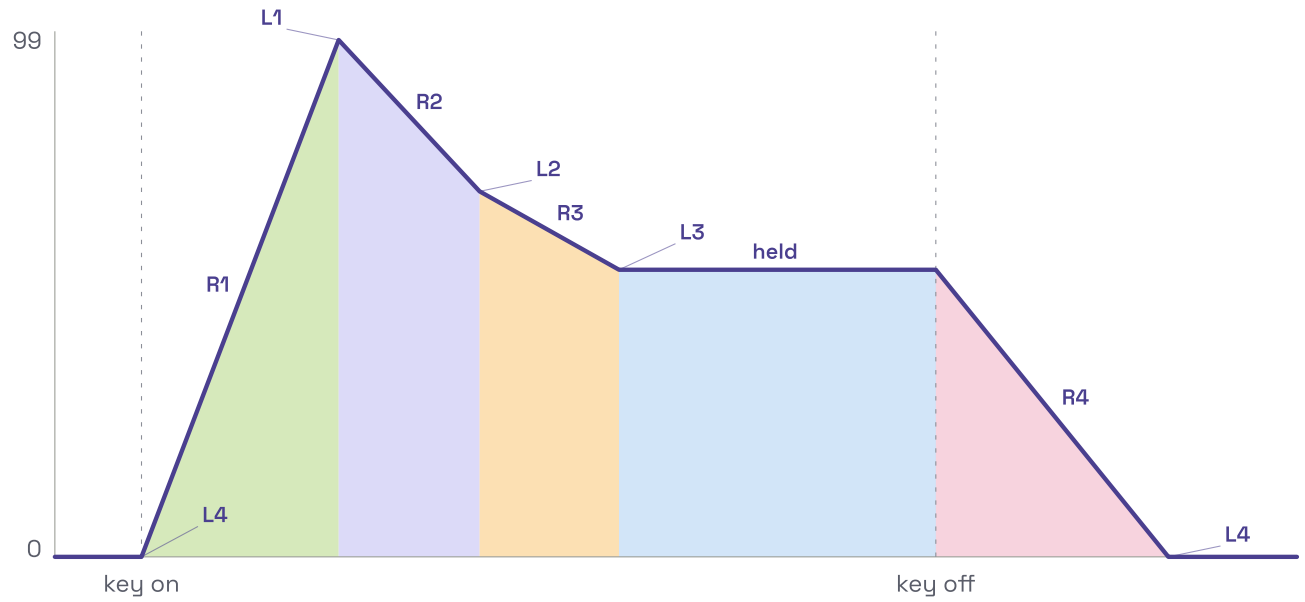


Figure 3.3 Four corners, L1 to L4, and four speeds, R1 to R4, for travelling between them. The note starts at L4, travels through L1 and L2, rests on L3 while you hold the key, and goes back to L4 when you let go.

Reading it from the left: before you press anything, the sound sits at L4. Press a key and it sets off at speed R1 towards L1. When it gets there it turns and heads at speed R2 towards L2, then at R3 towards L3. And there it stays, for as long as you keep the key down: a second, a minute, all afternoon.

Let go and it travels at R4 back to L4, where it started.

Four places to be, four speeds for getting between them. That's the whole system, and every sound Glass makes uses it. Every operator has one of these, so a sound has six of them.

Where to find it

The speeds and the corners are on two pages of whichever operator you've chosen:

OP1 › RATES R1 to R4 OP1 › LEVELS L1 to L4

> **R1** (encoder 1, 0-99) how fast the sound travels from where it starts up to L1.

> **R2** (encoder 2, 0-99) from L1 to L2.

> **R3** (encoder 3, 0-99) from L2 to L3, where it settles.

> **R4** (encoder 4, 0-99) after you let go: from wherever it was back to L4.

> **L1** (encoder 1, 0-99) the first corner, which the note heads for as soon as you press the key.

> **L2** (encoder 2, 0-99) the second corner.

> **L3** (encoder 3, 0-99) where it settles and stays while you hold the key down.

> **L4** (encoder 4, 0-99) where it sits before you play, and where it goes back to afterwards.

Both pages show the same picture of the shape, and each number sits on a chip the colour of its part of the picture: change the green number and the green part moves. That correspondence is worth more than anything else on these pages. The picture holds the note long enough to show every stage and then lets go, and a very long shape zooms out to fit (the readout under it, X1.6 and so on, says by how much).

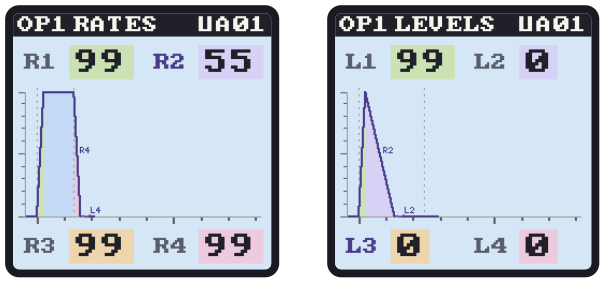


Figure 3.4 RATES and LEVELS share one picture, and each number sits on a chip the colour of its part of the shape. Here, the pluck: R2 55, and L2 and L3 at 0.

The bit that trips everyone up

Here's the thing to get straight before you touch anything, because it catches out people who've been doing this for years.

The rate numbers are not durations. They're speeds. A high rate means the sound moves quickly, which means that part of the note is *shorter*.

So turning R1 up makes the attack snappier, not longer. Turning R4 down makes the note hang around after you let go. It's the opposite of what most people expect the first time.

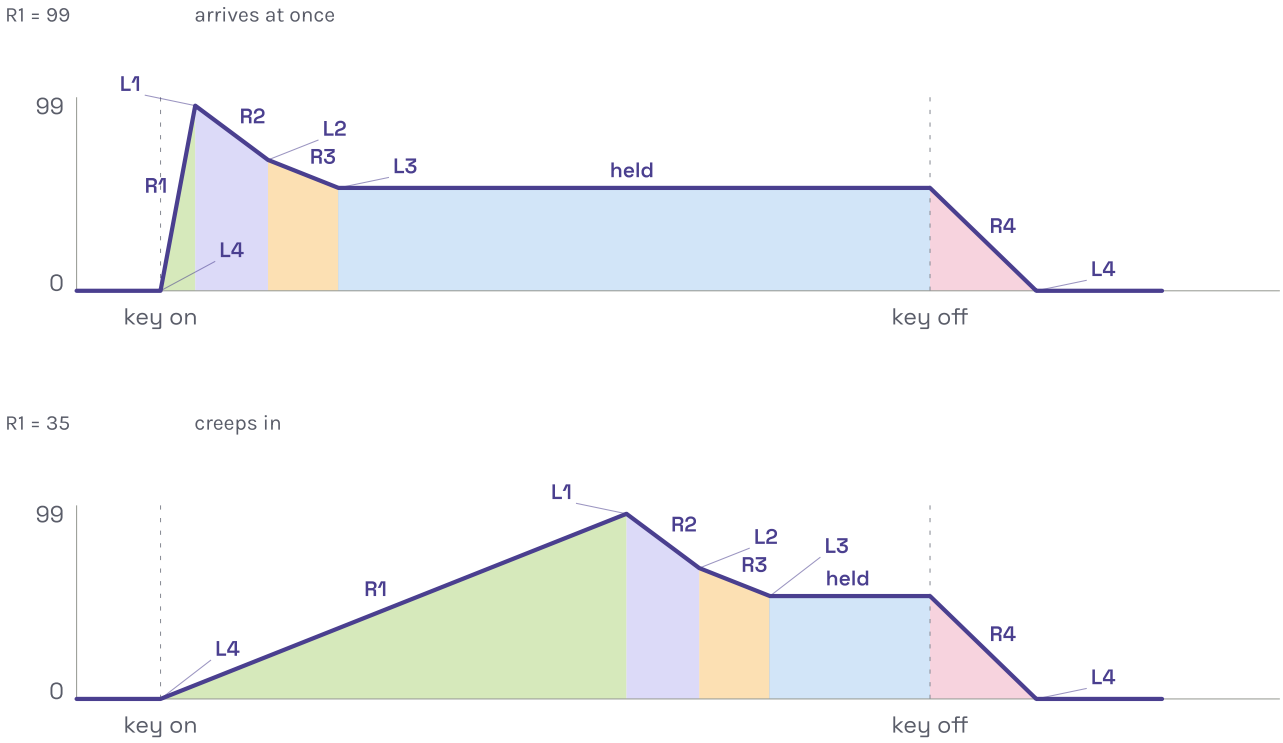


Figure 3.5 The same levels twice. Top, R1 at 99: the note arrives at once. Bottom, R1 at 35: it creeps in. A bigger rate is a faster move, so a shorter stage.

Try this

On SINE 1, ■ OP1 › ■ RATES › 1 R1 › **45**. Play a note and hold it.

Instead of arriving, the sound now creeps in. Turn R1 back up to 99 and it snaps in again. That one encoder is the difference between a bowed instrument and a struck one.

Where the note sits while you hold it

L3 is the level the sound settles at once it's finished travelling: the level you hear for as long as your finger stays down. It's the one corner whose length you don't set, because you set it yourself, with your hand.

L3 at 99 and a held note goes on at full volume, like an organ. L3 at 0 and the note dies away to nothing whether you hold the key or not, like a piano or a bell. That one choice, does the sound sustain or does it die away, accounts for most of the difference between the instrument families you already know by ear.

The strange one

L4 is where the sound sits before you press anything, and also where it goes back to after you let go. It's the same corner doing both jobs.

Which raises a question worth sitting with for a moment: what happens if L4 isn't zero? The answer is that the sound never actually stops. Once you've played a note it sits at whatever level you've set, quietly, all the time, whether you're playing or not: a drone under everything you do.

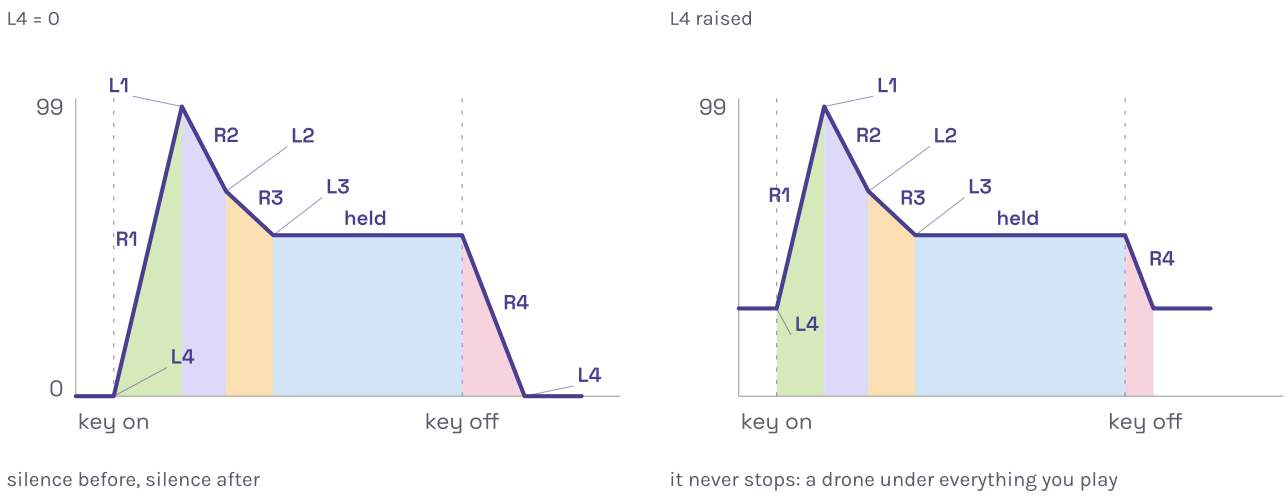


Figure 3.6 L4 is where the note starts and where it goes back to. Raise it off the floor and the note never stops.

Try this, then put it back

OP1 › LEVELS › 4 L4 › **80**. Play a note and let go. There it is: a hum that doesn't stop.

This is usually a mistake, and when a sound you're building won't shut up, L4 is the first place to look. But it's also a real effect that people use on purpose, so it's worth having heard it deliberately once rather than only ever meeting it by accident.

Now turn L4 back to 0, and the hum dies away.

Letting go early

Chapter 2 said this about every synth, and it's just as true here. Letting go of a key doesn't wait for anything: whatever the note was doing at that moment, still climbing towards L1 or halfway down to L2, it stops and heads straight for L4 at R4.

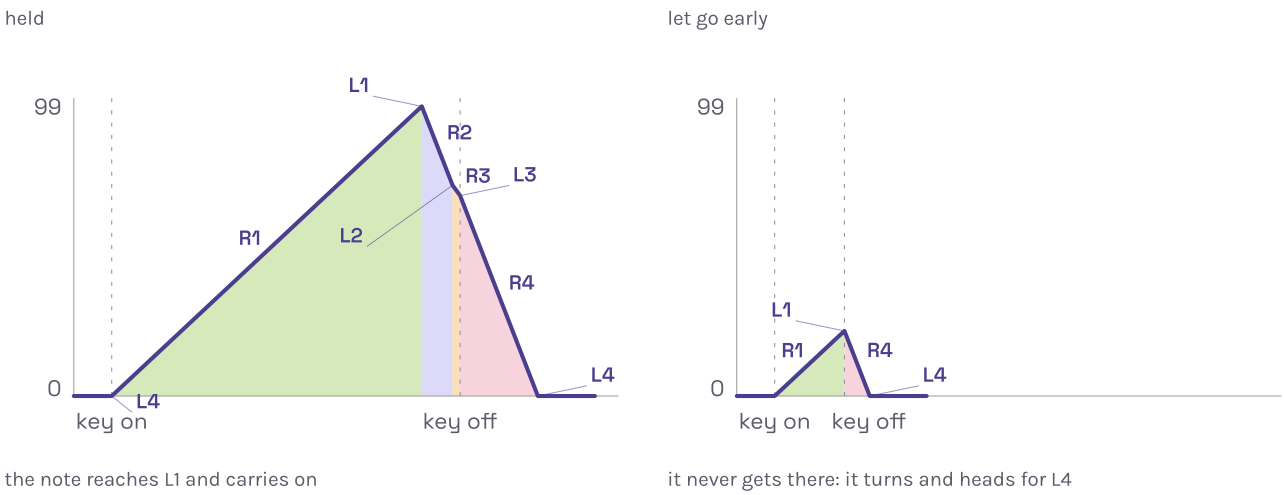


Figure 3.7 Letting go doesn't wait: wherever the note had got to, it turns and heads for L4 at R4.

So a sound with a slow attack never reaches full volume if you play it in short stabs. That isn't a fault, it's the character of the thing, and it's why slow sounds want long, held notes.

Four shapes worth knowing

Here are four envelopes that between them cover most of what you'll want. Put them in on OP1 of SINE 1 and listen. Every number is a starting point rather than a rule: nudge them until they sound right to you.

Organ

Instant on, full volume for as long as you hold it, instant off.

RATES	R1 99	R2 99	R3 99	R4 99
LEVELS	L1 99	L2 99	L3 99	L4 0

Pluck

Arrives at once and falls away to nothing, whether you keep holding or not.

RATES	R1 99	R2 55	R3 99	R4 99
LEVELS	L1 99	L2 0	L3 0	L4 0

Bell

The same fall, far more gradual: so gradual that letting go interrupts it.

RATES	R1 99	R2 25	R3 99	R4 30
LEVELS	L1 99	L2 0	L3 0	L4 0

Swell

Creeps up from silence, settles just below the top, fades gently when you let go.

RATES	R1 35	R2 60	R3 60	R4 45
LEVELS	L1 99	L2 90	L3 90	L4 0

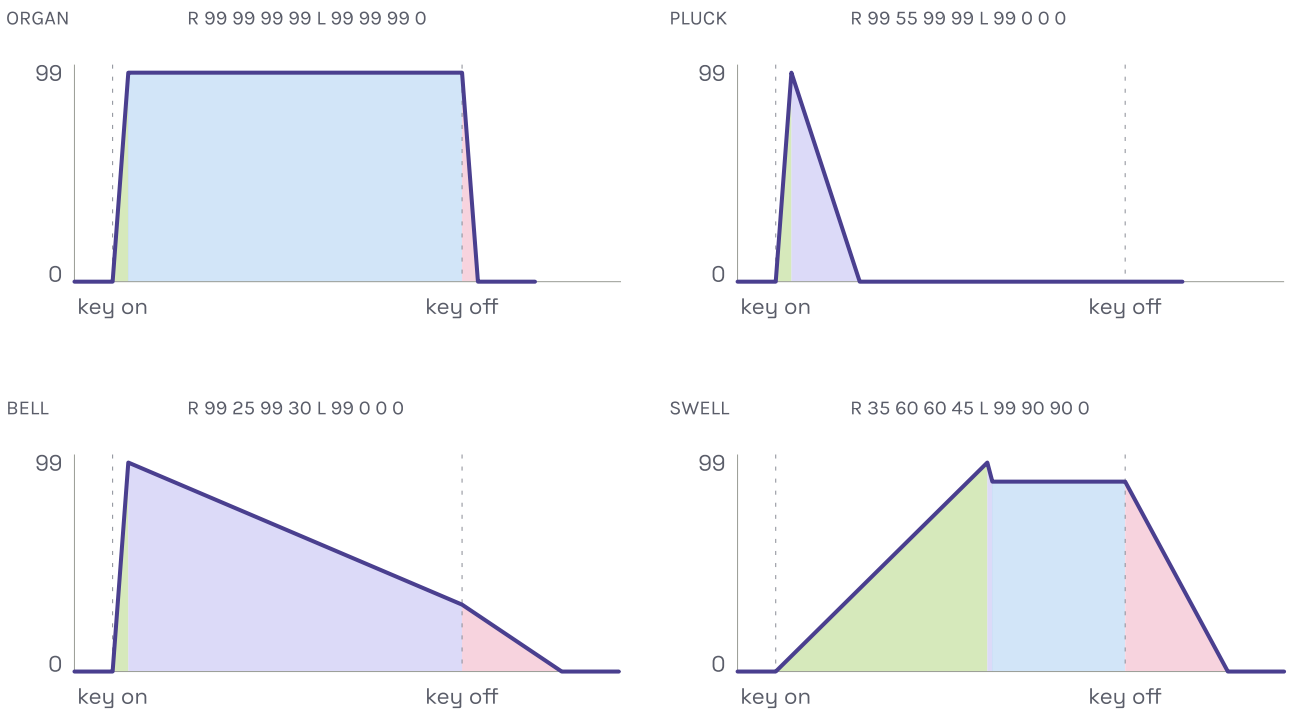


Figure 3.8 The four shapes from the recipes, each with the key held for the same time and let go at the dashed line.

Look at what the bell does

Every shape here is drawn with the key held for the same time and let go at the dashed line. Three of them have finished travelling by then and are sitting still. The bell hasn't: it's still on its way down when you let go, which is why its pink release is the only one you can see.

That's also why R4 does nothing at all on the pluck. By the time you let go the sound is already at zero, so there's nowhere left for it to travel.

Try this

Take the pluck and change one number at a time. Bring R2 down to 20: now it's a bell. Bring L2 up to 70: now it's an organ with a bump at the start. Bring R4 down to 20: now every note trails off after you let go, and playing fast makes them pile up on top of each other.

You aren't looking for the right answer here. You're building up a feel for which encoder does what, which is worth more than any table.

Build the pluck in UA02 and the swell in UA03, the two copies of SINE 1, and name them `PLUCK 1` and `SWELL 1`.

What you have so far

Three sounds: SINE 1, PLUCK 1 and SWELL 1. All of them one sine wave, all of them plain, but with completely different characters, because the shape of a note carries more information than the note itself.

3.6 Modulators

Everything so far has used one operator out of six. This section turns on a second one, and the result isn't what you'd expect, which is the whole reason this instrument exists. Chapter 2 said Glass has no filter, and makes its harmonics a completely different way. This is the way.

Go to SINE 1 and let's find out.

Turning up the one you can't hear

OP1 is the one you've been listening to. OP2 has been sitting at zero the whole time.

OP2 › OUTPUT › LEVEL

Play a note again and again, and slowly bring that from 0 up to about 70.

The sound doesn't get louder. It gets brighter: harder, buzzier, more like a real instrument and less like a test tone. Keep going to 99 and it turns harsh and metallic.

> ### This is the thing
>
> Same encoder, same page, same box as the one that changed the volume in section 3.4. Completely different result.
>
> The difference is the job OP2 has been given. It isn't connected to the output at all: it's connected to OP1. An operator wired to the output is a **carrier**, and you hear it. An operator wired to another operator is a **modulator**, and you never hear it directly; you only hear what it does to the one underneath. Hold *hold* OP1 to mute it and the sound vanishes, OP2 and all.

Why brightness, though?

Here's what's actually happening, and it's worth a minute, because everything else in FM synthesis follows from it.

A modulator pushes the carrier's pitch up and down. That's all it does. Turn the modulator's level up and it pushes harder.

If something pushes a note's pitch up and down a few times a second, you hear it as vibrato: a wobble. Speed the wobble up and at some point it stops sounding like a wobble at all. Your ear can't follow it any more, so it does something else with it: it hears a change in tone.

A modulator is doing vibrato several hundred times a second. Far too fast to hear as movement, so you hear it as brightness instead, and the harder it pushes, the brighter the result. That's the trick. That's the whole of it. Everything from here is deciding what to push with, how hard, and when.

What note is it pushing at?

The modulator has its own pitch, and it matters far more than the pitch of a carrier ever would. It's set on OP2's OSC page.

OP2 › OSC

COARSE (encoder 1, 0.5 and 1-31) a multiplier rather than a pitch. At 1 the modulator runs at the same frequency as the note you played; at 2 twice as fast, at 3 three times, and so on.

FINE (encoder 2, 0-99) fills in the gaps between the whole numbers. The screen shows the result under both boxes: X3.50 is three and a half times.

DETUNE (encoder 3, -7 to +7) a very small amount of not-quite-right, on purpose.

MODE (encoder 4, RATIO or FIXED) below.

Whole numbers

With OP2's LEVEL at about 70, hold a note and step 1 COARSE through 1, 2, 3, 4, 5 and on up to about 12. Listen to each one for a few seconds before you move on.

They all sound different, and they all still sound like a musical note: thin, hollow, reedy, nasal, brassy. It's the same instrument playing the same pitch with a different tone each time.

They stay musical because whole-number multiples of a note are exactly the harmonics chapter 2 talked about, the ones real instruments make. Twice the note is the octave above; three times is a fifth above that. Stack them up in different amounts and you get flutes, clarinets, oboes, trumpets: everything that vibrates in a well-behaved way.

Breaking it on purpose

Set COARSE to 3, then turn 2 FINE up to about 50, so the screen says X4.50 or so. Play something.

It stops sounding like a note and starts sounding like an object. Bells, gongs, metal bars, glass. Suddenly there's a percussion section in here that wasn't there a moment ago.

The frequencies you've made aren't whole-number multiples any more, so they don't line up with anything your ear expects from a vibrating string or a column of air. It hears metal instead. This is why FM was famous for bells and electric pianos long before it was famous for anything else: it's extremely good at the sounds other kinds of synth find awkward.

Close, but not quite

Put COARSE back to a whole number, 2 is a good one, FINE back to 0, and then move 3 DETUNE just a click or two off 0. Play a long held note, and wait.

The sound moves. It shimmers, or beats slowly, or seems to breathe, because two frequencies that are nearly but not quite in step drift in and out of line with each other. This is the difference between a sound that's obviously synthetic and one that seems alive. Real instruments are never perfectly in tune with themselves; strings stretch, air moves, wood warps. That's what DETUNE is for.

When the brightness changes

Now for the part that turns all this into actual instruments.

The modulator has an envelope of its own, on OP2's RATES and LEVELS pages, and it works exactly like OP1's. But because OP2's level changes the brightness rather than the volume, its envelope changes *how the brightness changes over time.*

Think about hitting a piano key hard: the sound is bright at the very start and then mellows, while the note is still ringing. That's not the volume changing, it's the tone. Almost every acoustic instrument does it.

Build an electric piano

Copy SINE 1 to UA04 (section 3.4 showed how), name it `E PIANO 1`, then:

OP1: the pluck, with R2 opened out a little.

RATES	R1 99	R2 40	R3 99	R4 99
LEVELS	L1 99	L2 0	L3 0	L4 0

OP2: ■ OP2 › ■ OUTPUT › 1 LEVEL › **75** , ■ OSC › 1 COARSE › **2** , and its own envelope set to fall away faster than OP1's.

RATES	R1 99	R2 50	R3 99	R4 99
LEVELS	L1 99	L2 0	L3 0	L4 0

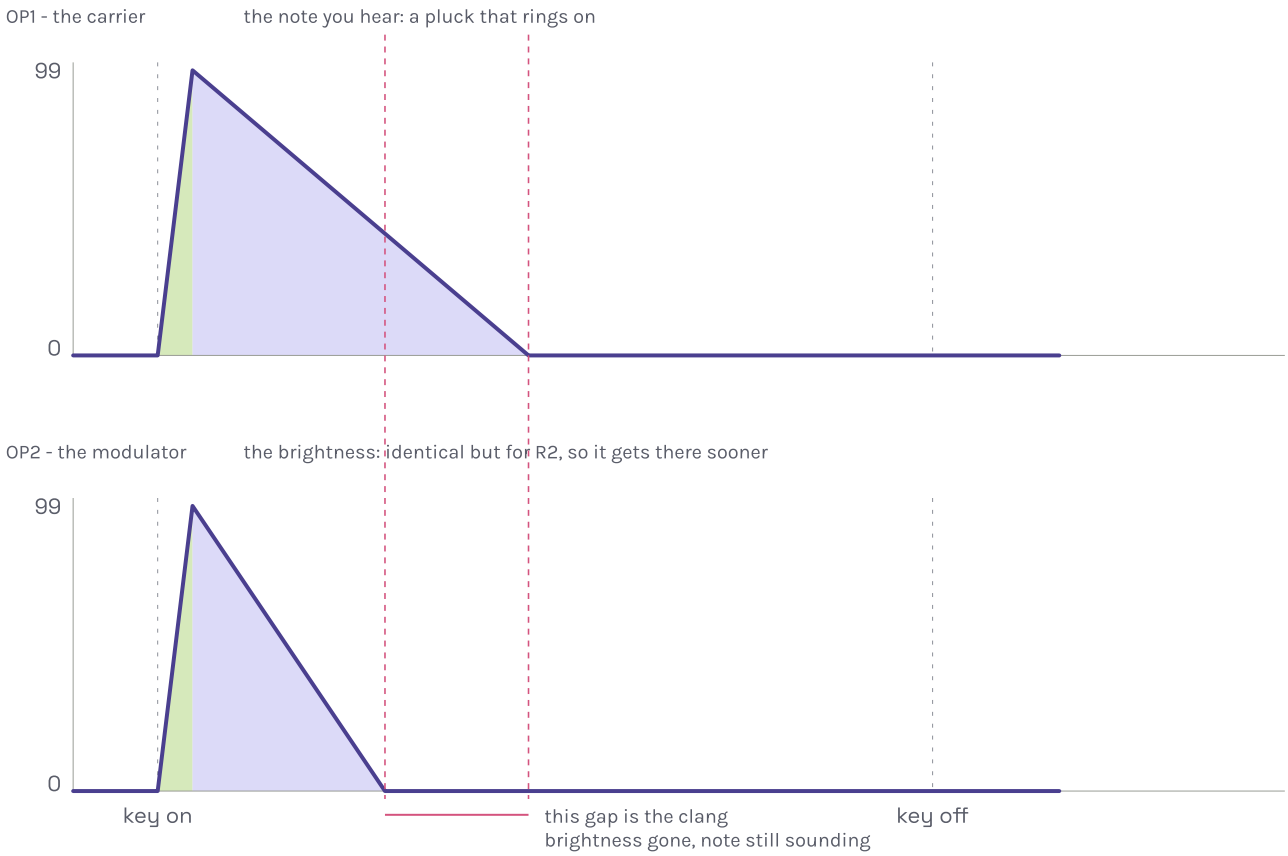


Figure 3.9 The electric piano's two envelopes. They differ only in R2: the brightness (OP2) dies away before the note (OP1) does, and the gap between them is the clang.

Play it. The note starts bright and clangy and settles almost at once into something soft and round underneath, and it should be recognisably an electric piano.

The two shapes are almost the same. The only number that differs is R2, and the whole instrument lives in the gap between them: the note is still sounding there, but the brightness has already gone. That gap is the clang. Go back to OP2's R2 and slow it down: the clang lasts longer. Speed it up and it becomes a click. That one number is the difference between a whole family of instruments.

Two envelopes, two jobs

The carrier's envelope shapes how loud the note is. The modulator's envelope shapes how bright it is. They're completely separate, and using them against each other, bright and quiet, or dull and loud, is where most good FM sounds come from.

Fixed instead of ratio

One more option on the OSC page: ■ OP2 › ■ OSC › 4 MODE, RATIO or FIXED.

Everything so far has been RATIO: the modulator's frequency follows the note you play, so the sound has the same character wherever you play it. Switch to FIXED and the operator stays at one frequency whatever key you press. COARSE then picks 1, 10, 100 or 1000 Hz, FINE goes up from there, and the screen shows the frequency in Hz.

That sounds like a mistake, and for most instruments it would be, but it's exactly right for anything where part of the sound doesn't move with the pitch: the thump of a bass drum, breath on a flute, the knock of a hammer. Real instruments are full of things that stay the same however high or low you play, and FIXED is how you get them. Put OP2 back to RATIO before you go on.

A second modulator

Adding a third operator is where it starts to get properly interesting, and it needs a different arrangement. Copy SINE 1 to UA05, name it `THREE OPS`, and on the copy:

■ VOICE › 1 ALGO › 16



Figure 3.10 Turning ALGO on the VOICE page shows the algorithm as you go. This is algorithm 16.

Algorithm 16 has one carrier, OP1, with OP2, OP3 and OP5 all feeding into it directly. For now leave OP5 at 0 and use the other two.

> #### Try this
>
> Set OP2 to LEVEL 65 with COARSE 1, and OP3 to LEVEL 50 with COARSE 7.
>
> Mute each one in turn, holding its OP button, and listen to what disappears. OP2 gives you the body of the sound; OP3 adds a thin, reedy edge on top. Together they make something neither could on its own.
>
> Now give them different envelopes, so the edge fades before the body does.

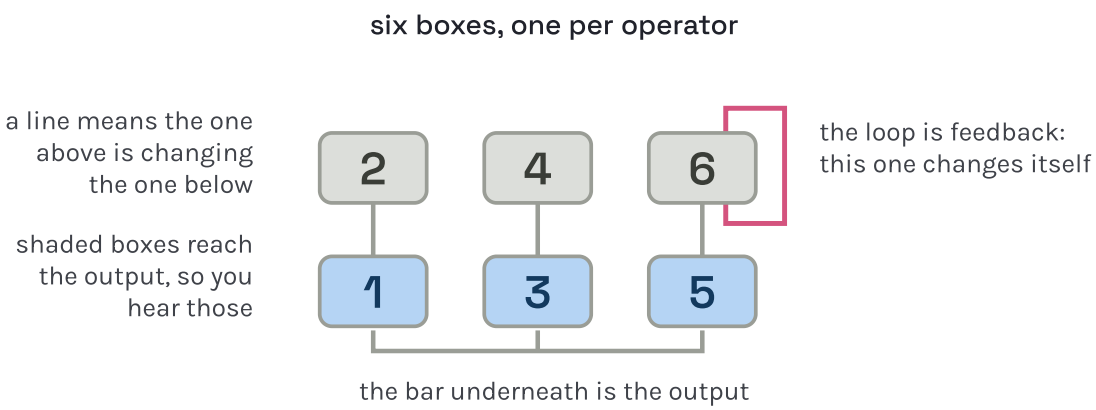
Two modulators side by side on one carrier behave pretty much as you'd hope: each adds its own character and you hear both. That makes them predictable and easy to work with, which is why this arrangement turns up in a great many factory sounds. There's another way to connect three operators, and it's much less well behaved. That's the next section.

3.7 Algorithms

You've been using algorithms since section 3.1, when you turned ALGO and listened to a sound fall apart and put itself back together. Now for what was actually happening: an algorithm is the wiring diagram, which operators are carriers and which modulate which.

Reading the picture

There are thirty-two of them, and turning ALGO on the VOICE page shows you the one you've landed on. It stays up until you push the encoder, change page or turn another encoder. Every diagram is made of the same parts:



Algorithm 5
three carriers, three modulators, one each

Figure 3.11 How to read an algorithm.

The whole chart is at the end of this section. It's meaningless until you know what the boxes mean, and alarming if you meet it cold, so leave it until you've read the rest.

Tall and thin, short and fat

Look down the chart and the shapes fall into two rough families, and you can predict a great deal from which family an algorithm belongs to.

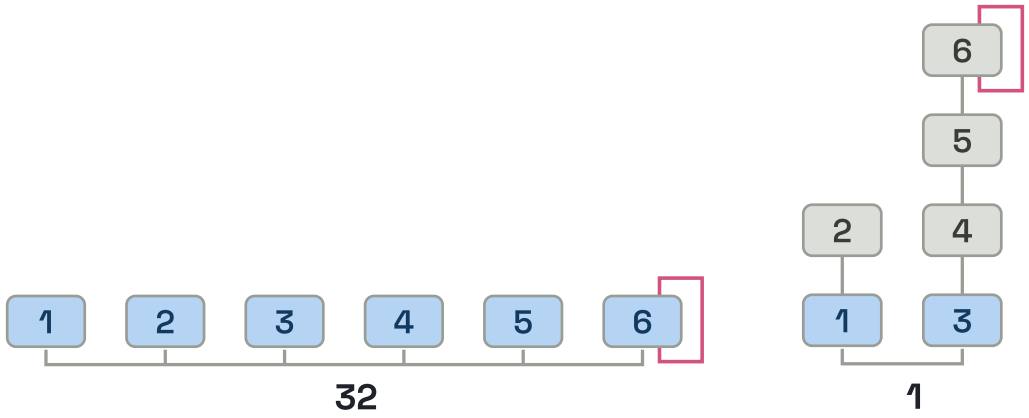


Figure 3.12 Short and fat, and tall and thin. Left, algorithm 32: six carriers and nothing modulating anything. Right, algorithm 1: a stack four deep beside a pair.

Short and fat means lots of carriers side by side. Algorithm 32 is the extreme: all six operators are carriers, none of them modulating anything. Six sine waves, added together, and nothing else. That's how a pipe organ works, separate pipes at different pitches sounding at once, and it's what these algorithms are good for: smooth, clean, blending sounds. Organs, strings, choirs, pads.

Tall and thin means operators stacked in a chain, each one modulating the next. Algorithm 1 has a stack four deep. Each stage multiplies the complexity of the one below, so the results are dense, bright, often metallic: bells, chimes, aggressive basses, sound effects.

Most of the thirty-two sit somewhere in between, and that's usually where you want to be: a couple of carriers for body, a stack for character.

> #### Try this
>
> Go back to UB01, your copy of the electric piano from section 3.1, and turn ALGO through them again, exactly as you did then. But this time watch the diagram as it goes past.
>
> You should now be able to see, before you hear it, roughly what's about to happen. That's the whole point of the chart.

Stacking modulators

In the last section you put two modulators side by side. Now for the other way. Stay on UA05, algorithm 16, and look at where OP4 goes: it doesn't feed OP1 at all. It feeds OP3, which feeds OP1.

> ### Try this
>
> Set OP3 to LEVEL 60 with COARSE 1, and put OP2 and OP5 at LEVEL 0. Familiar ground: one modulator, one carrier.
>
> Now bring OP4's LEVEL up slowly from 0.
>
> OP4 is modulating a modulator. You aren't making the sound brighter, you're making the *brightening* more complicated, and it gets strange much faster than it did before. Small moves make big differences.

Modulators side by side add up in ways you can predict. Modulators stacked on top of each other multiply in ways that are extremely hard to predict, even for people who've been doing this for decades. Which isn't a reason to avoid them. It's a reason to turn the encoder slowly and keep the ones you like.

Feedback

One operator in every algorithm has a pink loop drawn round it. That's **feedback**: an operator modulating itself.

■ VOICE ALGO, FEEDBK, OSCSYNC and TRANSP

> **ALGO** (encoder 1, 1-32) which of the thirty-two wirings the sound uses.

> **FEEDBK** (encoder 2, 0-7) how hard the looped operator modulates itself.

> **OSCSYNC** (encoder 3, ON or OFF) whether the operators start their waves afresh on every note (ON) or carry on from wherever they happened to be. The difference is in the very first instant of the note, and it's subtle.

> **TRANSP** (encoder 4, -24 to +24) moves the whole sound up or down in semitones. Unlike OCT- and OCT+, it's part of the sound.

Turn feedback up gradually and the operator's pure sine wave gets more and more jagged: first brighter, then buzzy, and at the top of the range it breaks down into noise. It's a small control with a big range, and it's how you get roaring basses, hissing cymbals and wind out of an instrument that only contains sine waves.

Try this

On algorithm 16 the loop is round OP6. OP6 modulates OP5, OP5 modulates OP1, and OP1 is the only carrier. So three operators have to be up before the loop can be heard at all: OP1, because it's the one that reaches your ears; OP5, because it's what OP6 modulates; and OP6 itself. Set OP5 and OP6 to LEVEL 70 and put OP2, OP3 and OP4 at 0.

Now set FEEDBK to 0 and step it up one number at a time. What's changing is OP6's own wave, and you hear it as a change in OP1's brightness.

Eight settings, and you can hear every step. There's no other control on Glass with that few positions and that much difference between them.

An honest word about all this

Here's something manuals tend not to say. People who are extremely good at FM, people who've made a living at it for thirty years, still spend a lot of their time turning encoders and going "ooh, what's that". The theory tells you roughly where to look. It doesn't tell you what you'll find. Nobody sits down and calculates their way to a good bell sound.

What the theory buys you is this: when something sounds nearly right, you know which encoder to reach for. And when something sounds wonderful by accident, you can work out why, which means you can do it again on purpose. That's genuinely all of it, and it's enough.

All thirty-two

Here they are. You're not meant to learn them, and nobody has. What you're meant to be able to do is glance at one and guess roughly what it'll sound like, which you now can.

Carriers are blue. The pink line is the feedback loop, and it's worth looking for: twenty of the thirty-two come in pairs or threes with identical wiring, differing only in where that loop sits.

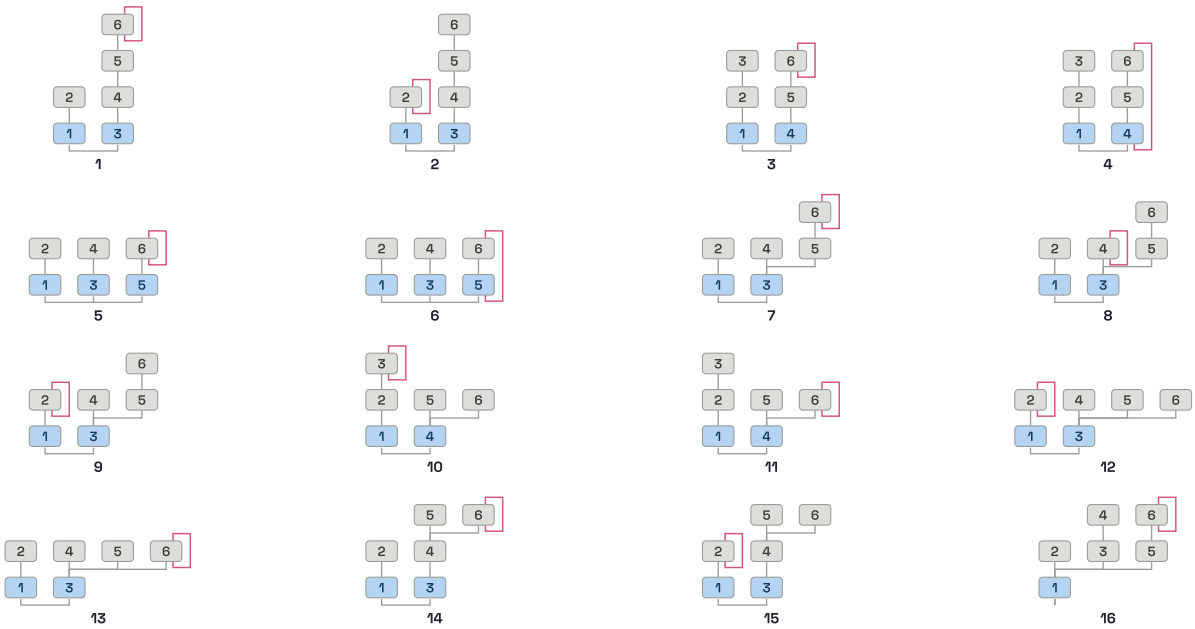


Figure 3.13 Algorithms 1 to 16. Carriers are blue; the pink loop is feedback.

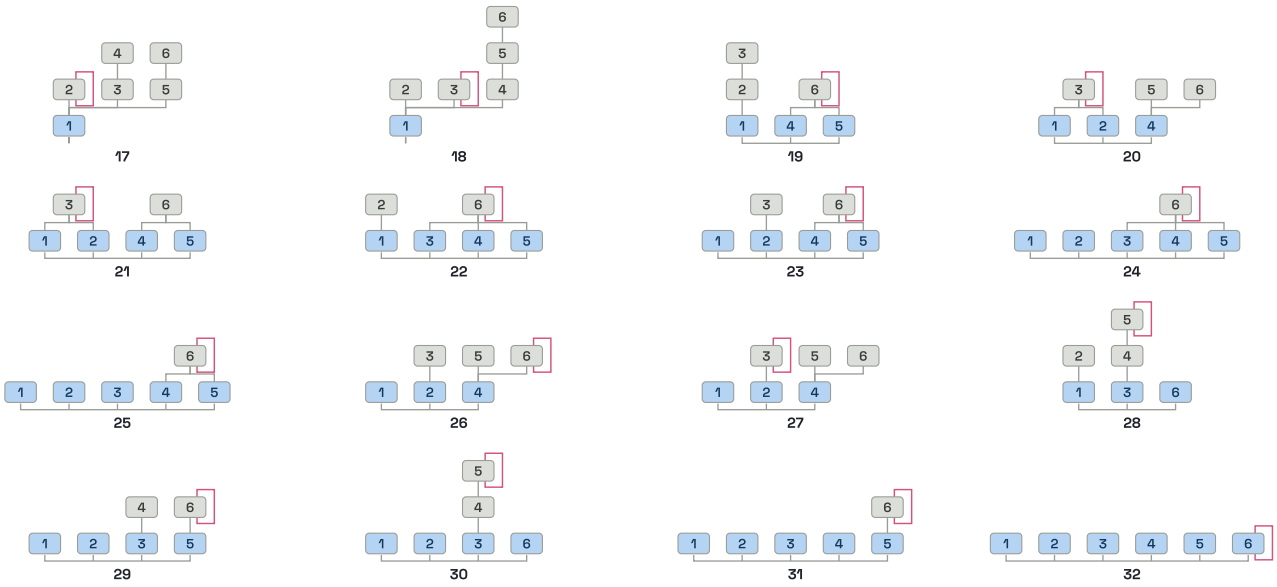


Figure 3.14 Algorithms 17 to 32, at the same scale.

Where you've got to

You can start a sound from nothing, shape how it grows and fades, make it bright or dull or metallic or alive, change how that brightness moves over time, and read the wiring diagram well enough to guess what an unfamiliar algorithm will do. That's the heart of Glass. The rest of the chapter is about pitch, movement, and what happens across the keyboard.

3.8 Pitch

Most of the pitch controls you've met already. On a carrier, COARSE and FINE move the note up or down (in octaves and fifths and so on, which is rarely what you want), and DETUNE nudges it. TRANSP on the VOICE page moves the whole sound in semitones. And a MIDI keyboard's pitch bend wheel bends the note, as far as SETUP's BEND says (section 3.12).

There's one more, and it's odder.

The pitch envelope

P-RATES and P-LEVELS hold a second kind of envelope, built exactly the same way as an operator's, with four rates and four levels. The difference is where it goes: instead of changing how loud the sound is, it changes what pitch it's at, all six operators together. It belongs to the sound, not to any one operator.

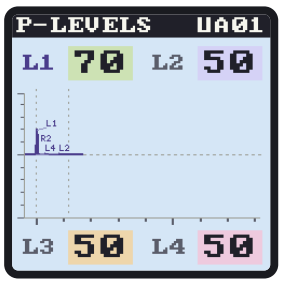


Figure 3.15 The pitch envelope's levels. 50 means no change, so here the note starts high (L1 70) and settles on its own pitch.

One thing to know before you touch it: on the pitch envelope, **50 means no change**. Above 50 bends the note up, below 50 bends it down. So a level of 0 isn't silence, it's a long way flat.

The other thing to know is that the numbers are wildly uneven. Going from 50 to 70 buys you about a fifth. The next ten steps buy another third or so, the ten after that most of an octave, and the last few run away entirely. Somewhere around 85 it stops being a pitch control and becomes a special effect. So the useful range is roughly 20 to 80, and it's worth knowing that before you spin the encoder, find yourself several octaves out and decide it's broken.

Try this, on a copy

Copy SINE 1 to a spare slot. On ■ P-LEVELS set L4 to 99 with L1, L2 and L3 at 50, and on ■ P-RATES set R1 to about 75. Play: a note that swoops down into place from above. Set R1 to about 40 instead, and you've got the sound used in every computer game until about 1985.

3.9 Movement

Every sound you've built so far does the same thing on every note: press a key, the envelopes run, and if you hold it the note sits there being exactly what it is. Chapter 2 (section 2.4) said what puts movement into a sound: the LFO, an oscillator too slow to hear, used to wobble something else. Glass's is on two pages.

The LFO page: what the wobble is like

■ LFO SPEED, DELAY, WAVE and KEYSYNC

SPEED (encoder 1, 0-99) how fast the wobble is. Human vibrato sits around five or six wobbles a second, which is about 35 to 40 here. The scale doubles roughly every twenty steps: about 2 a second at 10, 4 at 30 and 8 at 50, and past that too fast to count. Equal turns buy equal musical steps of speed, which is why it feels even all the way round.

DELAY (encoder 2, 0-99) how long the LFO waits after you press a key before it starts, then fades in. This is the one setting that makes vibrato sound like a person rather than a machine. Nobody plays a note with the vibrato already running: a violinist arrives on the note, and then it starts to move.

WAVE (encoder 3) the shape of the wobble, one of six.

KEYSYNC (encoder 4, ON or OFF) whether the LFO starts its wobble from the beginning on each note (ON) or runs on in the background (OFF). Off, every note catches the wobble at a different point, which sounds natural on single notes and a little chaotic on chords. On, every note starts its wobble together, which sounds deliberate on chords and a little mechanical on tunes.

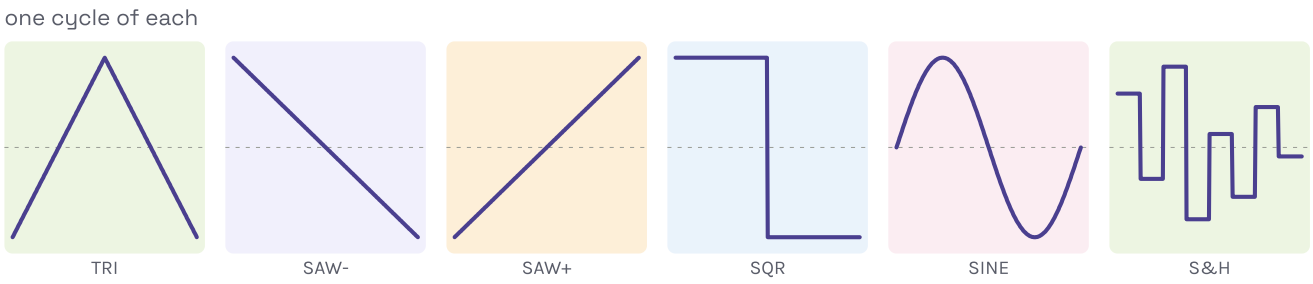


Figure 3.16 The LFO's six waves, by the names WAVE shows: one cycle of each.

TRI and SINE both go smoothly up and down, and for vibrato you won't be able to tell them apart; either is right. SQR jumps: straight to the top, straight to the bottom, nothing in between. On pitch that gives a trill between two notes; on volume, a hard on-off stutter. SAW+ climbs and drops back, SAW- falls and jumps

back; on pitch, the falling one is the sound of something powering off, over and over. S&H picks a random value, sits on it, then picks another. It's the sound of 1970s computers thinking, it has almost no use in a sound meant to be like an instrument, and it's enormous fun everywhere else.

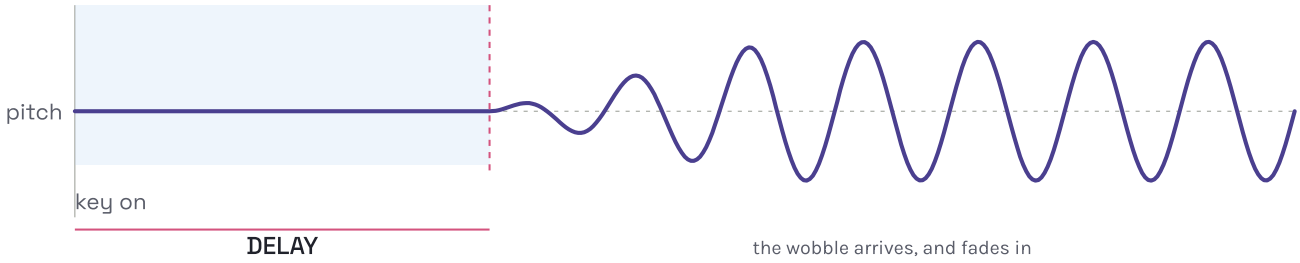


Figure 3.17 DELAY: the note starts steady, then the wobble arrives and fades in.

The MOD page: how far it moves things

MOD VIBRATO, P SENS and TREMOLO



Figure 3.18 The MOD page: how far the LFO moves the pitch (VIBRATO), how much of that gets through (P SENS), and how far it moves the level (TREMOLO). The fourth box is empty on purpose.

> **VIBRATO** (encoder 1, 0-99) how far the LFO pushes the pitch up and down. Small amounts are vibrato. Large amounts are a siren.

> **P SENS** (encoder 2, 0-7) how much of that vibrato this sound lets through. At 0 there's no vibrato at all, however high VIBRATO is. Every INIT VOICE starts at 3, so you'll rarely need it, but if VIBRATO seems to do nothing, look here. It sits right beside VIBRATO for exactly that reason.

> **TREMOLO** (encoder 3, 0-99) how far the LFO pushes the level up and down. Small amounts are a slight throb. Large amounts are a helicopter.

Try this first

On any of your sounds, ■ MOD › ◐ VIBRATO › `20` and ■ LFO › ◐ SPEED › `30`. Play a note and hold it.

Now wind VIBRATO slowly down towards 0 while the note is still sounding, and find the point where you stop noticing the wobble and just hear a slightly livelier instrument. That point is where you want to be most of the time. Then try a DELAY of about 40.

TREMOLO needs a second setting, on each operator

Vibrato moves the whole sound. Tremolo is chosen operator by operator: each operator's OUTPUT page has **AMP MOD** (encoder 3, 0-3), how much of the tremolo reaches that operator. It starts at 0 on every operator, so TREMOLO on its own does nothing at all. Turn AMP MOD up on the operators you want it on.

That's a feature rather than a nuisance. On a carrier, AMP MOD gives you a wobble in volume, which is plain tremolo. On a modulator it gives you a wobble in *brightness*: the sound shimmers rather than pulses, which is a much more musical effect, and the reason the setting is per operator.

At the very top of its range the LFO arrives at the edge of what you can hear as a pitch. Turn SPEED to 99 with VIBRATO high and the vibrato stops sounding like movement and starts sounding like a tone colour. That isn't a coincidence or a fault: it's the same thing the operators do to each other in section 3.6, happening slowly enough that you can hear it cross over.

A MIDI keyboard's mod wheel and aftertouch can bring in vibrato and tremolo too, while you play (section 3.12).

3.10 Playing across the keyboard

Play the bottom note of a piano and the top note of the same piano. They aren't the same sound an octave or six apart. The bottom note is long, dark and complicated. The top note is short, bright and thin. Nobody designed that on purpose: it falls out of the fact that one string is long and thick and the other short and tight.

An FM sound has no strings, so nothing falls out of anything. By default, the electric piano you built in section 3.6 does exactly the same thing at both ends of the keyboard, and that sameness is one of the main reasons a sound can be technically right and still sound fake.

This section is about the settings that break the sameness. It's the fiddliest part of Glass and gives the least reward per minute spent, which is why it's this far back. It's also the difference between a sound that's fine and a sound that's convincing. Everything here is *per operator*, and the interesting effects come from doing it to modulators rather than carriers.

How hard you play

On the OUTPUT page, **KEY VEL** (encoder 2, 0-7) is how much the force of your playing changes this operator's level. At 0, playing hard and playing softly sound the same from this operator. At 7, the difference is dramatic.

Velocity on a carrier is a volume control: play harder, that note is louder. Velocity on a modulator is a brightness control: play harder, that operator modulates more strongly, and the sound gets brighter as well as louder. That second one is what real instruments do. Hit a piano key harder and it doesn't just get louder, it gets harsher. Blow a trumpet harder and it gets brassier. Set velocity on your modulators, leave your carriers alone, and you get an instrument that changes character with how hard you play it.

> ### Glass's own keys can't show you this
>
> Glass's keys play every note at the same strength (velocity 100), however hard you press. To hear KEY VEL you need notes that come in harder and softer: a MIDI keyboard that senses how hard you play, plugged into the Root (section 1.10), or notes from Step, which can give every note its own velocity. Glass has no VELO setting on SETUP because it doesn't need one: each sound decides for itself, operator by operator, with KEY VEL.

Keyboard scaling

The KEYS page does one thing: it makes an operator get quieter, or louder, the further you play from a key you choose.

■ OP2 › ■ KEYS L DEPTH, R DEPTH, BREAK and CURVES

> **BREAK** (encoder 3, A-1 to C8) a note, which you choose: everything below it is the left side, everything above it the right side.

> **L DEPTH** (encoder 1, 0-99) how strong the effect is on the left side. At 0 that side does nothing, whatever curve you pick.

> **R DEPTH** (encoder 2, 0-99) the same, for the right side.

> **CURVES** (encoder 4) the shape each side takes, left then right, as two pairs of letters: -L, -E, +E or +L, so **−L+E** is -L on the left and +E on the right. All sixteen pairs are on the one encoder.

The minus curves make the operator quieter as you move away from BREAK; the plus curves make it louder. L is linear, changing evenly; E is exponential, changing slowly at first and then fast.

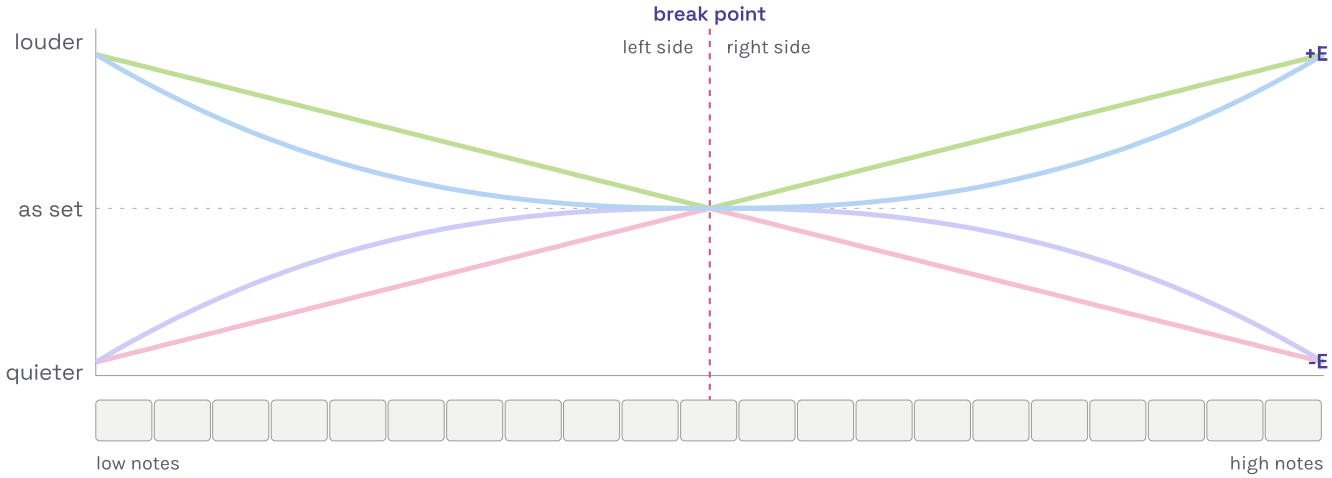


Figure 3.19 Keyboard scaling: an operator's level across the keys, either side of BREAK, for the four curves (-L, -E, +E, +L).

The classic use is a modulator with BREAK somewhere in the middle of the keyboard, a minus curve on the right and a moderate R DEPTH. Now the top of the keyboard is less brightly modulated than the bottom, which is what nearly every acoustic instrument does: high notes are purer, low notes richer.

Rate scaling, which is the one to set first

Keyboard scaling changes how loud an operator is at different pitches. **RATE SC** (OUTPUT, encoder 4, 0-7) changes how *fast* its envelope runs at different pitches.

Turn it up and notes at the top of the keyboard die away faster than notes at the bottom, which is the piano fact from the start of this section, made into a single encoder. It's the highest return on effort of anything in this section: if you're going to set one thing here and ignore the rest, set this. It reaches the operator envelopes and nothing else; the pitch envelope runs at the same speed wherever you play.

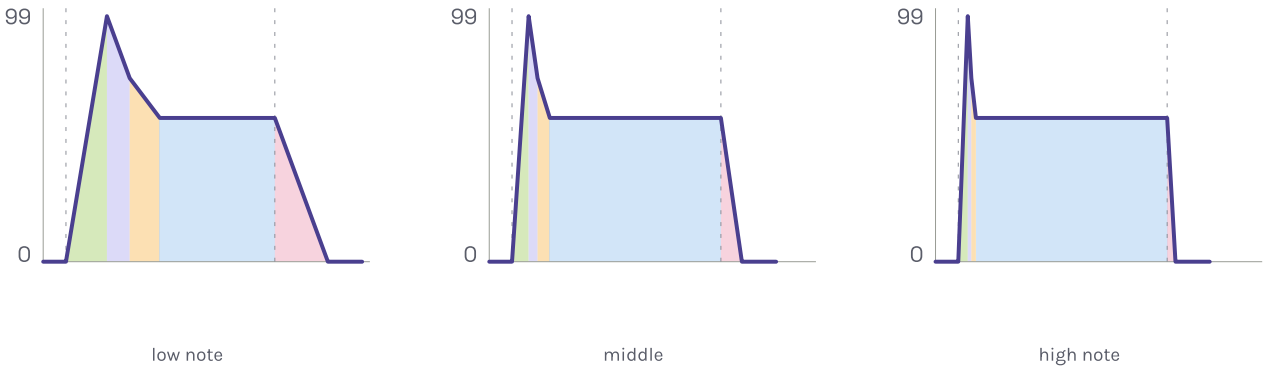


Figure 3.20 RATE SC: the same envelope played low, in the middle and high, getting shorter as it goes up.

Hear rate scaling on its own

Take your electric piano, E PIANO 1. Press OCT- once, play a low note and let it ring; then play a note two octaves up and let that ring. Roughly, time them. The high note rings on as long as the low one, or longer, which is the wrong way round for a piano.

Now ■ OP1 › ■ OUTPUT › ❹ RATE SC › 3 (OP1 is this sound's only carrier). Play the two notes again. The high note now dies away sooner, as it does on a real piano. If it still rings on, go higher.

An order to do this in

All of this is optional, all of it is fiddly, and it's easy to spend an hour here and end up with a sound that's worse. So, in order of how much they repay the effort:

1. RATE SC, on whichever operators are carriers. Notes get shorter as they get higher.
2. KEY VEL on your modulators. Playing harder gets brighter.
3. Keyboard scaling with a minus curve on the right, on your modulators. High notes get purer.
4. AMP MOD, only if you're already using TREMOLO and want the shimmer rather than the throb.

You don't have to do any of it, and a sound with none of it set is a perfectly good sound. It just won't change as you move around the keyboard, and now you'll be able to hear that it doesn't.

3.11 Effects

Glass has no effects of its own: every sound in it is the six operators and nothing else, which is exactly how the instruments it's built like always worked. The famous FM sounds of the 1980s were almost never heard dry, though. They went through chorus and reverb on the way to the record, and an electric piano with a little chorus is a different animal.

That's Haze's job. Join Haze onto the chain and it can send Glass into its chorus, echo and reverb, as much as you like (chapter 2, section 2.5). Try E PIANO 1 with a little chorus and a plate reverb.

3.12 Glass's SETUP

SETUP's THIS UNIT pages (chapter 1, section 1.9) have Glass's own settings: the ones that belong to the unit rather than to a sound. Encoder 1 on the first page is the MIDI channel, as on every unit.

> **BEND** (THIS UNIT 1, encoder 2, 0-12) how far a MIDI keyboard's pitch bend wheel bends the note, in semitones. Two is usual.

> **STEP** (THIS UNIT 1, encoder 3, ON or OFF) OFF bends smoothly; ON bends in whole semitones.

> **MODE** (THIS UNIT 1, encoder 4, POLY or MONO) POLY plays up to 16 notes at once. MONO plays one: the last key you're holding sounds, and a key played while another is held changes the pitch without starting the note again, which is how lead lines are played. Let go and it goes back to the key still held.

> **PORTA** (THIS UNIT 2, encoder 1, OFF or 1-99) glide: the pitch slides from one note to the next instead of jumping, and this is how long it takes. In POLY each note glides from the one before; in MONO it glides between keys held together.

> **WHEEL** (THIS UNIT 2, encoder 2, 0-99) how far a MIDI keyboard's mod wheel goes.

> **ROUTE** (THIS UNIT 2, encoder 3, OFF, PITCH, AMP or BOTH) what the mod wheel moves: vibrato (PITCH), tremolo (AMP) or both, through each sound's own P SENS and AMP MOD.

> **TOUCH** (THIS UNIT 2, encoder 4) the same choices for aftertouch, pressing harder on a key you're holding, if your keyboard sends it.

The last THIS UNIT page is RESET, as on every unit.

3.13 Forty years of free sounds

Here's the fact this section exists for.

Glass has every setting of the most famous FM synth of the 1980s, the DX7, and it can load that synth's sound banks: standard DX7 SysEx files of 32 sounds at a time. The DX7 came out in 1983, sold in enormous numbers, and was programmed by more or less everybody. People have been making banks of sounds for it, and giving them away, for over forty years. There are archives of thousands of them online, free, in a file format that hasn't changed. They were made for a synthesiser that cost as much as a car. All of them load into Glass, and play as they were made.

That's why the algorithms in section 3.7 are those thirty-two, in that order, and why every operator has four rates and four levels instead of the four simple stages other synths use.

From the card

Bank files go on the Root's memory card, in a folder called **GLASS**. Then press SOUNDS twice: the second press is the CARD page.

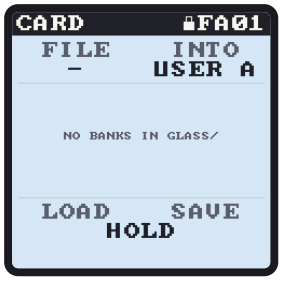


Figure 3.21 SOUNDS pressed again: the CARD page, for loading and saving whole banks of 32 on the Root's card.

> **FILE** (encoder 1) which bank file on the card, in alphabetical order.

> **INTO** (encoder 2, USER A-D) which of your banks it goes into.

> **LOAD** (encoder 3, hold) loads it. The file's 32 sounds replace all 32 in that user bank, so the screen asks you to hold.

> **SAVE** (encoder 4, hold) saves the user bank chosen in INTO as a new file on the card (USER-A-1.SYX, then USER-A-2.SYX and so on). It never writes over a file that's already there.

A bank always lands in one of your own banks; the factory banks never change, so you can always get back to them. And SAVE is how you keep a copy of your own sounds before you try something risky, such as a RESET to EVERYTHING.

> #### Before you fill all four banks
>
> Loading replaces every sound in the user bank you chose, including the ones you made. Save that bank to the card first if there's anything in it you want.
>
> Also expect a fair share of any forty-year-old bank to be unusable. They were made for a five-octave keyboard that sensed how hard you played, often for one particular record, and a lot of them are one thin sound that made sense in one arrangement. The hit rate is low, and the ones that hit are extraordinary.

Over MIDI

A computer can send Glass a bank or a single sound as SysEx, through the Root's USB socket or its MIDI IN, from an editor or librarian program. A bank doesn't go straight over anything: it waits at the top of the CARD page's list as `FROM MIDI`, and you load it from there as if it came from the card. A single sound goes into the sound you're on, if it's one of your own (on a factory sound the screen says `VOICE: PICK A USER SOUND`). Glass listens for SysEx on its own MIDI channel, or, with the Root's MIDI IN on FOLLOW, when it's the unit you touched last.

What to do with all this

Most of this chapter was about building sounds from nothing, which is the slow way and the way you actually learn. This section hands you a shortcut that skips all of it. Use both. Load a bank, find a sound in it that does something you can't explain, and then open it up, operator by operator, and work out what makes it do that. Every sound in every one of those archives is a worked example, made by somebody who knew what they were doing, and you now know how to read them.

DX7 is a trademark of Yamaha Corporation. Tessel isn't connected with Yamaha or endorsed by it; Glass loads sound files in that format.

> ### What you have at the end of this chapter
>
> SINE 1, PLUCK 1, SWELL 1, E PIANO 1 and THREE OPS in your first bank, and a working understanding of the only idea in FM that really matters: some operators are heard, some change the ones that are heard, and the difference between them is nothing but where they're plugged in. Plus a key to forty years of other people's sounds.

Chapter 4 - Volt

Start with something bright and buzzy, and take away what you don’t want.

Glass makes its sounds by adding: pure sine waves, pushing each other into brightness. Volt works the other way round, the way most synthesisers have since the 1960s. It starts with waves that are already full of harmonics, bright and buzzy, and then a filter takes away the ones you don't want. It's called **subtractive** synthesis, and it's the sound of more records than any other kind.

Volt is two instruments in one box, and every sound is one or the other:

- **CLASSIC** is the classic analogue synth of the early 1970s kind: three oscillators, a mixer, a filter that growls and whistles, and two contours. Basses, leads, brass, even drums.
- **ACID** is the little bass line machine of the early 1980s kind: one oscillator, a squelchy filter, and accent and slide. The sound of acid house.

Both are our own, but they work the classic way, with the kind of controls synths like these have always had, so what you learn here works on any synth of either kind you ever meet.

Volt plays one note at a time. That isn't a shortcoming: synths of both kinds were made that way, for basses and lead lines, and so much of their character comes from it that section 4.9 is about exactly that.

Subtractive synthesis: where it came from, and who uses it

Subtractive synthesis is the oldest way of making sounds on a synthesiser that's still in everyday use. Robert Moog's modular synthesisers of the 1960s worked this way: cabinets of oscillators, filters and envelopes joined by patch cables. In 1970 his company squeezed the most useful parts into one portable instrument with a keyboard, the Minimoog, and for the first time a synthesiser could go on stage. Kraftwerk, Parliament-Funkadelic, Gary Numan and countless others played it, and its fat basses and singing leads are still everywhere, from funk to the whining leads of 1990s West Coast hip hop.

Most of the famous synthesisers since have worked the same way: the Sequential Prophet-5, Roland's Juno-106 and SH-101, Korg's MS-20, the ARP Odyssey, and a great many made today. Learn it once and you can sit down at almost any synthesiser in the world and make it do something.

Volt's other half, ACID, makes the kind of sound the TB-303 made famous. Its story is in the introduction: a failed bass machine that became acid house, from Phuture's "Acid Tracks" in Chicago in 1987 to the acid house summer in Britain a year later, and it's still the sound of acid techno today.

Subtractive synthesis is good at warm, fat and round: basses, leads, brass, strings, pads, and drums, if you're clever. Volt isn't a Minimoog or a TB-303. But it works the way they do, so what you learn here will serve you on them, and on every subtractive synth you ever meet.

4.1 Find a sound you like

Volt comes with two factory banks: FAC A, the CLASSIC sounds, and FAC B, the ACID ones.

> **Try this**
>
> ■ SOUNDS › ❷ SOUND and go through FAC A, then ❶ BANK › FAC B and go through that. Play two or three notes on each, and try a few of them lower down with ■ OCT-: Volt is a bass synth as much as a lead one. Write down the codes of the ones you like.

The first seven in FAC A are worth a closer look, because they're classic sounds people have been making on synths like this for fifty years: PROG LEAD, SINE LEAD, FUNK QUACK and DISCO BASS, then three that might surprise you: KICK DRUM, SYNTH TOM and SNARE DRUM. Drums, made with the same three oscillators and a filter as everything else. Once you've read this chapter, open them up and see how.

As you go through, look at the CLASSIC and ACID buttons, top and second in the right-hand column. Whichever the sound is, its button is lit.

4.2 How Volt is laid out

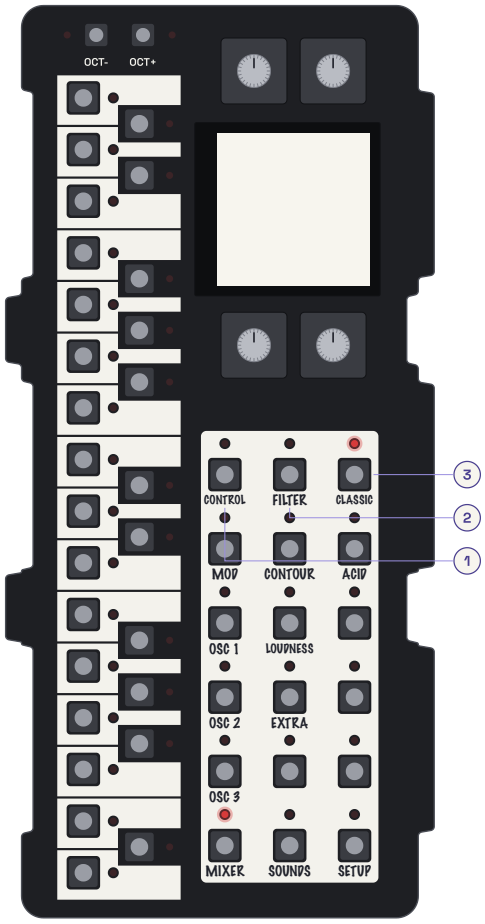


Figure 4.1 Volt's buttons. 1: CLASSIC's controllers and sound sources, in the order the sound goes through them. 2: what happens to the sound next: the filter, the two contours, EXTRA, then SOUNDS. 3: CLASSIC and ACID, then SETUP.

1. **The left column** is CLASSIC's controllers and sound sources, top to bottom: CONTROL, MOD, OSC 1, OSC 2, OSC 3 and MIXER, in the order the sound goes through them.
2. **The middle column** is what happens to the sound after that: FILTER, CONTOUR (the filter's own contour), LOUDNESS (the loudness contour), EXTRA, then SOUNDS.
3. **The right column** is CLASSIC and ACID, then SETUP at the bottom.

The readings on the screen are the classic ones, as a synth of this kind would mark its knobs: most knobs read 0 to 10, an oscillator's range is in feet (16', 8'...), waves are drawn as pictures, and times are in milliseconds and seconds.

Some controls have a little on/off switch that goes with them, and on Volt **pushing the encoder works that switch**. You'll see it as a small chip under the value, saying ON or OFF. On the MIXER page, for instance, each channel has a level and an on/off switch: turn to set the level, push to switch it on or off.

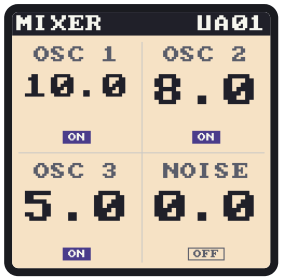


Figure 4.2 The MIXER page on INIT SOUND: a level for each oscillator and the noise, and under each the switch that pushing its encoder works (ON or OFF).

CLASSIC and ACID

On one of your own sounds, pressing CLASSIC or ACID switches the sound to that mode and shows its pages. A sound keeps both halves, a CLASSIC one and an ACID one, so switching back and forth never loses anything. ACID has two pages: press ■ ACID again for the second, ACID 2.

Press a CLASSIC page's button while the sound is ACID (or the other way round) and Volt shows you that page for as long as you hold the button, with a note saying which mode the sound is in. Let go and you're back where you were. Nothing on it changes while you look.



Figure 4.3 Holding FILTER on an ACID sound: CLASSIC's page, shown while the button's held, changing nothing.

Factory sounds work as on every unit: turn anything and you'll hear it change, the star in the top bar says so, and choosing the sound again puts it back (chapter 1, section 1.8).

4.3 The journey of a sound

Before you touch anything, here's the whole of CLASSIC in one picture. Once you've got this, every page makes sense.

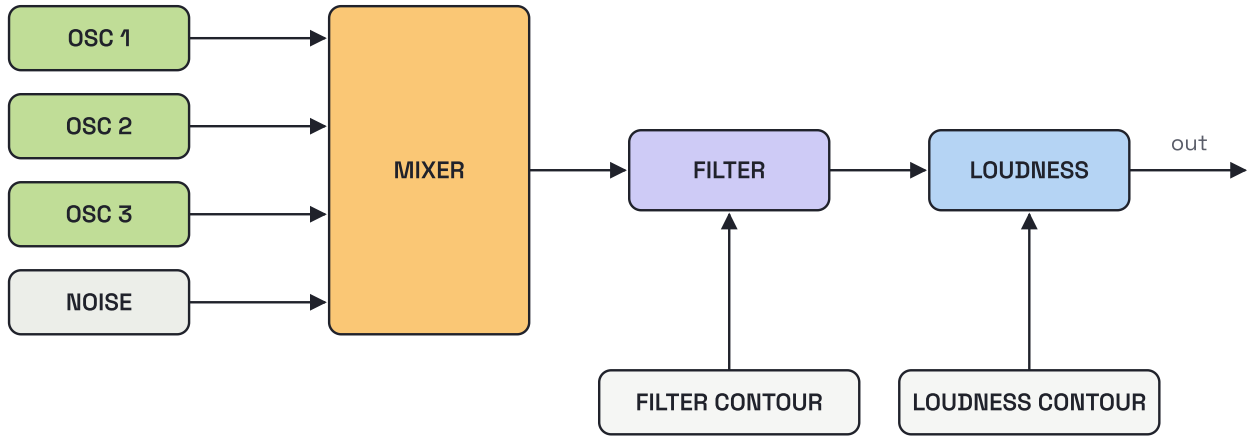


Figure 4.4 CLASSIC from start to finish: three oscillators and noise, mixed, through the filter (opened and closed by the filter contour, the CONTOUR page), then shaped by the loudness contour (the LOUDNESS page).

1. Three **oscillators** make the raw sound: three waves, at pitches you choose. There's a **noise** source too.
2. The **mixer** decides how much of each goes on.
3. The **filter** takes away the top of the sound: as much or as little as you like, and its own contour can open and close it on every note.
4. The **loudness** contour shapes how loud each note is, from the moment the key goes down to the moment it's gone.

The rest of CLASSIC, the CONTROL and MOD pages, changes how those four behave while you play.

A sound to work on

Your own slots start as INIT SOUND, a plain, useful bass. It's a good sound, but it has all three oscillators on and the filter half closed, so to hear what each part does we'll strip it right down to one oscillator and an open filter first.

SOUNDS › 1 BANK › USER A › 2 SOUND › UA01

1 MIXER › *push* 2 and *push* 3: OSC 2 and OSC 3 switch off, so only OSC 1 is left.

2 FILTER › 1 CUTOFF › +5.0, 2 EMPHASIS › 0.0 and 3 AMOUNT › 0.0: the filter wide open, doing nothing.

3 LOUDNESS › 3 SUSTAIN › 10.0: the note stays at full loudness while you hold it.

Play it. That's one sawtooth, raw: bright, buzzy and rather rude. Name it ONE OSC on the SOUNDS page (4 NAME: turn for the letter, push for the next one).

The sounds you'll build

This chapter builds ten sounds of your own, one in each section, each in its own slot of USER A. Every one starts as a copy of something you've already made, mostly ONE OSC, so you're never more than a few steps from a sound that works. When a section says "start from a copy of ONE OSC in UA04", that means:

SOUNDS › 2 SOUND › UA01 › 3 COPY TO › UA04 › *hold* 3

Copying moves you onto the copy, so you're ready to work. By the end of the chapter, USER A will hold:

Slot	Sound	Section	What it shows
UA01	ONE OSC	4.3	one raw oscillator, the place to start
UA02	THREE SAWS	4.4	three oscillators a touch apart, and the mixer's growl
UA03	WHISTLE	4.6	the filter playing on its own
UA04	PLUCK BASS	4.7	the filter contour snapping open
UA05	BRASS	4.7	a slow filter contour: the "bwah"
UA06	WOBBLE	4.8	oscillator 3 moving the filter
UA07	SOLO LEAD	4.9	glide, vibrato and a ringing release
UA08	KICK	4.11	a drum from a whistling filter
UA09	SNARE	4.11	a drum from noise
UA10	MY ACID	4.12	the squelch

If you ever lose your way in this chapter, go back to ONE OSC.

Reading the times

Volt's ATTACK and DECAY boxes show a time, not a number: 2 MS, 200 MS, 1.0 S. The recipes in this chapter give them the same way, so turn until the screen says what the recipe says. Each click is a small step, and the steps get bigger as the times get longer, as on the dials of the synths Volt follows.

4.4 The oscillators

■ OSC 1 RANGE and WAVE ■ OSC 2 and ■ OSC 3 RANGE, FREQ and WAVE

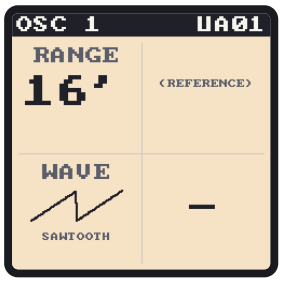


Figure 4.5 OSC 1's page: RANGE in feet, and the wave drawn as a picture. OSC 1 is the one the others are tuned against, so it has no FREQ.

RANGE (encoder 1, LO, 32', 16', 8', 4', 2') the octave, in feet, as organ stops are named: 8' is the note you play, 16' an octave below, 4' an octave above. The bigger the number, the lower the note. LO is far below hearing as a note, and section 4.8 is about what it's for.

FREQ (OSC 2 and OSC 3, encoder 2, -7.0 to +7.0) tunes the oscillator up or down by up to seven semitones, against OSC 1, which is the reference and has no FREQ of its own.

WAVE (encoder 3) the shape of the wave, drawn on the screen: TRIANGLE, SHARKTOOTH, SAWTOOTH, SQUARE, WIDE and NARROW.

OSC 3 CTL (OSC 3, encoder 4, ON or OFF) whether oscillator 3 follows the keys (ON), or stays at one pitch whatever you play (OFF). Section 4.8.

The whole instrument can be tuned too: ■ CONTROL › 1 TUNE, up to two and a half semitones either way.

The six waves

On ONE OSC, hold a note and step ■ OSC 1 › 3 WAVE through all six.

TRIANGLE is the soft one: few harmonics, flute-like. SHARKTOOTH is half triangle, half sawtooth, a little brighter. SAWTOOTH has every harmonic there is, which is why it's the one you'll start from most: the brightest and buzziest, the raw material for brass and strings and most basses. SQUARE has every other harmonic, which gives it that hollow, woody sound. WIDE and NARROW are squares that spend longer up than down; the narrower, the thinner and more nasal, down to a reedy buzz.

Put it back on SAWTOOTH before you go on: the next sound starts from ONE OSC as it was.

More than one oscillator

Now the best thing about three oscillators: playing them together. This is the second sound of your own.

Build THREE SAWS

Start from a copy of ONE OSC in UA02. First, bring OSC 1 up to the note you play: ■ OSC 1 › 1 RANGE › `8'`.

Now switch OSC 2 on, at the same pitch: ■ MIXER › *push* 2, then ■ OSC 2 › 1 RANGE › `8'` and 3 WAVE › `SAWTOOTH`. Set its FREQ to 2 FREQ › `+0.0` for a moment. Play a long note: two sawtooths in perfect step just sound like one louder one.

Now turn FREQ up a single click, to +0.1, and play a long note again. Wait.

The sound starts to move: it swirls, or beats slowly, as the two oscillators drift in and out of step. The further apart, the faster the beating; too far and it just sounds out of tune. That slow swirl is the sound of two oscillators slightly apart, and it's in nearly every big analogue sound you've ever heard. Volt's oscillators also wander a few cents on their own, as analogue oscillators always do (DRIFT on the EXTRA page, section 4.10), so even exactly in tune they're never quite still.

THREE SAWS, the third saw

Switch OSC 3 on (■ MIXER › *push* 3), then ■ OSC 3 › 1 RANGE › `8'`, 3 WAVE › `SAWTOOTH` and 2 FREQ › `-0.1`: a click below the note, where OSC 2 is a click above. Now one oscillator sits in tune and the other two lean either side of it.

Last, the mixer: ■ MIXER and set all three levels to 7.0. (Why 7 and not 10 is the next section.)

THREE SAWS

ONE OSC, plus:

OSC 1	`RANGE 8'`	`WAVE SAWTOOTH`	
OSC 2	`RANGE 8'`	`FREQ +0.1`	`WAVE SAWTOOTH`
OSC 3	`RANGE 8'`	`FREQ -0.1`	`WAVE SAWTOOTH`
MIXER	`OSC 1 7.0 ON`	`OSC 2 7.0 ON`	`OSC 3 7.0 ON`

Play a chord's worth of notes one after another, low down and high up. It's big and bright and alive, and still completely raw: the filter is wide open. Name it `THREE SAWS`.

Two other tricks while you're here. OSC 2's FREQ at +7.0 puts it a fifth above: a chord on every key, an old trick for leads and stabs. And an oscillator's RANGE an octave below the others gives a bass a floor to stand on. INIT SOUND uses that one: two sawtooths a touch apart, and a square an octave below.

4.5 The mixer

■ MIXER OSC 1, OSC 2, OSC 3 and NOISE

Four levels, 0 to 10, one for each oscillator and one for noise, each with its own switch on the encoder's push. Noise is the hiss of chapter 2: every pitch at once, so no pitch at all. Mix a little in for breath, or use it on its own for wind, waves and the snares of drums (section 4.11 makes one). ■ EXTRA › 2 NOISE chooses its colour: WHITE, the bright hiss, or PINK, darker and softer.

The mixer is part of the sound

On classic analogue synths, turning the mixer up too far overloads the filter after it, and the sound gets thicker and growlier. Players soon found it out, and it's a big part of why those synths sound the way they do. Volt does the same: past about 5, it growls. So the mixer isn't only a balance: 10 on all three oscillators is a growl, 4 is clean, and you choose.

Hear the growl

On THREE SAWS, hold a low note and turn all three mixer levels from 7.0 up to 10.0 one after another, then down to 4.0. It gets a bit louder as you go up, but listen past the loudness: at 10 the sound is thicker and rougher, at 4 it's smooth and polite. Neither is right. Leave it at 7.0, and remember that the mixer is the first place to make a sound fatter.

4.6 The filter

Here's where the sound gets its character. Chapter 2 (section 2.2) said a filter takes harmonics away. This one is a **low-pass** filter: it lets the low part of the sound through and cuts the high part, and it cuts steeply, which is why it sounds so smooth.

● FILTER CUTOFF, EMPHASIS, AMOUNT and KEYBOARD

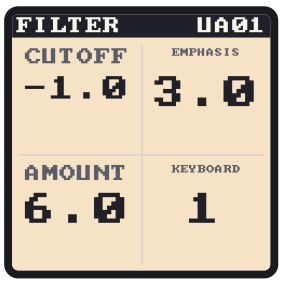


Figure 4.6 The FILTER page.

CUTOFF (encoder 1, -5.0 to +5.0) where the filter starts to cut. +5 is wide open, and the sound comes through as the mixer made it. Turn it down and the top goes, then the middle, until at -5 there's hardly anything left.

EMPHASIS (encoder 2, 0-10) the filter's resonance: it makes the filter ring at the cutoff, so the harmonics right there stand out. A little gives an edge; a lot thins out the bass and gives that vowel-like "wow". From about 7, the filter starts to **whistle** at its own pitch.

AMOUNT (encoder 3, 0-10) how far the filter's contour opens the filter on each note. Section 4.7.

KEYBOARD (encoder 4, OFF, 1, 2, 1+2) how much the cutoff follows the keys. OFF, it stays put, so high notes come out duller than low ones; 1 follows a third of the way, 2 two thirds, 1+2 all the way, so every note is equally bright.

Try this

On THREE SAWS, hold a low note and turn CUTOFF slowly down from +5 to -5 and back. That one control is most of what subtractive synthesis is. Listen to how the three saws' swirl stays, even as the brightness goes: the filter takes away harmonics, not movement.

Now set CUTOFF to about 0 and turn EMPHASIS up slowly while you sweep CUTOFF again. Around 4 or 5 the sweep turns into a "wow". Past 7, you'll hear the whistle come in on top of the note.

Put CUTOFF back to +5.0 and EMPHASIS to 0.0 afterwards: the next sounds start from THREE SAWS as it was.

A sound with no oscillators

That whistle is the filter ringing on its own, and it doesn't need the oscillators at all. Players have used it for fifty years, for eerie, pure leads like a theremin's or a singing saw's.

Build WHISTLE

Start from a copy of ONE OSC in UA03.

1. ■ MIXER › *push* ❶ until OSC 1 says OFF. Now nothing at all goes into the filter. Play a note: silence.

2. ■ FILTER › ❷ EMPHASIS › `10.0`. Play a note: a pure, high tone, from nothing. That's the filter.

3. It plays the same note whatever key you press, because the filter doesn't know about the keys. ❹ KEYBOARD › `1+2`, and it follows them all the way: now you can play a tune on it. ❶ CUTOFF moves the whole thing up or down; put it at 0.0.

4. To make it sing: ■ LOUDNESS › ❶ ATTACK › `200 MS`, so each note fades in a little, and ■ CONTROL › *push* ❷ to switch GLIDE on, with ❷ GLIDE › `4.0`, so it slides from note to note. Play joined-up notes, each key down before you let go of the last.

5. It's a quiet sound, so ■ CONTROL › ❹ VOLUME › `10.0`.

WHISTLE

ONE OSC, plus:

MIXER	`OSC 1 OFF`
FILTER	`CUTOFF +0.0` `EMPHASIS 10.0` `KEYBOARD 1+2`
LOUDNESS	`ATTACK 200 MS`
CONTROL	`GLIDE 4.0 ON` `VOLUME 10.0`

Name it `WHISTLE`. A sound made of nothing but a filter, ringing.

4.7 The two contours

Chapter 2 (section 2.3) called the shape of a note its envelope, and drew it with attack, decay, sustain and release. Synths of this kind often call their envelopes **contours**, and have two of them: one for loudness, and one for the filter. Volt has both.

LOUDNESS ATTACK, DECAY and SUSTAIN CONTOUR the same, for the filter

> **ATTACK** (encoder 1, 2 MS to 14 S) how long the note takes to get to the top.
>
> **DECAY** (encoder 2, 2 MS to 14 S) how long it then takes to fall to the sustain level.
>
> **SUSTAIN** (encoder 3, 0-10) the level it holds while your finger stays down.

Where's the release?

There's no RELEASE knob. Volt has a switch instead, the classic way: CONTROL › 3 DECAY, ON or OFF.

- **ON**: when you let go, the note dies away at the speed DECAY says.
- **OFF**: when you let go, the note stops.

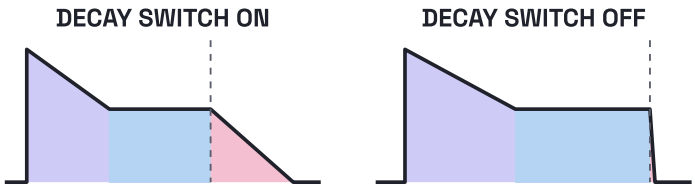


Figure 4.7 The DECAY switch, on the CONTROL page, is the release: ON, a note dies away at its DECAY speed after you let go (the dashed line); OFF, it stops.

It works on both contours at once. It's a quirk, and it's kept, because it's how the instrument behaves: a long DECAY with the switch ON gives a sound that rings on after you let go, which is lovely for leads and a nuisance for tight basses, so flip it OFF for those.

Four shapes

On ONE OSC, try these on the LOUDNESS page, with CONTROL's DECAY switch ON. Hold each note a second or two, then let go.

Organ

On at once, full while held, gone when you let go (almost: DECAY is short).

`ATTACK 2 MS` `DECAY 10 MS` `SUSTAIN 10.0`

Piano-ish

Struck, and dying away whether you hold it or not.

`ATTACK 2 MS` `DECAY 775 MS` `SUSTAIN 0.0`

Bowed

Swells in, holds, and fades when you let go.

`ATTACK 775 MS` `DECAY 600 MS` `SUSTAIN 10.0`

Pluck

Short and snappy.

`ATTACK 2 MS` `DECAY 346 MS` `SUSTAIN 0.0`

Then try the piano-ish one with the DECAY switch OFF, and play short notes: each one stops dead when you let go. Put LOUDNESS back to ATTACK 2 MS, DECAY 775 MS, SUSTAIN 10.0 when you're done, and the switch ON.

The filter contour

Now the part that makes analogue synths sound alive. CONTOUR is the same shape, but it moves the *filter*: on every note, it opens the filter from wherever CUTOFF has it, up by as much as AMOUNT says, and closes it again. Bright at the start, darker as it goes: chapter 2's "brightness over time".

Build PLUCK BASS

Start from a copy of ONE OSC in UA04. It's a bass, so play it low: press ■ OCT- once (that only moves the keys, it isn't part of the sound). Then set:

PLUCK BASS

ONE OSC, plus:

FILTER	CUTOFF −3.0	EMPHASIS 3.0	AMOUNT 7.0
CONTOUR	ATTACK 2 MS	DECAY 346 MS	SUSTAIN 0.0
LOUDNESS	ATTACK 2 MS	DECAY 775 MS	SUSTAIN 0.0
CONTROL	VOLUME 10.0		

Play a bass line, short notes. Every note snaps open and closes down: a plucked, rubbery bass. Name it `PLUCK BASS`.

Now change one thing at a time, and put it back each time. AMOUNT up to 10: a bigger snap. CONTOUR's DECAY longer, to 600 MS: a slower "yow". EMPHASIS up to 6.0: the snap gets a squelch. CUTOFF up to 0.0: brighter all the way through, so the snap matters less. That last one is worth noticing: CUTOFF and AMOUNT work together, one setting where the filter rests and the other how far each note opens it.

A slow filter: brass

PLUCK BASS opens its filter at once and lets it fall. Turn that round, so the filter opens *slowly* at the start of each note, and you get the sound synths have been using for brass since the 1970s: the "bwah" of a horn section, the note getting brighter as it's blown.

Build BRASS

This one starts from THREE SAWS, because brass wants the thickness of three oscillators:

SOUNDS › 2 SOUND › UA02 › 3 COPY TO › UA05 › *hold* 3

1. Close the filter most of the way, and give its contour plenty to do:
   FILTER › 1 CUTOFF › −2.0 , 2 EMPHASIS › 1.0 , 3 AMOUNT › 6.0 .
2. The slow opening: CONTOUR › 1 ATTACK › 200 MS , 2 DECAY › 600 MS , 3 SUSTAIN › 6.0 . Each note now climbs to full brightness over a fifth of a second, falls back a little, and holds there.
3. The loudness follows a little behind: LOUDNESS › 1 ATTACK › 10 MS , 2 DECAY › 600 MS , 3 SUSTAIN › 9.0 .

BRASS

THREE SAWS, plus:

FILTER	CUTOFF −2.0	EMPHASIS 1.0	AMOUNT 6.0
CONTOUR	ATTACK 200 MS	DECAY 600 MS	SUSTAIN 6.0
LOUDNESS	ATTACK 10 MS	DECAY 600 MS	SUSTAIN 9.0

Play slow, held notes, and listen to the start of each: "bwah". Name it **BRASS**. Then try CONTOUR's ATTACK at 600 MS: a slower, softer blow, more like a swelling string section. And notice that because the DECAY switch is ON, every note dies away in 600 MS when you let go, as a horn's does.

Two contours, two jobs

LOUDNESS shapes how loud the note is. CONTOUR shapes how bright it is. PLUCK BASS and BRASS are almost the same sound with the filter contour turned round: fast-then-falling for a pluck, slow-then-holding for a horn. Most of the classic sounds are one of those two shapes, or the swell.

4.8 Oscillator 3 as a modulator

Chapter 2 (section 2.4) said movement in a sound usually comes from an LFO, an oscillator too slow to hear. Volt has no LFO. It has something cleverer: oscillator 3 can stop being a sound and become the thing that moves the other two.

1 ■ OSC 3 › ❶ RANGE › **LO** : so slow it's no longer a note.

2 ❹ OSC 3 CTL › **OFF** : it stops following the keys, so the wobble is the same speed on every note.

3 ■ MIXER › *push* ❸ until OSC 3 is OFF: you don't want to hear it, only use it.

Then the MOD page decides where it goes and how much:

■ MOD MOD MIX, OSC MOD, FILT MOD and WHEEL

> **MOD MIX** (encoder 1, 0-10) what does the moving: at 0, oscillator 3; at 10, noise; in between, some of each. Noise makes the movement random.

> **OSC MOD** (encoder 2, ON or OFF) on: it moves the pitch of oscillators 1 and 2. That's vibrato.

> **FILT MOD** (encoder 3, ON or OFF) on: it moves the filter's cutoff. That's a wah, or a wobble.

> **WHEEL** (encoder 4, 0-10) how much. On a synth like this it's usually the modulation wheel beside the keys; here it's an encoder, and a MIDI keyboard's mod wheel moves it too (section 4.14).

Build WOBBLE

Start from a copy of ONE OSC in UA06, and set oscillator 3 up as above, with ■ OSC 3 › 3 WAVE › `TRIANGLE`.

First, vibrato: ■ MOD › 2 OSC MOD › `ON` and 4 WHEEL › `1.5`. Hold a note: the pitch wavers. ■ OSC 3 › 2 FREQ sets how fast. That's one use; but this sound is going to be the other.

Turn OSC MOD OFF and 3 FILT MOD › `ON`. Close the filter so it has room to move: ■ FILTER › 1 CUTOFF › `-1.0` and 2 EMPHASIS › `5.0`. Now ■ MOD › 4 WHEEL › `5.0`. The tone goes round and round: a wobble.

WOBBLE

ONE OSC, plus:

OSC 3	`RANGE LO`	`WAVE TRIANGLE`	`OSC 3 CTL OFF`
MOD	`FILT MOD ON`	`WHEEL 5.0`	
FILTER	`CUTOFF -1.0`	`EMPHASIS 5.0`	

Name it `WOBBLE`. Then play with it: OSC 3's WAVE on SQUARE and the wobble jumps between two places instead; on SAWTOOTH it sweeps up and drops; MOD MIX up to 10 and it goes random, like a sample-and-hold. OSC 3's FREQ up and down: from a slow sweep to a fast flutter.

Oscillator 3 doesn't have to be slow, either. Put it back in the audio range, OSC 3 CTL ON, and OSC MOD makes it modulate the other two at an audible pitch: the clangs and bells of chapter 3, Glass's own trick, done the analogue way.

4.9 Playing it

Volt plays one note at a time, and the way it does it is part of the sound.

The lowest key wins. Hold two keys in CLASSIC and you hear the lower one. So hold a high note and tap lower ones below it: each tap sounds while you hold it, and the high note comes back when you let go, an old trick for trills. (ACID is the other way round: the last key you pressed wins.)

Joined-up notes don't start again. Play a new key before letting go of the last, and the pitch changes but the contours carry on where they were, so there's no new snap of the filter. Play them separately and every note snaps. How you join your notes is part of the playing.

Glide. ■ CONTROL › *push* 2 switches glide on, and 2 GLIDE sets how long the pitch takes to slide from one note to the next.

> **VOLUME** (CONTROL, encoder 4, 0-10) the sound's own level. Use it to make your sounds sit together: a bass and a lead that should be equally loud.

As classic synths of its kind did, CLASSIC plays every note at the same loudness, however hard you hit a MIDI keyboard. If you'd rather it followed how hard you play, turn VELO on in SETUP (section 4.14).

Build SOLO LEAD

Everything so far, in one lead line. Start from a copy of THREE SAWS in UA07 (■ SOUNDS › ❷ SOUND › `UA02` › ❸ COPY TO › `UA07` › *hold* ❸).

1. Oscillator 3 becomes the vibrato, as in WOBBLE: ■ OSC 3 › ❶ RANGE › `LO` , ❸ WAVE › `TRIANGLE` , ❹ OSC 3 CTL › `OFF` , and ■ MIXER › *push* ❸ to OFF. Then ■ MOD › ❷ OSC MOD › `ON` and ❹ WHEEL › `1.5` : a gentle waver.
2. A little edge from the filter: ■ FILTER › ❶ CUTOFF › `+1.0` , ❷ EMPHASIS › `2.0` , ❸ AMOUNT › `3.0` .
3. Glide: ■ CONTROL › *push* ❷ to ON, and ❷ GLIDE › `3.0` .
4. A ringing release: ■ LOUDNESS › ❷ DECAY › `600 MS` , with CONTROL's DECAY switch ON.

SOLO LEAD

THREE SAWS, plus:

OSC 3	`RANGE LO` `WAVE TRIANGLE` `OSC 3 CTL OFF`
MIXER	`OSC 3 OFF`
MOD	`OSC MOD ON` `WHEEL 1.5`
FILTER	`CUTOFF +1.0` `EMPHASIS 2.0` `AMOUNT 3.0`
CONTROL	`GLIDE 3.0 ON` `DECAY ON`
LOUDNESS	`DECAY 600 MS`

Name it `SOLO LEAD` . Now play it properly: a slow tune, high up ((■ OCT+)), some notes joined, some apart. Joined, the pitch glides and the filter doesn't snap; apart, each note starts fresh. Hold a high note and tap a lower one underneath for a trill. Let a last note go and hear it ring. That's how a lead line on a synth like this is played: the hands do half the work.

4.10 The EXTRA page

■ EXTRA EXTERNAL, NOISE and DRIFT



Figure 4.8 The EXTRA page: EXTERNAL (the output fed back into the mixer), the noise colour, and DRIFT.

EXTERNAL (encoder 1, 0-10, push ON or OFF) classic synths of this kind had a mixer input for an outside sound. Players discovered that plugging a synth's own output back into it overloaded the filter even harder, for a fatter, dirtier sound, and it became a famous trick. EXTERNAL is that trick: Volt's output fed back into its own mixer, as much as you like.

NOISE (encoder 2, WHITE or PINK) the noise colour (section 4.5).

DRIFT (encoder 3, ON or OFF) on: each oscillator wanders slowly in pitch by a few cents, as analogue oscillators always do. It's what keeps a held chord of oscillators from sounding frozen. Off for a sound that must be dead still.

Try this

On BRASS, push ■ EXTRA › *push* 1 to switch EXTERNAL on, and turn it up slowly while you hold a low note. Somewhere past 5 the brass turns into something much nastier. Back to OFF afterwards, unless you like it: plenty of people do.

4.11 Drums from three oscillators

Section 4.1 pointed at three factory sounds that might have surprised you: KICK DRUM, SYNTH TOM and SNARE DRUM. Long before drum machines, people made drum sounds on synths like this one, and the tricks are all things you've now done: a whistling filter, a burst of noise, and very short contours. Build them yourself, and you'll understand drum machines too (Strike's chapter takes it further).

A kick drum

A kick drum is a low thump that drops in pitch as it dies. WHISTLE was a filter ringing on its own; a kick is a filter ringing very low, pinged by a short, sharp contour.

Build KICK

Start from a copy of ONE OSC in UA08.

1. A soft, low oscillator to give the thump some body: ■ OSC 1 › ❶ RANGE › `32'` and ❸ WAVE › `TRIANGLE`, and ■ MIXER › ❶ OSC 1 › `7.0`.

2. The filter ringing low, the same pitch on every key: ■ FILTER › ❶ CUTOFF › `-3.5`, ❷ EMPHASIS › `10.0`, ❹ KEYBOARD › `OFF`.

3. The ping: ❸ AMOUNT › `6.0`, and ■ CONTOUR › ❷ DECAY › `45 MS` with ❸ SUSTAIN › `0.0`. On every note the filter jumps up and falls straight back, and its whistle falls with it: that's the drop in pitch.

4. Short: ■ LOUDNESS › ❷ DECAY › `263 MS` and ❸ SUSTAIN › `0.0`. And loud: ■ CONTROL › ❹ VOLUME › `10.0`.

KICK

ONE OSC, plus:

OSC 1	`RANGE 32'`	`WAVE TRIANGLE`		
MIXER	`OSC 1 7.0 ON`			
FILTER	`CUTOFF -3.5`	`EMPHASIS 10.0`	`AMOUNT 6.0`	`KEYBOARD OFF`
CONTOUR	`DECAY 45 MS`	`SUSTAIN 0.0`		
LOUDNESS	`DECAY 263 MS`	`SUSTAIN 0.0`		
CONTROL	`VOLUME 10.0`			

Play short notes: thump, thump. Name it `KICK`. Now tune it: CUTOFF higher and it's a tom; up to about +0.0 and KEYBOARD to 1+2, and you've got a tom you can play tunes on, which is what the factory's SYNTH TOM is. CONTOUR's DECAY longer for a boomier kick with a longer drop; LOUDNESS's DECAY longer for a boom that hangs on.

A snare drum

A snare is two things at once: the drum's skin, a short tone, and the snares underneath it, a burst of rattle. On a synth, the rattle is noise.

Build SNARE

Start from a copy of ONE OSC in UA09.

1. The skin, quietly: ■ OSC 1 › ❶ RANGE › `8'` , ❸ WAVE › `TRIANGLE` , and ■ MIXER › ❶ OSC 1 › `3.0` .
2. The snares: ■ MIXER › ❹ NOISE › `10.0` and push it to ON.
3. A filter that snaps shut on the noise: ■ FILTER › ❶ CUTOFF › `+1.0` , ❷ EMPHASIS › `1.0` , ❸ AMOUNT › `5.0` , ❹ KEYBOARD › `OFF` ; ■ CONTOUR › ❷ DECAY › `45 MS` , ❸ SUSTAIN › `0.0` .
4. Short and loud: ■ LOUDNESS › ❷ DECAY › `200 MS` , ❸ SUSTAIN › `0.0` , ■ CONTROL › ❹ VOLUME › `10.0` .

SNARE

ONE OSC, plus:

OSC 1	`RANGE 8'`	`WAVE TRIANGLE`		
MIXER	`OSC 1 3.0 ON`	`NOISE 10.0 ON`		
FILTER	`CUTOFF +1.0`	`EMPHASIS 1.0`	`AMOUNT 5.0`	`KEYBOARD OFF`
CONTOUR	`DECAY 45 MS`	`SUSTAIN 0.0`		
LOUDNESS	`DECAY 200 MS`	`SUSTAIN 0.0`		
CONTROL	`VOLUME 10.0`			

Name it **SNARE** . Try the noise PINK (■ EXTRA › ❷ NOISE) for a softer, darker snare; OSC 1's level up for more ring; LOUDNESS's DECAY at 10 MS and the noise alone for a hi-hat's tick.

Play KICK and SNARE in turn, on the beat and the off-beat, and you've got the start of a drum part, from a bass synth. Now open the factory's KICK DRUM and SNARE DRUM: you'll find you can read every setting.

4.12 ACID

Now the other instrument: one oscillator, a filter that squelches, and two ingredients hidden in the notes, accent and slide. The sound was made famous by producers turning the knobs while a pattern played, and that's still how it's done.

Every sound of your own has an ACID half waiting, so start from a copy of ONE OSC in UA10 and press ■ ACID. The CLASSIC button goes out, ACID lights, and the sound is now an ACID sound. Its ACID half has never been touched, so it starts from INIT SOUND's: a perfectly good acid bass to start from.

■ ACID TUNING, CUTOFF, RESONANCE and ENV MOD ■ ACID ACID 2 DECAY, ACCENT, WAVEFORM and VOLUME

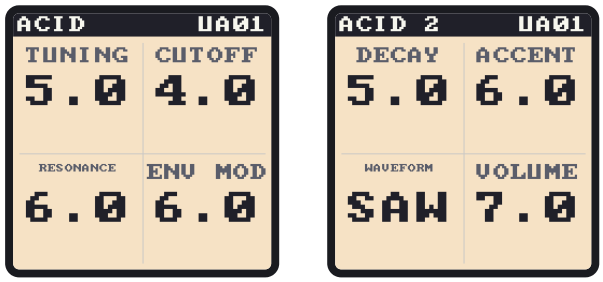


Figure 4.9 ACID's two pages. Press ACID for the first, again for ACID 2.

> **TUNING** (ACID, encoder 1) the pitch, fine.

> **CUTOFF** (ACID, encoder 2, 0-10) where the filter sits.

> **RESONANCE** (ACID, encoder 3, 0-10) how much it rings: the squelch.

> **ENV MOD** (ACID, encoder 4, 0-10) how far each note sweeps the filter open. Turning it up also lowers where the filter rests, so the sweep gets wider at both ends.

> **DECAY** (ACID 2, encoder 1, 0-10) how long that sweep takes to close.

> **ACCENT** (ACID 2, encoder 2, 0-10) how much harder accented notes hit.

> **WAVEFORM** (ACID 2, encoder 3, SAW or SQUARE) the only choice of wave. The square is made from the saw, so it isn't quite a perfect square, and that's part of its sound.

> **VOLUME** (ACID 2, encoder 4, 0-10) the sound's level.

Notice what isn't there: no attack, no sustain, no loudness contour. On ACID the loudness of a note is fixed: it comes in, holds while the key's down, and stops. DECAY only shapes the filter. Everything you do to an ACID sound, you do with four knobs: CUTOFF, RESONANCE, ENV MOD and DECAY.

Build MY ACID

Play a run of short notes, low down, all the way through this: the squelch is in the movement.

1. ■ ACID › ❸ RESONANCE › `8.0`. The filter starts to ring on every note: the squelch arrives.
2. ❷ CUTOFF › `2.5` and ❹ ENV MOD › `7.0`: the filter rests lower and jumps further, so each note goes "yow".
3. ■ ACID 2 › ❶ DECAY › `4.0`: a little tighter, so fast lines stay clear.
4. ❹ VOLUME › `8.0`.

MY ACID

ONE OSC's ACID half, with:

ACID	`CUTOFF 2.5`	`RESONANCE 8.0`	`ENV MOD 7.0`
ACID 2	`DECAY 4.0`	`WAVEFORM SAW`	`VOLUME 8.0`

Name it `MY ACID`. Now turn the four, one at a time, while you play: CUTOFF and ENV MOD against each other, the filter's resting place and how far it jumps. RESONANCE up and down: more or less squelch. DECAY down to 1.0 for short, tight blips; up to 8.0 for long, slow "yowww"s. Turning these while a pattern plays is how acid house was made, and nobody has ever needed more.

Then the other wave: ■ ACID 2 › ❸ WAVEFORM › **SQUARE**, CUTOFF down to 1.5, RESONANCE to 3.0, ENV MOD to 3.0, DECAY to 6.0. A deep, hollow, rubbery bass, from the same four knobs. Same instrument, nothing like the squelch.

And a reminder of the trick from section 4.2: press ■ CLASSIC on MY ACID and it's ONE OSC again, unchanged; press ACID and it's your acid sound. Each slot keeps both.

Accent and slide

ACID's two secret ingredients don't live on knobs: they're in the notes. Any note can be **accented** or **slid**.

- **Accent**: an accented note is louder, with a sharper, shorter sweep, and a run of accents builds up, each one squelchier than the last (how much, RESONANCE decides). ACCENT sets how strong it is.
- **Slide**: a slid note glides from the one before, quickly, without starting again: a sliding, rubbery line.

Volt's own keys play every note unaccented. Accents come from a MIDI keyboard played hard (velocity 100 or more) or from Step, the sequencer, which can mark any note ACCENT or SLIDE. Slide comes from joined-up playing: press the next key before letting go of the last, on Volt's keys or a MIDI keyboard.

Try this

On MY ACID, play a line of short notes, then the same line again with every second note joined to the one before it (the next key down before you let go). The joined ones slide: that rubbery "bwoop" between notes is half of acid's sound, and you've just played it with your fingers.

ACID wants a sequencer

You can play ACID from the keys, and it's a fine bass. But the sound everyone knows comes from a pattern of notes, accents and slides going round and round while you turn the knobs. That's what Step is for, and the Music Production manual starts with exactly that.

4.13 Effects

Volt has no effects of its own; nor did classic synths of either kind. With Haze joined on, though, Volt gets chorus, echo and reverb, and one more thing that belongs to it more than to anything: **distortion**. Acid bass lines were very often driven hard into a distortion pedal, and Haze's ACID DRIVE preset is that. On its own with one unit, Volt's nearest thing is its own mixer overload (section 4.5) and EXTERNAL (section 4.10).

4.14 Volt's SETUP

SETUP's THIS UNIT page has Volt's own settings, beside the MIDI channel:

> **BEND** (encoder 2, 0-12) how far a MIDI keyboard's pitch bend wheel bends the note, in semitones. Two is usual.

> **WHEEL** (encoder 3) what moves the WHEEL encoder on the MOD page: a MIDI keyboard's MOD WHEEL, BREATH, EXPRESSION (a pedal), AFTERTOUCH, or OFF, so only the encoder moves it.

> **VELO** (encoder 4, OFF or ON) OFF, the classic way: every CLASSIC note is equally loud. ON: CLASSIC's loudness follows how hard a MIDI keyboard or Step plays. (ACID always uses how hard you play, for its accents.)

What you have at the end of this chapter

Ten sounds of your own: ONE OSC, THREE SAWS, WHISTLE, PLUCK BASS, BRASS, WOBBLE, SOLO LEAD, KICK, SNARE and MY ACID. Between them they use every page on Volt, and they show the idea behind most of the synthesisers ever made: start with something bright, and take away what you don't want. Nearly every factory sound is one of yours with a few numbers changed; open some up and see if you can tell which.

Chapter 5 - Strike

Every drum is a tone, a hiss or a clang, and how quickly it dies.

Strike is a drum machine, and every drum in it is synthesised: there are no recordings inside it at all. Each kick, snare and cymbal is made the moment you hit it, from the same few things chapter 2 was about, a pitch, a tone, a shape. That means every drum can be retuned, lengthened, shortened and brightened, and it means you'll understand exactly what you're changing.

Drum synthesis: where it came from, and who uses it

The first drum machines were rhythm boxes for home organs: Wurlitzer's Sideman, in 1959, played a handful of fixed patterns, and for twenty years most of those that followed did the same, to keep a singer or an organist in time. Roland's CR-78 of 1978 was one of the first you could program with your own patterns, and its soft, ticking sounds are on Blondie's "Heart of Glass" and Phil Collins's "In the Air Tonight".

Then came the TR-808 (its story is in the introduction), and in 1983 the TR-909, whose punchy kick and snare were the backbone of Chicago house and Detroit techno, and still are. In Britain, Simmons built electronic drum kits played with sticks, and their pitch-bending toms are one of the sounds of 1980s pop. At much the same time, the first drum machines that played recordings of real drums arrived, and for a while synthesised drums looked old-fashioned. They came back, because a synthesised drum can be bent into any shape a recording can't.

Hip hop, electro, house, techno, trap, drum and bass, synth-pop: all of them were built on synthesised drums. Strike isn't any of the machines above. Its kits are our own, each with a character of its own. But a kick, a snare or a hi-hat is made the same way on all of them, and what you learn here works on any drum machine, or on any synth, when you want to make a drum from scratch.

5.1 Find a kit you like

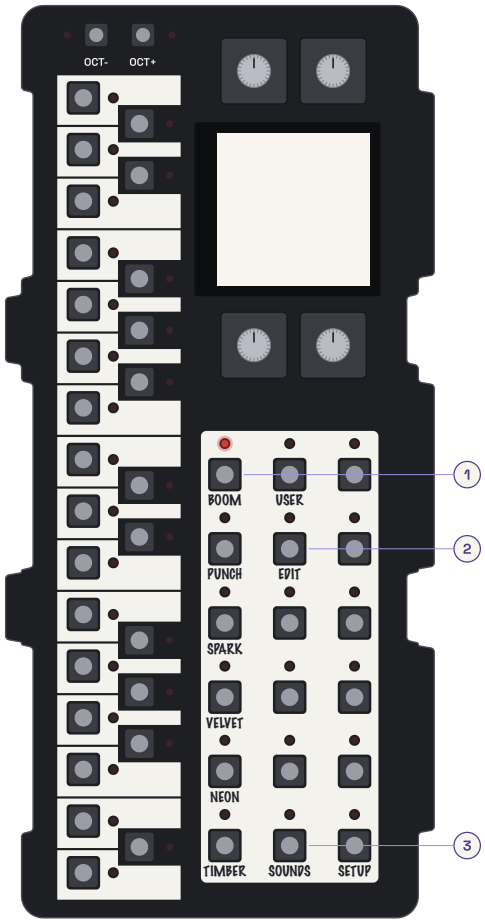


Figure 5.1 Strike's buttons. 1: the six factory kits. 2: EDIT, with USER above it. 3: SOUNDS, beside SETUP. The white keys are the drums, the same kind of drum on each key in every kit.

The left-hand column is six kit buttons. Press one and that kit is ready on the keys:

> **BOOM** a long, deep, booming kick, a snappy snare, handclaps, and hats and cymbal made of a metallic hiss. The sound of electro and hip hop, and of a great deal of pop since.

> **PUNCH** a short, punchy kick with a click at the front, a fat snare, and bright, sizzling hats, crash and ride. House and techno.

> **SPARK** small, bright and tight: a little kick, a crisp snare, two toms, hats and a cymbal. Lovely with acid bass lines.

> **VELVET** soft, ticking, old-fashioned: a gentle kick and snare, bongos and conga, cowbell, claves, maracas, a tambourine, a güiro and a metallic beat. Early synth-pop, and anything lo-fi.

> **NEON** electronic drums as a drummer played them in the 1980s: a kick, a snare and three toms, all with a pitch that bends (section 5.8). It has no hats or cymbals, on purpose.

> **TIMBER** an acoustic-inspired kit: drums made by working out, roughly, how real drums, cymbals and shakers vibrate. It's still a synthesised kit, with an acoustic flavour, not the sound of a real drummer, and it's at its best when you want something a little less electronic.

> **Try this**
>
> Press each kit and play down the white keys, top to bottom. Then play a simple beat on each: KICK, then SNARE, then KICK, KICK, SNARE, with CL HAT between. You'll hear six completely different bands.

The keys

Strike's 16 white keys are the drums, and **each key is the same kind of drum in every kit**: the kick is always the top key, the snare always the third, and so on. So a beat played on one kit plays on any other. The black keys do nothing, and Strike has no OCT buttons, because its keys are drums, not notes. When you play a key, the screen shows that drum's name.

Key	Drum	Key	Drum
1	KICK	9	LO TOM
2	PERC (a kit's extra drum)	10	MID TOM
3	SNARE	11	HI TOM
4	TAMB (tambourine)	12	CRASH
5	RIM	13	RIDE
6	CLAP	14	COWBELL
7	CL HAT (closed hi-hat)	15	CLAVES
8	OP HAT (open hi-hat)	16	MARACAS

A kit only has the drums it was made with, so some keys are silent in some kits, and the screen says `NOT IN THIS KIT`. A kit may also put a drum of its own on a key: VELVET's CLAP key is a güiro, for instance, and the screen gives it its own name.

> **For the sound of a real drum kit**
>
> Strike makes every drum by synthesis, so none of its kits is a real kit. For the genuine sound of real drums, recorded, you want Taste, the sampler, which plays recordings of real instruments.

Like a real hi-hat, the closed hat stops the open one: play OP HAT, then CL HAT straight after, and the open one is cut short.



Figure 5.2 Playing BOOM: the screen names the drum you hit.

5.2 What a drum is made of

Every drum in Strike is built from chapter 2's pieces. Once you know which, every setting makes sense.

- **A kick** is a low tone, often close to a pure sine, that starts a little higher and falls quickly to its own pitch, and dies away. Sometimes there's a click at the very front, the beater hitting the drum. Pitch, and a very short shape.
- **A snare** is two things at once: a short tone, the drum itself, and a burst of noise, the rattle of the metal wires under it. How much of each is most of the difference between snares.
- **Hi-hats and cymbals** have no pitch you could sing. They're a clang of several square waves at odd, unrelated pitches, plus noise, with the low part filtered away. The closed hat dies at once; the open hat and the cymbals ring.
- **A clap** is noise, in a few quick bursts, like several people not quite clapping together.
- **Toms, congas, bongos, cowbell and claves** are tones that ring and die, each at its own pitch.

So when you change a drum, you'll nearly always be changing one of three things: its pitch (TUNE), how long it lasts (DECAY), or what it's made of (TONE, SNAPPY and their like). That's what sections 5.5 to 5.8 are about.

5.3 Looking inside a kit

Press (■ EDIT), then play a key: the screen shows that drum's settings, up to four, one on each encoder, and the key's light stays on so you know which drum you're on. The top bar says which drum and which kit it came from. Play another key to look at another drum. Press EDIT again to go back to playing.



Figure 5.3 EDIT, then KICK, on BOOM: the kick's four settings. The top bar says which drum, and which kit it's from.

Each drum has only the settings that suit it, so the boxes differ from drum to drum and kit to kit. On BOOM the kick has LEVEL, TUNE, DECAY and TONE, the snare LEVEL, TUNE, TONE and SNAPPY, and the closed hat just LEVEL. SPARK and VELVET have only LEVEL on every drum: they were made that way, simple boxes with nothing to tweak, and their charm is in their sound as it is. An empty box means that drum hasn't got that setting.

On a factory kit EDIT works too, as every unit's factory sounds do: you hear the change, the star in the top bar says it isn't kept, and choosing the kit again puts it back. Copy it to one of your own kits to keep it. Holding a key to swap its drum (below) works only on your own kits.

5.4 A kit of your own

Your own kits live on the SOUNDS page, as every unit's sounds do (chapter 1, section 1.8): FAC A holds the six factory kits, and USER A has room for 32 of your own. Every one of your slots starts as INIT KIT, which is BOOM's drums.



Figure 5.4 The SOUNDS page: FAC A holds the six kits; USER A, your own 32.

The USER button, at the top of the middle column, always takes you back to the last of your own kits you played, so it works like a seventh kit button. To start from a different factory kit, choose it and copy it: for example ■ PUNCH, then ■ SOUNDS › 3 COPY TO › UA02 › *hold* 3.

In your own kit, EDIT does what you ask, and pushing an encoder puts that setting back to the factory's value. While a setting is away from the factory's, the screen shows the factory's under it.



Figure 5.5 In your own kit, EDIT changes things. TUNE has been turned to 55; under it, the factory's 50. Push the encoder to put it back.

The kits you'll build

This chapter builds four kits of your own, each for a different kind of music, and each one shows something new:

Slot	Kit	Section	What it shows
UA01	DEEP KIT	5.5	tuning: a long, deep kick for hip hop and trap
UA02	TIGHT KIT	5.6	decay: short, tight drums for techno
UA03	ELECTRO	5.7 and 5.8	swapping drums: NEON's drums with BOOM's hats and cymbal, and drums that bend
UA04	MIXED KIT	5.7 and 5.9	a drum from five kits, then balancing a kit

How to change a drum, every time

Every recipe in this chapter works the same way: press ■ EDIT, play the drum's key, and turn its encoders until the boxes say what the recipe says. The recipes list each drum by its key name, and its settings in encoder order. Press EDIT again when you're done, to go back to playing. A setting you don't see in a recipe stays as it is.

5.5 Tuning drums

Start on your first kit, and give it a name: ■ SOUNDS › ❶ BANK › USER A › ❷ KIT › UA01, then ❹ NAME : DEEP KIT. Press SOUNDS again to close the page. It's INIT KIT, so it starts as BOOM.

A deep kick

Press ■ EDIT, then the KICK key. Turn ❸ DECAY up to 80 and ❷ TUNE down to 30, and play: a long, deep boom that you'll feel more than hear, the kick that sits under hip hop and trap. It now rings for about a second and a half, three times as long as before. Bring ❹ TONE down to 20 to take the edge off the front, so it's all boom.

Bass players tune their notes to a kick like this one, so TUNE matters: play the kick and a bass note together, and turn TUNE until they agree.

The snare

Play the SNARE key. ❹ SNAPPY is the noise, the rattle of the wires: down to 10 for a dull, boxy drum, up to 90 for a crack. ❸ TONE changes the body of the drum under it. For DEEP KIT, tune it down a little to sit with the kick, and give it plenty of crack: TUNE 40, SNAPPY 75.

DEEP KIT

INIT KIT (BOOM's drums), with:

KICK	LEVEL 90	TUNE 30	DECAY 80	TONE 20
SNARE	TUNE 40	SNAPPY 75		
CLAP	LEVEL 85			

Play a slow beat: KICK, then CLAP and SNARE together, KICK, KICK, CLAP and SNARE. The clap layered on the snare is an old trick for a fatter backbeat.

While you're in DEEP KIT, try the toms: on BOOM's three tom keys, ❹ TYPE switches each between TOM and CONGA, the same circuit with two characters, and TUNE sets each one's pitch. Other kits' drums have different settings: PUNCH's kick has ATTACK instead of TONE, how much click there is at the front. TIMBER's kick has BEATER (how hard the beater is), and its snare SNARES: at 0, the wires are switched off, and you've got a deep, ringing drum.

Tune the kit, not just the drums

Drums have pitches, and they sound best when the pitches get on with each other and with the song. Tune the kick to the key of your bass line, the toms a pleasant step apart, and the whole kit sounds deliberate. It's the oldest trick of drummers with real drums, and it costs you nothing here.

5.6 How long a drum lasts

A drum is nearly all attack and decay: it starts at once and dies away, and how quickly it dies is most of its shape. Where a drum has DECAY, it's the most useful setting it has. DEEP KIT's kick showed the long end; techno lives at the short end, where every drum is out of the way before the next one arrives.

Build TIGHT KIT

This one starts from PUNCH, the punchy kit with the click at the front. Copy it into your second slot: ■ PUNCH, then ■ SOUNDS › 3 COPY TO › `UA02` › *hold* 3, and name it `TIGHT KIT`.

1. The kick: ■ EDIT, KICK, 3 DECAY › `25`. It's over in under a fifth of a second now: a thump, not a boom. 4 ATTACK › `70` for more click, so it cuts through. 2 TUNE › `45`.
2. The hats: CL HAT, 3 DECAY › `15`: a short tick. OP HAT, 3 DECAY › `30`: an open hat that closes quickly, so it doesn't wash over everything.
3. The snare: SNAPPY up to 80, for a sharp crack.

TIGHT KIT

PUNCH, with:

KICK	`TUNE 45`	`DECAY 25`	`ATTACK 70`
SNARE	`SNAPPY 80`		
CL HAT	`DECAY 15`		
OP HAT	`DECAY 30`		

Play four kicks to a bar, an open hat between each pair, and the snare on the second and fourth: a techno beat. Now play the same beat on PUNCH itself, then on DEEP KIT. Same notes, three different decays, three different kinds of music.

> **Try this**
>
> Back on DEEP KIT: an open hat with a short DECAY is half closed, with a long one it washes. Try OP HAT's DECAY at 15, then 70. And a cymbal with a short DECAY is a choke, a crash that's grabbed: CRASH's 3 DECAY › **15**. Leave them where you like them best.

5.7 Swapping drums between kits

Here's the quickest way to make a kit nobody else has: take each drum from wherever you like. In one of your own kits, **hold a key down** for a moment, and the screen says where that drum comes from: `SNARE FROM BOOM`. Turn any encoder while you hold it, and the drum changes to that key's drum from another kit, or NONE, to silence the key. The kits come round in the order of their buttons, BOOM, PUNCH, SPARK, VELVET, NEON, TIMBER, then NONE, and a kit that hasn't got that drum is skipped.

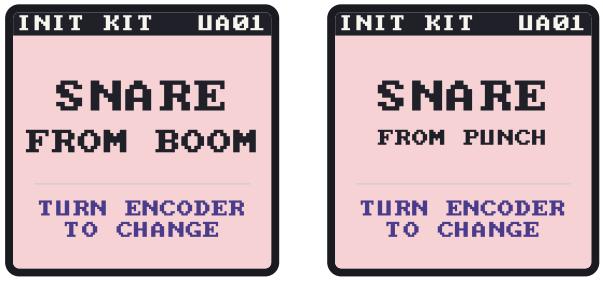


Figure 5.6 Holding SNARE in your own kit: where it's from (left), and after one turn of an encoder, PUNCH's snare instead (right).

A swapped drum arrives with its factory settings, and its own set of knobs. When you play a key in one of your kits, the screen says where it's from.

Build ELECTRO

NEON's drums are made for bending, and it has no hats or cymbals at all. BOOM has a fine metallic hi-hat and cymbal. Put them together, and you've got a kit for electro and 1980s pop that neither kit is on its own.

Go to your third slot, ■ SOUNDS › ❷ KIT › `UA03`, and name it `ELECTRO`. It starts as BOOM, so the hats and cymbal are already there.

1. Hold the KICK key and turn an encoder until the screen says `KICK FROM NEON` (four clicks on). Let go.
2. The same for SNARE, then LO TOM, MID TOM and HI TOM: each one FROM NEON.
3. Leave CL HAT, OP HAT and CRASH as they are: FROM BOOM.

Play a beat: NEON's kick and snare, BOOM's hats over them. Then the three toms in a quick roll, top to bottom. Two kits' worth of character in one.

Build MIXED KIT

Now take it further: a kit from five. In your fourth slot, UA04, named `MIXED KIT`: hold KICK and turn until it says FROM PUNCH. Hold SNARE: leave it FROM BOOM. Hold CL HAT and OP HAT: FROM TIMBER. Hold the three toms: FROM NEON. Hold COWBELL: FROM VELVET. Five kits, one band.

It won't sit together yet: some drums will be much louder than others. Section 5.9 fixes that.

5.8 Drums that bend

NEON's drums have one setting the others don't: ❸ BEND, how fast the pitch falls after the hit. A little, and the kick and toms get a natural thud. A lot, and they swoop down with that "pew" that every electronic drum kit of the 1980s had. Their settings are LEVEL, TUNE, BEND and DECAY.



Figure 5.7 NEON's kick: TUNE, BEND (how fast the pitch falls) and DECAY.

ELECTRO, bending

On ELECTRO, press ■ EDIT and play the MID TOM key. Turn 3 BEND up to 90 and play it: "pew". Then 4 DECAY up to 85, so you hear the whole swoop. Do the same to the LO TOM and HI TOM, and play the three in a quick roll.

ELECTRO

INIT KIT, with KICK, SNARE and the three toms FROM NEON, and:

LO TOM	`BEND 90`	`DECAY 85`
MID TOM	`BEND 90`	`DECAY 85`
HI TOM	`BEND 90`	`DECAY 85`
KICK	`BEND 80`	

The kick with a little more bend gets a deeper thud. Now try the opposite: BEND all the way down to 0 on a tom, and it's a plain, steady tone, nothing like a drum. The fall in pitch is what makes it sound hit.

5.9 Loudness and accents

Every drum has LEVEL, on encoder 1, 0 to 99, and getting the levels right is what turns a pile of drums into a kit. A good rule to start from: the kick and snare loudest, the hats well under them, everything else somewhere between.

MIXED KIT, balanced

On MIXED KIT, press ■ EDIT and play a beat you know with one hand while you set each drum's ❶ LEVEL with the other. Start here, then trust your ears:

MIXED KIT

The drums of section 5.7, with:

KICK	LEVEL 90
SNARE	LEVEL 80
CL HAT	LEVEL 50
OP HAT	LEVEL 50
TOMS	LEVEL 70
COWBELL	LEVEL 45

Then tune it, as in section 5.5: the toms a step apart, the cowbell somewhere that doesn't fight the kick. The kit becomes one band rather than five.

Strike's own keys play every hit at the same strength. From a MIDI keyboard, or pads that sense how hard you hit them, harder is louder. From Step, any step can be marked ACCENT, and an accented hit is louder by as much as SETUP says:

> **ACCENT** (SETUP, THIS UNIT, OFF or +1 to +10 dB) how much louder an accented step is: +6 to start with. Accents are what make a programmed beat feel played.

5.10 Effects

Strike has no effects of its own. With Haze joined on, though, a drum kit is transformed by a room: a little plate reverb behind the snare, an echo on a rim shot, and Haze's PUMP can make one of your synths duck in time with Strike's kick, the breathing sound of so much dance music.

5.11 Playing Strike from Step or a keyboard

Strike takes MIDI channel 10 automatically, the channel drums have always used, so it's usually STRIKE 10. Notes arrive by the General MIDI drum map that almost every keyboard, computer and drum pad follows: C2 (note 36) is the kick, D2 the snare, F#2 the closed hat and so on, whatever kit you're on. So a beat programmed for any General MIDI drum kit plays on Strike as it is.

With Step, the sequencer, you play the drums on Strike's own keys while Step's ENTRY page is open, and Step writes them in. That's the Music Production manual's job, and its first chapter starts there.

> ### What you have at the end of this chapter
>
> Four kits of your own: DEEP KIT for hip hop and trap, TIGHT KIT for techno, ELECTRO for the 1980s, and MIXED KIT from five kits, tuned and balanced. And the three questions to ask of any drum: what's its pitch, how long does it last, and what's it made of?

Chapter 6 - Ember

Six voices, one filter each, and a chorus that makes them sing.

Ember plays chords. That's what sets it apart: Volt and Strike play one sound at a time, Glass plays many but builds each one from sine waves. Ember is a **polyphonic** synthesiser, a polysynth: up to six notes at once, each with its own oscillator, filter and envelope, all following the same settings. Press three keys and you've got three complete synthesisers playing together.

It works the subtractive way: start with a wave that's bright and full of harmonics, take away what you don't want with a filter, and shape it with an envelope. If you've read Volt's chapter you'll know the idea; this chapter teaches it again from the start, Ember's way, so it works on its own.

Polysynths: where they came from, and who uses them

The first synthesisers played one note at a time. Playing chords meant a synthesiser for every note, and early polyphonic instruments were huge and expensive: Yamaha's CS-80 of 1977, the instrument behind Vangelis's *Blade Runner*, weighed about as much as a person. In 1978 Sequential's Prophet-5 brought five voices and, for the first time, a memory that kept every setting of a sound, so a player could build a sound once and call it back on stage.

In the early 1980s Roland's Juno-6, Juno-60 and then the Juno-106 (1984) made polysynths cheap: one oscillator per voice, digitally controlled so it stayed in tune, and a built-in chorus that made even a plain sound wide and warm. They sold in their thousands, and second-hand they became one of the sounds of house music, as well as of countless pop records. Lush pads, glassy strings, fat brass stabs, rubbery basses: the polysynth's whole vocabulary.

Ember isn't any of those instruments. But it's a six-voice polysynth of that early-1980s kind, and it works the way they do, so what you learn here will serve you on any of them, and on any synthesiser with a filter and an envelope.

6.1 Find a sound you like

Ember has one factory bank, FAC A: 32 sounds, from pads and strings to basses, organs and stabs.

> **Try this**
>
> ■ SOUNDS › 2 SOUND and go through all 32. This time, play chords: three or four keys at once, and hold them. Ember is made for it. Write down the codes of the ones you like.
>
> Listen out for FA26, SHIMMER, and FA02, STRINGS: both lean on the chorus, which is section 6.8.

6.2 How Ember is laid out

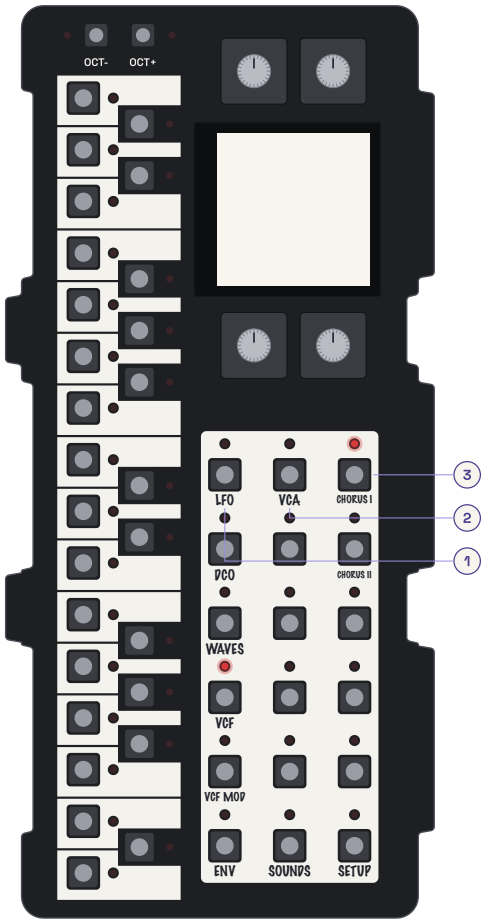


Figure 6.1 Ember's buttons. 1: the left column, the sound's path in order: LFO, DCO, WAVES, VCF, VCF MOD and ENV. 2: VCA, and SOUNDS at the bottom. 3: CHORUS I and II, then SETUP.

1 **The left column** is the sound's path, top to bottom, in the order the sound goes through it: LFO, DCO (the oscillator), WAVES, VCF (the filter), VCF MOD (what moves the filter) and ENV (the envelope).

The middle column has VCA, the loudness, at the top and SOUNDS at the bottom.

3 **The right column** has the two CHORUS buttons, then SETUP.

Most settings read 0 to 10, like the sliders on a synthesiser of this kind; a few are switches, ON or OFF, or a short list of choices. Every setting has its own encoder: turn it.

6.3 The journey of a sound

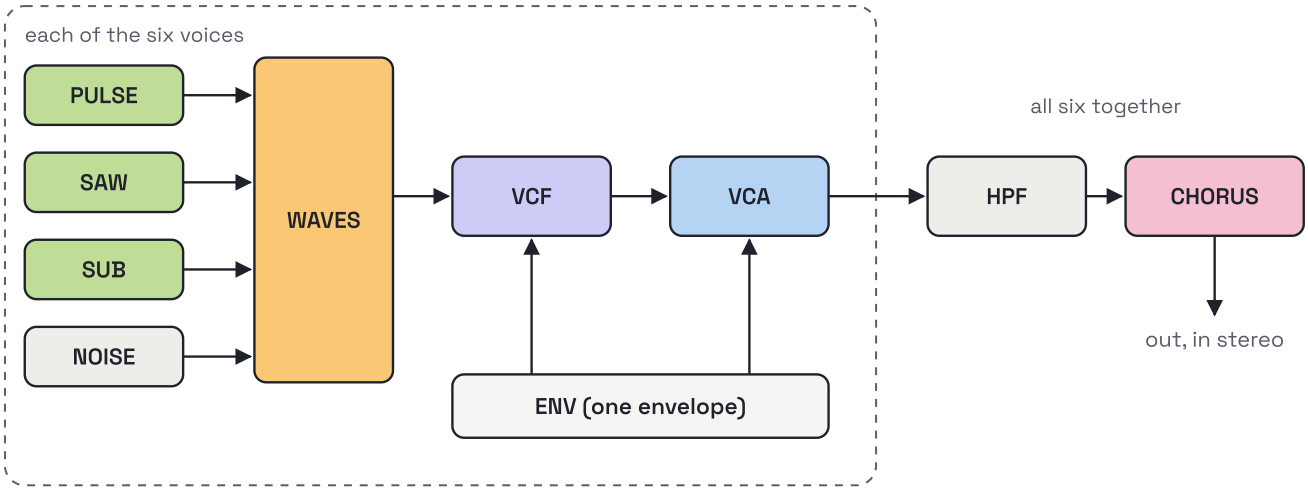


Figure 6.2 Ember from start to finish, for each of its six voices: the oscillator's waves, mixed, through the filter and the VCA, both moved by one envelope; then, for all six together, the high-pass filter and the chorus.

Each of Ember's six voices makes its sound the same way:

1 The **oscillator** makes the raw waves: a pulse, a sawtooth, a **sub**-oscillator an octave lower, and noise. WAVES mixes them.

2 The **filter** takes away the top of the sound.

3 The **VCA** sets how loud it is.

4 One **envelope** shapes both: it opens and closes the filter, and it shapes the loudness.

Then all six voices are added together and go through two more things they share: a **high-pass filter**, which can thin out the bass, and the **chorus**.

A sound to work on

Your own slots start as INIT SOUND, and on Ember that's already the plainest sound there is: one sawtooth, the filter wide open, full loudness while the key is down. A perfect blank page.

SOUNDS › 1 BANK › USER A › 2 SOUND › UA01

Play a chord. Bright, buzzy, rather harsh: that's a sawtooth, raw. Name it SAW 1 on the SOUNDS page (4 NAME : turn for the letter, push for the next one).

The sounds you'll build

This chapter builds seven sounds of your own. One of them, WARM PAD, grows section by section, a piece at a time, the way you'd build any sound; the others are made in a single section each. Every one starts as a copy of SAW 1. When a section says "start from a copy of SAW 1 in UA03", that means:

SOUNDS › 2 SOUND › UA01 › 3 COPY TO › UA03 › *hold* 3

Copying moves you onto the copy, so you're ready to work. By the end, USER A will hold:

Slot	Sound	Sections	What it shows
UA01	SAW 1	6.3	the raw sawtooth, the place to start
UA02	WARM PAD	6.4 to 6.8	every part of Ember, added one at a time
UA03	STAB	6.6	the envelope snapping the filter open
UA04	SYNTH BASS	6.6	the sub-oscillator, and the high-pass filter's bass lift
UA05	ORGAN	6.6	the VCA on GATE
UA06	SWEEP	6.7	the LFO moving the filter
UA07	BELL KEYS	6.8	both choruses at once

6.4 The oscillator

Ember has one oscillator per voice, set on two pages.

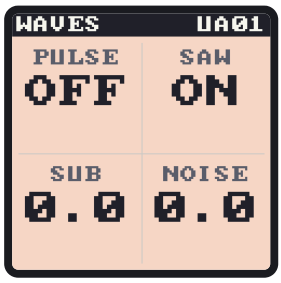


Figure 6.3 The WAVES page on INIT SOUND: the pulse and the sawtooth switched on or off, the sub-oscillator and noise as levels.

> **PULSE** (WAVES, encoder 1, ON or OFF) a pulse wave: a square whose width you can change (below). Hollow and woody.

> **SAW** (WAVES, encoder 2, ON or OFF) the sawtooth: every harmonic there is, the brightest, the one most sounds start from. Pulse and saw can be on together.

> **SUB** (WAVES, encoder 3, 0-10) a square wave an octave below the note, for weight: the floor under a bass or a pad.

> **NOISE** (WAVES, encoder 4, 0-10) hiss, for breath, wind and the snap of a stab.

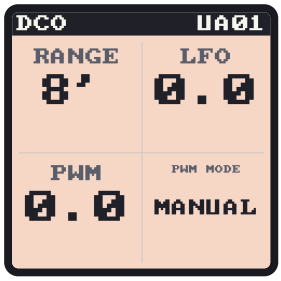


Figure 6.4 The DCO page: the octave, vibrato from the LFO, and the pulse's width (by hand or moved by the LFO).

> **RANGE** (DCO, encoder 1, 16', 8', 4') the octave, in feet: 8' is the note you play, 16' an octave below, 4' an octave above.

> **LFO** (DCO, encoder 2, 0-10) vibrato: how much the LFO moves the pitch (section 6.7).

> **PWM** (DCO, encoder 3, 0-10) the pulse's width: at 0 a square, turning up makes it narrower, thinner and more nasal.

> **PWM MODE** (DCO, encoder 4, LFO or MANUAL) MANUAL: PWM sets the width and it stays put. LFO: the LFO sweeps the width back and forth, and PWM sets how far: the pulse seems to move and thicken, almost as if there were two oscillators.

> ### Try this
> On SAW 1, switch SAW off and PULSE on, and hold a chord while you turn PWM from 0 up to about 8: the sound gets thinner and more nasal. Now set PWM MODE to LFO and PWM to about 5: the sound comes alive, moving slowly. Put SAW 1 back as it was afterwards (SAW ON, PULSE OFF, PWM 0.0, PWM MODE MANUAL): everything else starts from it.

With only one oscillator per voice, Ember gets its thickness in other ways: the sub-oscillator underneath, the moving pulse width, and above all the chorus (section 6.8). The pad you're about to start uses all three.

WARM PAD, part 1: the waves

Start from a copy of SAW 1 in UA02, and name it `WARM PAD`.

1. Both waves at once: ■ WAVES › ❶ PULSE › `ON`, with SAW still ON. Hold a chord: brighter and fuller than either alone.
2. A moving pulse: ■ DCO › ❹ PWM MODE › `LFO` and ❸ PWM › `5.0`. Listen to the chord start to breathe.
3. Weight underneath: ■ WAVES › ❸ SUB › `4.0`.

Hold a low chord. Still harsh, and it starts and stops like a switch, but there's a lot going on in it now. The next three sections tame it.

6.5 The filter

The filter is where the sound gets its tone. It's a **low-pass** filter: it lets the low part of the sound through and cuts the high part, steeply.

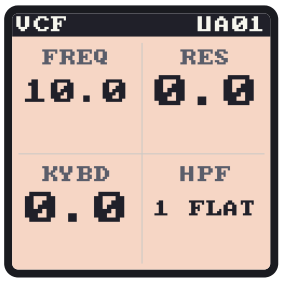


Figure 6.5 The VCF page on INIT SOUND: the filter wide open.

FREQ (VCF, encoder 1, 0-10) where the filter starts to cut. 10 is wide open; turn it down and the sound darkens, until there's almost nothing left.

RES (VCF, encoder 2, 0-10) resonance: the filter rings at FREQ, so the harmonics there stand out. A little gives an edge, more gives a "wow", and near the top the filter whistles on its own.

KYBD (VCF, encoder 3, 0-10) how much the filter follows the keys. At 0 it stays put, so high notes come out duller than low ones; at 10 it follows them all the way.

HPF (VCF, encoder 4, 0-3) the high-pass filter, after all six voices: 0 lifts the bass a little, 1 is flat (does nothing), 2 and 3 cut the bass away, for thinner, brighter sounds that sit above a bass line.

Try this

On SAW 1, hold a chord and turn FREQ slowly from 10 down to 2 and back. Then set FREQ to about 4 and turn RES up while you sweep again. Try HPF at 3 on a full chord: the low end goes, and what's left is bright and airy. Put SAW 1 back afterwards: FREQ 10.0, RES 0.0, HPF 1.

WARM PAD, part 2: the filter

Back on WARM PAD (UA02): ■ VCF › ❶ FREQ › `5.5`, ❷ RES › `1.0` and ❸ KYBD › `3.0`. The harshness has gone and it's warm; the little resonance keeps it from sounding dull, and KYBD keeps high chords from going muddy.

6.6 One envelope

Ember has one envelope, and it does two jobs at once: it shapes how loud each note is, and it can open and close the filter.



Figure 6.6 VCF MOD (how far the envelope and the LFO move the filter) and ENV, the one envelope that shapes both the filter and the loudness.

■ ENV A, D, S and R

- **A** (encoder 1, 0-10) attack: how long the note takes to get to the top.
- **D** (encoder 2, 0-10) decay: how long it takes to fall to the sustain level.
- **S** (encoder 3, 0-10) sustain: the level it holds while the key is down.
- **R** (encoder 4, 0-10) release: how long it takes to die away after you let go.

The loudness follows the envelope as long as ■ VCA › ❶ VCA › `ENV`. Set it to GATE instead and the loudness ignores the envelope: on while the key is down, off when you let go, like an organ. The envelope then works on the filter alone. ❷ LEVEL is the sound's own volume.

Four shapes

On SAW 1, with VCA on ENV, try these on the ENV page, and put A 0.0, D 0.0, S 10.0, R 1.0 back afterwards:

Organ

`A 0.0` `D 0.0` `S 10.0` `R 1.0`

Piano-ish

`A 0.0` `D 5.0` `S 0.0` `R 4.0`

Pad

`A 5.0` `D 6.0` `S 8.0` `R 6.0`

Pluck

`A 0.0` `D 3.0` `S 0.0` `R 2.0`

WARM PAD, part 3: the shape

On WARM PAD: the pad shape, ■ ENV › ❶ A › `5.0`, ❷ D › `6.0`, ❸ S › `8.0`, ❹ R › `6.0`. Now each chord swells in over about a second, and fades away when you let go. Play slow chords, one into the next, and let them overlap as they fade.

The envelope on the filter

Now the part that makes a synth sound alive. On the VCF MOD page:

> **ENV** (VCF MOD, encoder 1, 0-10) how far the envelope opens the filter on each note, from wherever FREQ has it.

> **POLARITY** (VCF MOD, encoder 2, NORMAL or INVERT) NORMAL opens the filter; INVERT closes it instead, so the note starts dark and brightens as it fades.

> **LFO** (VCF MOD, encoder 3, 0-10) how far the LFO moves the filter (section 6.7).

WARM PAD, part 4: brightening as it swells

On WARM PAD, VCF MOD › ENV › `2.0` : a touch of the envelope on the filter, so each chord gets a little brighter as it swells in, and darker as it fades. Small, but it's the difference between a pad that sits there and one that breathes.

Build STAB

Start from a copy of SAW 1 in UA03, and set:

STAB

SAW 1, plus:

VCF	FREQ 3.0	RES 4.0	KYBD 5.0	
VCF MOD	ENV 5.0			
ENV	A 0.0	D 3.5	S 1.0	R 3.0
VCA	LEVEL 10.0			

Play short chords: every one snaps open and closes, the stab of a hundred house records. Name it **STAB**. Then try ENV higher for a brighter snap, D longer for a slower "yow", and PULSE on with PWM MODE on LFO for a thicker one.

WARM PAD and STAB use the same envelope, turned round: slow and held for the pad, fast and falling for the stab. Nearly every polysynth sound is one of those, or somewhere between.

Build SYNTH BASS

Polysynths play basses too, and the sub-oscillator is the reason. Start from a copy of SAW 1 in UA04.

1. An octave down, with a square underneath that: ■ DCO › ❶ RANGE › `16'`, ■ WAVES › ❶ PULSE › `ON` and ❸ SUB › `8.0`.

2. Dark, with a pluck from the envelope: ■ VCF › ❶ FREQ › `2.5`, ❷ RES › `3.0`, ❸ KYBD › `5.0`, and ■ VCF MOD › ❶ ENV › `4.0`.

3. The shape: ■ ENV › ❶ A › `0.0`, ❷ D › `4.0`, ❸ S › `3.0`, ❹ R › `2.0`.

4. The bass lift: ■ VCF › ❹ HPF › `0`. HPF 0 isn't a cut at all: it gently lifts the lowest notes. On a bass, that's what you want.

SYNTH BASS

SAW 1, plus:

DCO	`RANGE 16'`			
WAVES	`PULSE ON`	`SAW ON`	`SUB 8.0`	
VCF	`FREQ 2.5`	`RES 3.0`	`KYBD 5.0`	`HPF 0`
VCF MOD	`ENV 4.0`			
ENV	`A 0.0`	`D 4.0`	`S 3.0`	`R 2.0`

Play single notes, low down, a bass line. Name it `SYNTH BASS`. Then play a chord on it and hear why basses are usually one note at a time: a low chord turns to mud.

Build ORGAN

The VCA on GATE: loudness on while the key is down, off when you let go, nothing in between, like an organ's. Start from a copy of SAW 1 in UA05.

1. A hollow square and its octave below: ■ WAVES › ❶ PULSE › `ON`, ❷ SAW › `OFF`, ❸ SUB › `6.0`. PWM stays at 0.0 and PWM MODE on MANUAL, so the pulse is a square.
2. The filter fairly open, and following the keys all the way, so every note has the same tone: ■ VCF › ❶ FREQ › `7.0`, ❸ KYBD › `10.0`.
3. ■ VCA › ❶ VCA › `GATE`, and ❷ LEVEL › `5.5`.
4. Press ■ CHORUS II: the faster chorus, for the swirl of an organ's rotating speaker.

ORGAN

SAW 1, plus:

WAVES	`PULSE ON`	`SAW OFF`	`SUB 6.0`
VCF	`FREQ 7.0`	`KYBD 10.0`	
VCA	`VCA GATE`	`LEVEL 5.5`	
CHORUS	`II ON`		

Name it `ORGAN`. Play chords, and short ones: every note starts and stops dead, exactly as long as your finger. Now turn the ENV page's settings: nothing changes, because on GATE the loudness ignores the envelope, and VCF MOD's ENV is at 0, so the filter does too. Give the envelope a job on the filter alone: ■ ENV › ❷ D › `2.0` and ❸ S › `0.0`, then ■ VCF MOD › ❶ ENV › `3.0`. Each note still lasts exactly as long as your finger, but now starts with a flash of brightness: an organ's key click. Keep it or put ENV back to 0.0, as you like

One envelope, two jobs

Because one envelope shapes both the filter and the loudness, a long attack makes a note swell in loudness and brightness together, and a short decay makes both snap. That's most of the classic polysynth sounds. When you want the loudness to stay put while the filter moves, set the VCA to GATE, as ORGAN does.

6.7 Movement

Ember's LFO, one slow wobble shared by every voice, can move three things: the pitch (vibrato), the pulse's width (PWM MODE LFO, section 6.4) and the filter.

LFO RATE and DELAY

> **RATE** (encoder 1, 0-10) how fast it wobbles.

> **DELAY** (encoder 2, 0-10) how long it waits after you start playing before it fades in. The wait starts again with the first key of each new phrase, after every key has been let go, so a held chord starts steady and only then begins to move.

How much it moves each thing is set where that thing is: **LFO** on the DCO page for vibrato, **PWM** with PWM MODE on LFO for the pulse, and **LFO** on the VCF MOD page for the filter.

Build SWEEP

Start from a copy of SAW 1 in UA06.

First, vibrato, to hear the delay: LFO › 1 RATE › `5.5`, 2 DELAY › `6.0` and DCO › 2 LFO › `1.5`. Hold a note: it starts straight, then the vibrato arrives. That's for leads and strings; this sound is going to be the other kind.

Now DCO › 2 LFO › `0.0` and LFO › 2 DELAY › `0.0`. Close the filter so it has room to move: VCF › 1 FREQ › `4.0` and 2 RES › `5.0`. Then VCF MOD › 3 LFO › `6.0`, and LFO › 1 RATE › `3.0`.

SWEEP

SAW 1, plus:

LFO	`RATE 3.0`	`DELAY 0.0`
VCF	`FREQ 4.0`	`RES 5.0`
VCF MOD	`LFO 6.0`	

Hold a chord: the tone goes round and round, slowly opening and closing. Name it `SWEEP`. Try the RATE faster for a wah-wah, slower for a sweep you only notice after a while.

> **WARM PAD, part 5: a slow drift**
>
> On WARM PAD, the LFO is already sweeping the pulse width (PWM MODE LFO). Slow it right down, so the pad drifts instead of wobbling: LFO › 1 RATE › `3.0`. It was already 3.0 if you never changed it; turn it either way and listen to what it does to the pad.

A MIDI keyboard's mod wheel adds vibrato too, as much as SETUP's WHEEL allows (section 6.10).

6.8 The chorus

Here's Ember's secret ingredient, and the reason chapter 2 said a chorus is "half the secret of many famous synth sounds". A chorus copies the sound, sways the copy's timing back and forth very slightly, and plays it beside the original. On Ember the copy sways one way on the left and the other way on the right, so the sound spreads wide, warm and shimmering, as if it were several instruments, not one.

Two buttons control it, each on or off, lit while on:

- **CHORUS I** on its own: a slow, gentle sway, about once every two seconds. Wide and warm.
- **CHORUS II** on its own: faster, a little less deep. Richer, more obviously moving.
- **Both on** (I+II): something else again, a fast, shallow shimmer, nearer to vibrato than to chorus. Lovely on pads and electric pianos.

Switch the chorus on and you'll also hear a faint hiss in quiet moments. That's on purpose: the circuits these choruses grew from always hissed, and on dark, quiet sounds the hiss breathing in time with the sway is part of the swirl.

WARM PAD, part 6: the chorus

On WARM PAD, hold a chord and press ■ CHORUS I. That's it: the pad goes wide and warm, and that's the sound people mean when they talk about this kind of synth. Press CHORUS I again to switch it off and listen to how flat the pad was without it; then on again.

Last, the level: with everything on, the pad is loud, so ■ VCA › ❷ LEVEL › `5.0`.

WARM PAD

SAW 1, plus everything from sections 6.4 to 6.8:

WAVES	`PULSE ON` `SAW ON` `SUB 4.0`
DCO	`PWM 5.0` `PWM MODE LFO`
VCF	`FREQ 5.5` `RES 1.0` `KYBD 3.0`
VCF MOD	`ENV 2.0`
ENV	`A 5.0` `D 6.0` `S 8.0` `R 6.0`
VCA	`LEVEL 5.0`
CHORUS	`I ON`

Six changes, one at a time, and you've built a pad. Compare it with the factory's WARM PAD, FA01: they're close cousins.

Build BELL KEYS

Now the third chorus. Start from a copy of SAW 1 in UA07.

1. A thin pulse: ■ WAVES › ❶ PULSE › `ON`, ❷ SAW › `OFF`, ■ DCO › ❸ PWM › `3.0` (PWM MODE stays MANUAL).
2. Struck, and ringing away: ■ ENV › ❶ A › `0.0`, ❷ D › `6.0`, ❸ S › `2.0`, ❹ R › `5.0`.
3. Bright at the strike and mellowing as it rings: ■ VCF › ❶ FREQ › `6.0`, ❸ KYBD › `5.0`, ■ VCF MOD › ❶ ENV › `3.0`.
4. Both choruses: press ■ CHORUS I and ■ CHORUS II, so both are lit. And ■ VCA › ❷ LEVEL › `5.5`.

BELL KEYS

SAW 1, plus:

WAVES	`PULSE ON` `SAW OFF`
DCO	`PWM 3.0`
VCF	`FREQ 6.0` `KYBD 5.0`
VCF MOD	`ENV 3.0`
ENV	`A 0.0` `D 6.0` `S 2.0` `R 5.0`
VCA	`LEVEL 5.5`
CHORUS	`I ON` `II ON`

Name it `BELL KEYS`, and play chords, held and let go. That quick shimmer is I+II. Switch one chorus off and then the other, and hear how each sounds on its own on the same sound.

The chorus belongs to the sound, so each sound keeps its own setting. With Haze joined on, Ember can also send into Haze's echo and reverb, as every synth can (chapter 2, section 2.5).

6.9 Playing chords

Ember has six voices, so six notes at once. Play a seventh, and the voice that has been playing longest gives way to the new note, fading out quickly so the change doesn't click. Ember also uses its voices in turn, so a note you've let go of carries on ringing through its release while the next notes start on other voices.

> **Sustain pedal.** From a MIDI keyboard, the sustain pedal holds notes as a piano's does.

> **How hard you play.** Ember's own keys play every note at the same loudness. From a MIDI keyboard, turn VELO on in SETUP (section 6.10) and harder is louder.

> **LEVEL** (VCA, encoder 2, 0-10) the sound's own volume, to make your sounds sit together. Each step is about 2 dB, which is why the recipes above set it: a full chorused pad with four notes held is much louder than a single plucked note.

Try this

On WARM PAD, play a chord and, while it's still fading, a second chord: both ring together, because the second chord takes free voices. Then hold four notes and play three more, one by one: you'll hear the first notes give way. Six is plenty for chords; it's a thing to know when the release is long.

6.10 Ember's SETUP

SETUP's THIS UNIT pages have Ember's own settings, beside the MIDI channel:

> **BEND** (encoder 2, 0-12) how far a MIDI keyboard's pitch bend wheel bends the notes, in semitones.

> **VCF BEND** (encoder 3, 0-10) how far the same wheel moves the filter too: bend the wheel and the sound brightens or darkens.

> **WHEEL** (encoder 4, 0-10) how much vibrato the mod wheel can add.

> **VELO** (second page, encoder 1, OFF or ON) OFF: every note equally loud. ON: harder is louder, from a MIDI keyboard or from Step.

What you have at the end of this chapter

Seven sounds of your own: SAW 1, WARM PAD, STAB, SYNTH BASS, ORGAN, SWEEP and BELL KEYS. Between them, every page on Ember, and the shape of nearly every polysynth sound there is: a wave, a filter, one envelope moving both, and a chorus to make it sing. Open the factory's STRINGS, BRASS and HOUSE ORGAN, and see how close you already are.

Chapter 7 - Axis

Four waves at the corners of a diamond, and a point that moves between them.

Every synth so far makes a sound and then changes it. Axis makes four sounds at once and moves between them. Each voice has four oscillators, A, B, C and D, sitting at the four corners of a diamond, and a point on the diamond decides how much of each you hear. Put the point at a corner and you hear that oscillator alone; in the middle, all four equally; anywhere else, a blend. Then let an envelope move the point while the note plays, and the sound travels: from a bell to a voice to a buzz and back again, on every note. That's **vector synthesis**.

Vector synthesis: where it came from, and who uses it

Vector synthesis began with Sequential Circuits' Prophet VS in 1986: four digital waves per voice, mixed by a joystick, the joystick's moves remembered by an envelope, then an analogue filter. Its waves were made by computer, with a gritty, glassy, slightly metallic sound that nothing else had, and its evolving pads turned up in film music and on records of the late 1980s. Sequential didn't last much longer, but its ideas did: some of the same people went on to Korg's Wavestation (1990), which strung waves together in sequences, and Yamaha put vector mixing into its SY22 and TG33. Software versions are still made today.

Axis isn't any of those instruments. It's a vector synth of the late-1980s kind and works the way they do, so what you learn here will serve you on any of them. But its waves are its own: all 128 were made for Axis, from our own recipes, and none is copied from anywhere. They're in families of sixteen, so you can move through related sounds, and section 7.8 has a trick that the vector synths of the 1980s couldn't do.

7.1 Find a sound you like

Axis has one factory bank, FAC A, with 32 sounds.

> **Try this**
>
> ■ SOUNDS › 2 SOUND and go through all 32. Hold the notes this time, for a few seconds each: a lot of Axis's sounds change as they go, and you'll miss it in short notes. FA01, VECTOR PAD, travels round all four corners for ever; FA12, WAVE SWEEP, sweeps through bright, resonant waves; FA14, VOWELSCAN, changes vowel as each note opens.

7.2 How Axis is laid out

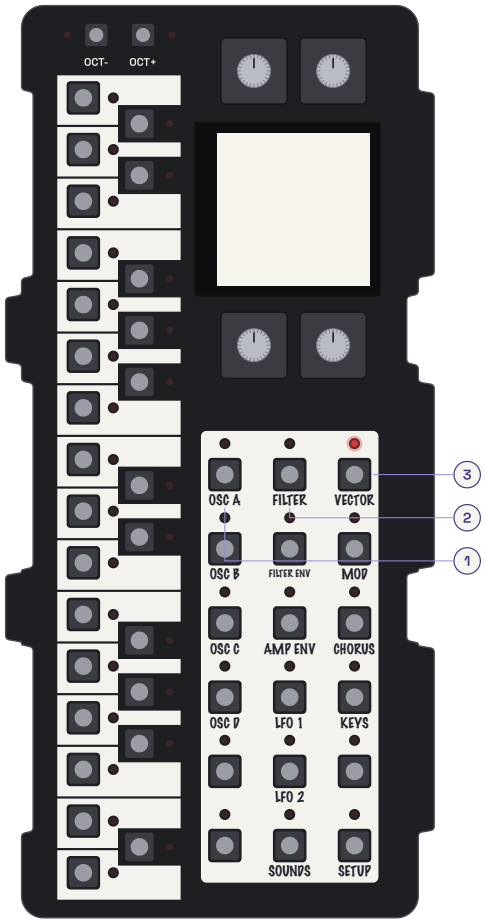


Figure 7.1 Axis's buttons. 1: the four oscillators, A to D. 2: the filter, the envelopes, the LFOs, then SOUNDS. 3: VECTOR (pressed again, its envelope), MOD, CHORUS, KEYS, then SETUP.

1 **The left column** is the four oscillators: OSC A, OSC B, OSC C and OSC D.

2 **The middle column** is what happens to the sound: FILTER, FILTER ENV, AMP ENV (the loudness), LFO 1, LFO 2, then SOUNDS.

3 **The right column** is VECTOR (press it again for its envelope), MOD, CHORUS, KEYS, then SETUP.

A sound to work on

Your own slots start as INIT SOUND: four sawtooths, all the same, with the point in the middle, so it sounds like one plain sawtooth. Give its corners four different waves, and you've got something to explore:

SOUNDS › 1 BANK › USER A › 2 SOUND › UA01

1 OSC A › 1 WAVE › 0 SINE : the pure one.

2 OSC B › 1 WAVE › 3 SQUARE : hollow.

3 OSC C › 1 WAVE › 65 AH HI : a voice.

4 OSC D › 1 WAVE › 96 BELL 1 : a bell.

Name it CORNERS on the SOUNDS page (4 NAME : turn for the letter, push for the next one).

The sounds you'll build

This chapter builds seven sounds of your own. Most start as a copy of CORNERS, so you'll always know what's at each corner. When a section says "start from a copy of CORNERS in UA03", that means:

SOUNDS › 2 SOUND › UA01 › 3 COPY TO › UA03 › *hold* 3

Copying moves you onto the copy, so you're ready to work. By the end, USER A will hold:

Slot	Sound	Section	What it shows
UA01	CORNERS	7.2	four different waves, one at each corner
UA02	WIDE PAD	7.4 and 7.5	waves from different families, tuned apart, and the filter
UA03	PULSER	7.6	a rhythm from a looping envelope
UA04	TRAVELLER	7.7	the vector envelope going round the diamond
UA05	DRIFTER	7.7	two LFOs through the MOD list
UA06	SCANNER	7.8	WAVE SCAN
UA07	UNISON LEAD	7.10	all eight voices on one note

> ### Reading the numbers
>
> Axis shows every setting as it is: times in MS and S, rates in HZ, the cutoff in HZ and KHZ, levels from 0.0 to 10.0, X and Y from -100 to +100 (two at a click). The recipes give them the same way, so turn until the screen says what the recipe says. Times and rates move in small steps at the bottom and bigger ones at the top, so a recipe's 0.10 S or 1.02 S is exactly a value the screen can show.

7.3 Four waves on a diamond

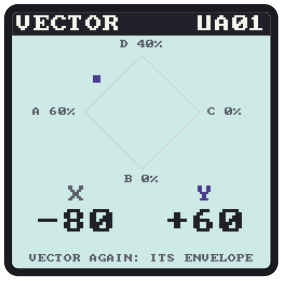


Figure 7.2 The VECTOR page: the diamond, with A on the left, D at the top, C on the right and B at the bottom, the point (moved with encoders 3 and 4), and how much of each corner you're hearing.

The VECTOR page is the diamond itself. A is on the left, C on the right, D at the top and B at the bottom. The square dot is the point, and beside each corner is how much of it you're hearing.

> **X** (VECTOR, encoder 3, -100 to +100) left and right: towards A, or towards C.

> **Y** (VECTOR, encoder 4, -100 to +100) down and up: towards B, or towards D.

The point is on the bottom encoders on purpose: your hands stay below the screen and you can watch the diamond while you turn. Move the point beyond the diamond's edge and the two corners nearest it share the sound between them.

> **Try this**
>
> On CORNERS, hold a chord and turn X slowly all the way left (the sine), then right (the voice); then Y down (the square) and up (the bell). Then find places in between that you like: a little voice in the bell, a little square in the sine. Every point on the diamond is a different sound.
>
> Put X and Y back to 0 when you're done: the copies of CORNERS start with the point in the middle.

7.4 The oscillators

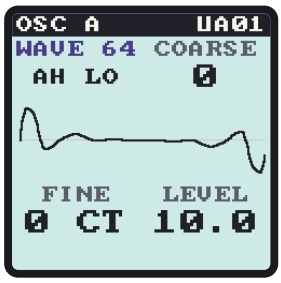


Figure 7.3 An oscillator's page: the wave by number and name, drawn; COARSE, FINE and LEVEL.

Each corner's oscillator has its own page, all four alike:

> **WAVE** (encoder 1, 0-127) which of Axis's 128 waves it plays: its number and name (64 AH LO), and the wave drawn.

> **COARSE** (encoder 2, -24 to +24) its pitch, in semitones: 12 is an octave up, 7 a fifth.

> **FINE** (encoder 3, -100 to +100) a fine adjustment, in cents (a hundredth of a semitone): a little apart from another corner and they beat slowly against each other.

> **LEVEL** (encoder 4, 0-10) how loud this corner is when the point is on it.

The waves

The 128 waves come in eight families of sixteen, and the waves in a family are related, so turning WAVE moves you through sounds that belong together:

Waves	Family	What they sound like
0-15	Plain shapes	Sine, triangle, sawtooth, square, pulses: the basics.
16-31	Organ	Pipe and electric organ registrations, from a soft flute to a full organ.
32-47	Hollow	Odd harmonics: clarinets, reeds, woody and nasal sounds.
48-63	Octaves and even	Octave stacks, even harmonics, whistles and sparkle.
64-79	Vocal	Vowels, low and high: AH, EH, EE, OH, OO and more.
80-95	Resonant	A sawtooth with one bright peak that climbs through the family.
96-111	Bells and metal	Bells, glass, chimes, vibes, marimba, gongs.
112-127	Digital	Dense digital waves, sync sounds, folded and clipped waves, buzz and grit.

The waves are played at a fairly low resolution (12 bits), as vector synths of this kind did, which gives them a slight grain and glassiness. That's part of the sound, not a fault.

> **Try this**
>
> On CORNERS, hold a note and turn OSC C › 1 WAVE slowly from 64 to 79, through the vocal family: one voice singing sixteen vowels. Then put it back to 65.

A pad from four families

CORNERS uses four very different waves, so the point decides what you hear. A pad wants the opposite: four waves that blend into one, tuned a little apart so they move against each other, with the point in the middle to hear them all.

WIDE PAD, part 1: the corners

This one starts from INIT SOUND itself, not CORNERS: go to UA02 and name it `WIDE PAD`.

1. A and C, the left and right, a soft sawtooth each, pulled a few cents apart:
   OSC A › ❶ WAVE › `8 SOFT SAW` , ❸ FINE › `−8 CT` ;
   OSC C › ❶ WAVE › `8 SOFT SAW` , ❸ FINE › `+8 CT` . Hold a chord: a slow, warm beating.

2. B, the bottom, a hollow wave an octave down for body:
   OSC B › ❶ WAVE › `33 HOLLOW 2` , ❷ COARSE › `−12` .

3. D, the top, a glassy shine an octave up, quieter: OSC D › ❶ WAVE › `100 GLASS 1` , ❷ COARSE › `+12` , ❹ LEVEL › `6.0` .

Leave the point in the middle, X 0 and Y 0, so all four play equally. Bright and big, but buzzy: the filter's next.

7.5 The filter

After the diamond, each voice goes through a resonant low-pass filter, which takes away the top of the sound.

CUTOFF (FILTER, encoder 1) where the filter starts to cut, shown in Hz: all the way up, it's wide open.

RES (FILTER, encoder 2, 0-10) resonance: the filter rings where it cuts, giving an edge, then a "wow".

ENV (FILTER, encoder 3, -100 to +100) how far the FILTER ENV opens it (or, below 0, closes it) on each note.

KEYBOARD (FILTER, encoder 4, 0-10) how much the filter follows the keys; 8.5 to start, so high notes stay nearly as bright as low ones.

WIDE PAD, part 2: filter, shape and chorus

1. Take the buzz off: FILTER › ❶ CUTOFF › `2.3 KHZ` and ❷ RES › `1.5`.

2. A slow swell and a long fade: on AMP ENV, ❶ POINT › `1` and ❷ TIME › `1.02 S` (the time to reach full loudness), then ❶ POINT › `4` and ❷ TIME › `1.22 S` (the time to fade after you let go). Section 7.6 explains these pages properly.

3. CHORUS, and press it again so it's lit: on.

4. KEYS › ❹ LEVEL › `10.0`: a soft pad needs its full level.

WIDE PAD

INIT SOUND, with:

OSC A	`WAVE 8 SOFT SAW`	`FINE −8 CT`	
OSC B	`WAVE 33 HOLLOW 2`	`COARSE −12`	
OSC C	`WAVE 8 SOFT SAW`	`FINE +8 CT`	
OSC D	`WAVE 100 GLASS 1`	`COARSE +12`	`LEVEL 6.0`
FILTER	`CUTOFF 2.3 KHZ`	`RES 1.5`	
AMP ENV	`POINT 1 TIME 1.02 S`	`POINT 4 TIME 1.22 S`	
CHORUS	`ON`		
KEYS	`LEVEL 10.0`		

Play slow chords and let them fade into each other. Then move the point while one holds: towards D and the glass comes up; towards B and the body thickens. The same pad, four ways.

7.6 Envelopes with five points

Axis's envelopes are more than attack, decay, sustain and release. Each has five **points**, 0 to 4, each with its own level, and a **time** to travel to each one from the last. One of them is the **sustain** point, where the note waits while you hold the key; when you let go, it carries on through the points after it.

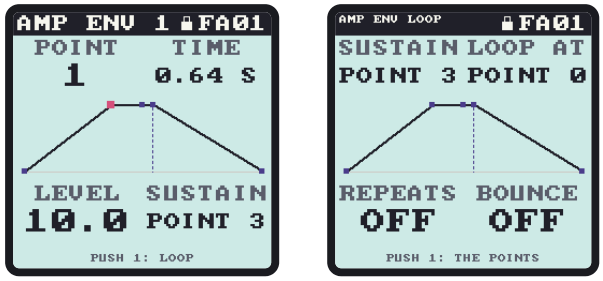


Figure 7.4 An envelope's page (here the AMP ENV): each point's TIME and LEVEL, and which point it holds at (SUSTAIN). Push encoder 1 for its loop settings (right).

AMP ENV (the loudness) and FILTER ENV (the filter, by FILTER's ENV): POINT, TIME, LEVEL and SUSTAIN

> **POINT** (encoder 1, 0-4) which point the other boxes are about. The picture marks it.

> **TIME** (encoder 2) how long it takes to get to this point from the one before, from 4 ms to 41 seconds. (Point 0 is where it starts, so it has no time.)

> **LEVEL** (encoder 3, 0-10) this point's level.

> **SUSTAIN** (encoder 4, point 1-4) the point it holds at while the key is down.

INIT SOUND's AMP ENV is the plainest shape: up to full at point 1 almost at once, along to point 3, held there, then down to silence at point 4 when you let go. Five points let you draw much more: a swell that overshoots and settles, a pluck with a second, softer bump, a note that fades and then rises again while you hold it.

Loops

Push encoder 1 on any envelope page and the boxes become its loop settings (push again for the points):

> **SUSTAIN** (encoder 1) the sustain point, as before.

> **LOOP AT** (encoder 2, point 0-3) where a loop goes back to.

> **REPEATS** (encoder 3, OFF, 1-7, FOR EVER) how many times it goes round, from LOOP AT to the sustain point, before settling.

> **BOUNCE** (encoder 4, ON or OFF) ON: back and forth between the two, instead of jumping back to the start.

Build PULSER

A rhythm made by the envelope alone. Start from a copy of CORNERS in UA03, and go to ■ AMP ENV.

1. Point 1: ❶ POINT › `1`, ❷ TIME › `0.10 S`, ❸ LEVEL › `10.0`.
2. Point 2: ❶ POINT › `2`, ❷ TIME › `0.10 S`, ❸ LEVEL › `2.0`. The note dips.
3. Point 3: ❶ POINT › `3`, ❷ TIME › `0.10 S`, ❸ LEVEL › `6.0`. It comes back up, not quite as far. SUSTAIN stays at point 3.
4. The loop: push encoder 1, then ❷ LOOP AT › `POINT 1` and ❸ REPEATS › `FOR EVER`.
5. It's a single-note sound with gaps in it, so ■ KEYS › ❹ LEVEL › `10.0`.

PULSER

CORNERS, with an AMP ENV of:

POINT 1	`TIME 0.10 S`	`LEVEL 10.0`
POINT 2	`TIME 0.10 S`	`LEVEL 2.0`
POINT 3	`TIME 0.10 S`	`LEVEL 6.0`
LOOP	`SUSTAIN POINT 3`	`LOOP AT POINT 1` `REPEATS FOR EVER`
KEYS	`LEVEL 10.0`	

Name it **PULSER**. Hold a note: it pulses, loud-quiet-softer, loud-quiet-softer, for as long as you hold it. Each time round it jumps from point 3 back to point 1. Now push encoder 1 and turn ❹ BOUNCE › `ON`: instead of jumping back it goes back the way it came, 1-2-3-2-1-2-3, and the rhythm changes shape. Then try the times: all three at 0.40 S for a slow throb, or point 2's at 0.40 S and the others short for a swing.

7.7 The vector envelope, and movement

Here's the heart of Axis. Press ■ VECTOR VECTOR ENV (VECTOR again) and the diamond comes back with five points on it: this envelope moves the **point on the diamond** while the note plays, so the sound travels between the corners. It works like the other envelopes, five points with a time to each, a sustain point and a loop, except that each point is a place on the diamond, its X and Y, instead of a level.

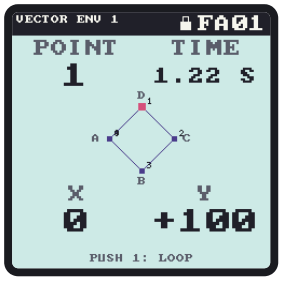


Figure 7.5 The VECTOR envelope on FA01, VECTOR PAD: its five points drawn on the diamond, the path the sound takes.

■ VECTOR VECTOR ENV POINT, TIME, X and Y (push encoder 1 for its loop)

Build TRAVELLER

A sound that goes round all four corners, for ever. Start from a copy of CORNERS in UA04, and press ■ VECTOR VECTOR ENV. Set the five points to the four corners, a second apart, and back to where it started:

TRAVELLER

CORNERS, with a VECTOR ENV of:

POINT 0	`X 0`	`Y +100`	`the bell (D)`	
POINT 1	`TIME 1.02 S`	`X +100`	`Y 0`	`the voice (C)`
POINT 2	`TIME 1.02 S`	`X 0`	`Y −100`	`the square (B)`
POINT 3	`TIME 1.02 S`	`X −100`	`Y 0`	`the sine (A)`
POINT 4	`TIME 1.02 S`	`X 0`	`Y +100`	`the bell again`
LOOP	`SUSTAIN POINT 4`	`LOOP AT POINT 0`	`REPEATS FOR EVER`	
KEYS	`LEVEL 5.0`			

X and Y move two at a click, so +100 is fifty clicks; turn fast. Name it `TRAVELLER`, hold a chord, and watch the point on the screen go round as the sound goes from bell to voice to square to sine and back. Every note goes round on its own, from the moment it starts, so play a second chord a little after the first and they'll be in different places on the diamond at once. That's FA01, VECTOR PAD, made by you.

Then try BOUNCE ON in the loop settings: round and back again instead of round and round. And the times: all four at about 4 S for a slow, evolving pad, or at 0.10 S for a strange, warbling buzz.

The LFOs and the MOD list

Axis has two LFOs, each with its own page:

> **RATE** (LFO 1 or 2, encoder 1) how fast, in Hz.

> **SHAPE** (encoder 2) TRIANGLE, SAW UP, SAW DOWN, SQUARE or RANDOM (a new random value each time round).

> **DELAY** (encoder 3) how long it waits after a note starts before fading in.

An LFO on its own moves nothing. What it moves, and how much, is set on the MOD page: a list of six connections, each one a SOURCE moving a DESTINATION by an AMOUNT.

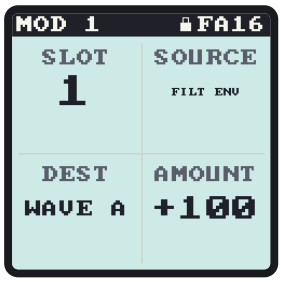


Figure 7.6 The MOD page on FA16, SCAN BASS: slot 1 sends the filter envelope to WAVE A, the WAVE SCAN.

> **SLOT** (MOD, encoder 1, 1-6) which of the six connections the other boxes show.

> **SOURCE** (encoder 2) what does the moving: LFO 1, LFO 2, FILT ENV, AMP ENV, VELOCITY (how hard a key is played), WHEEL (the mod wheel), KEY (which note), PRESSURE (pressing harder on a held key), or NONE.

> **DEST** (encoder 3) what it moves: PITCH, CUTOFF, RES, LEVEL, X, Y, PAN, and the WAVE destinations of section 7.8.

> **AMOUNT** (encoder 4, -100 to +100) how much, either way.

> #### Some sources need a MIDI keyboard
>
> Axis's own keys play every note at the same strength and have no mod wheel, so VELOCITY, WHEEL and PRESSURE only move when you play from a MIDI keyboard (one that sends aftertouch, for PRESSURE). The LFOs, the two envelopes and KEY all work from Axis's own keys.

Build DRIFTER

Two LFOs, two jobs. Start from a copy of CORNERS in UA05.

1. A vibrato that arrives late: MOD › 1 SLOT › `1`, 2 SOURCE › `LFO 1`, 3 DEST › `PITCH`, 4 AMOUNT › `+3`. Then on LFO 1: 1 RATE › `5.0 HZ` and 3 DELAY › `0.48 S`. Hold a note: it starts still, then wavers.

2. A point that wanders: MOD › 1 SLOT › `2`, 2 SOURCE › `LFO 2`, 3 DEST › `X`, 4 AMOUNT › `+60`; and LFO 2 › 1 RATE › `0.20 HZ`, once every five seconds.

3. KEYS › 4 LEVEL › `7.0`.

DRIFTER

CORNERS, with:

MOD 1	`LFO 1` `PITCH` `+3`
MOD 2	`LFO 2` `X` `+60`
LFO 1	`RATE 5.0 HZ` `DELAY 0.48 S`
LFO 2	`RATE 0.20 HZ`
KEYS	`LEVEL 7.0`

Name it `DRIFTER`, and hold a chord for a long time, watching the point: it drifts from the sine side to the voice side and back on its own, while the vibrato wavers on top. Change LFO 2's SHAPE to RANDOM, and it jumps to a new place every five seconds instead.

TRAVELLER and DRIFTER both move the point, in different ways: the vector envelope draws a path that starts again with every note; an LFO sways back and forth whatever the notes do. Put both on one sound and you've got movement inside movement.

7.8 WAVE SCAN

Here's something the vector synths of the 1980s couldn't do. The MOD list has five more destinations: **WAVE A**, **WAVE B**, **WAVE C**, **WAVE D** and **WAVES** (all four together). Send a source there and it pushes that oscillator's wave along its family of sixteen, blending smoothly from each wave into the next, never leaving the family. At AMOUNT +100, a source at full moves it fifteen waves: the whole family.

Build SCANNER

Start from a copy of CORNERS in UA06.

1. Corner A gets the first of the resonant family: ■ OSC A › ❶ WAVE › `80 RESO 3`.
2. Listen to A alone: ■ VECTOR › ❸ X › `-100`.
3. The scan: ■ MOD › ❶ SLOT › `1`, ❷ SOURCE › `FILT ENV`, ❸ DEST › `WAVE A`, ❹ AMOUNT › `+100`.
4. Slow the filter envelope down, so you can hear it work: ■ FILTER ENV, ❶ POINT › `1` and ❷ TIME › `0.64 S`, then ❶ POINT › `2` and ❷ TIME › `1.02 S`.
5. ■ KEYS › ❹ LEVEL › `10.0`.

SCANNER

CORNERS, with:

OSC A	`WAVE 80 RESO 3`
VECTOR	`X -100` `Y 0`
MOD 1	`FILT ENV` `WAVE A` `+100`
FILTER ENV	`POINT 1 TIME 0.64 S` `POINT 2 TIME 1.02 S`
KEYS	`LEVEL 10.0`

Name it **SCANNER**, and hold a note: the resonant peak sweeps up through the family as the filter envelope rises, then falls back part of the way as it settles at point 2's level. Notice that the filter itself isn't moving: FILTER's ENV is still 0. The FILTER ENV is only being used as a source, to scan the waves. Change its times and the sweep follows.

An LFO makes a steady sweep that never stops; KEY moves the wave further along the higher you play; FILT ENV and AMP ENV sweep on every note. With a MIDI keyboard, WHEEL scans by hand (FA17, DRAWBARS, pulls in organ stops this way) and PRESSURE scans as you press. Seven other factory sounds scan too: SCAN PAD, VOWELSCAN, RESO SWEEP, SCAN BASS, BELL SCAN, FOLD BASS and COMPUTER. Open their MOD pages to see how.

7.9 Chorus

Axis has a stereo chorus of its own: press ■ CHORUS to see its page, and press it again to switch it on or off (lit while on). WIDE PAD has it on.

> **CHORUS** (encoder 1, ON or OFF) the same switch.

> **RATE** (encoder 2) how fast it sways, in Hz.

> **DEPTH** (encoder 3, 0-10) how deep.

> **SPREAD** (encoder 4, 0-10) how far apart the eight voices sit, left and right, so a chord is wide even with the chorus off. (PAN in the MOD list moves a voice as it plays.)

> ### Try this
> On WIDE PAD, hold a chord and switch the chorus off and on: off, it's smaller and plainer; on, it fills the room. Then try DEPTH at 10.0 and RATE faster for a seasick wobble, and back.

With Haze joined on, Axis can also send into Haze's echo and reverb, as every synth can.

7.10 Playing it

■ KEYS MODE, DETUNE, GLIDE and LEVEL

> **MODE** (encoder 1, POLY, UNISON or MONO) POLY: up to eight notes at once. UNISON: all eight voices on one note, slightly apart, for one enormous sound (FA27, SUPER SAW). MONO: one note at a time, joined-up notes gliding without starting again.

> **DETUNE** (encoder 2, 0-10) in UNISON, how far apart the voices are.

> **GLIDE** (encoder 3, OFF or a time) how long the pitch takes to slide from one note to the next.

> **LEVEL** (encoder 4, 0-10) the sound's own volume. Eight voices on one note are loud, which is why UNISON sounds want it low.

Build UNISON LEAD

This one starts from INIT SOUND: four sawtooths. Go to UA07 and name it `UNISON LEAD`.

1. Turn the level down first, because the next step makes it very loud: KEYS › 4 LEVEL › `3.5`.
2. All eight voices on one note: 1 MODE › `UNISON`, 2 DETUNE › `3.0`. Play a note: one enormous, swirling sawtooth.
3. Glide between notes: 3 GLIDE › `0.18 S`.
4. An edge from the filter, opening on each note: FILTER › 1 CUTOFF › `1.6 KHZ`, 2 RES › `3.0`, 3 ENV › `+40`.

UNISON LEAD

INIT SOUND, with:

KEYS	`MODE UNISON`	`DETUNE 3.0`	`GLIDE 0.18 S`	`LEVEL 3.5`
FILTER	`CUTOFF 1.6 KHZ`	`RES 3.0`	`ENV +40`	

Play a tune high up. One note at a time, every one huge, sliding into the next. DETUNE up to 6.0 for a wilder swirl, down to 1.0 for a tighter, more focused one. Then go back to CORNERS and try UNISON there: a unison of four different waves.

From a MIDI keyboard, harder is louder while SETUP's VELO is ON (as it starts); VELOCITY in the MOD list can make harder brighter, too.

7.11 Axis's SETUP

BEND (THIS UNIT, encoder 2, 0-12) how far a MIDI keyboard's pitch bend wheel bends the notes, in semitones.

VELO (THIS UNIT, encoder 3, ON or OFF) ON: harder is louder; OFF: every note the same.

What you have at the end of this chapter

Seven sounds of your own: CORNERS, WIDE PAD, PULSER, TRAVELLER, DRIFTER, SCANNER and UNISON LEAD. Between them, every page on Axis, and the idea behind every vector synth: not one sound changed, but four sounds and a path between them.

Chapter 8 - Taste

Any sound in the world, on a key: then chop it, bend it and play it.

Every synth so far makes its sounds from scratch. Taste doesn't make sounds at all: it **records** them, and plays them back from the keys. A drum hit, a chord from another unit, a word, rain on a window: once it's in Taste, it's on a key, and you can play it higher and lower, chop it into pieces, play it backwards, stretch it, loop it and filter it. That's a **sampler**, and a short recording is a **sample**.

Taste plays samples two ways, and every sound of Taste's holds one of each:

- **A KIT** puts a different sample on every key: 27 keys, 27 samples. Drum kits, chopped-up loops, a word on each key.
- **An INSTRUMENT** puts one sample across all the keys, played higher up the keyboard and lower down it, so you can play tunes and chords with it.

Sampling: where it came from, and who uses it

People were cutting up recorded sound long before computers: in the 1940s and 50s, composers of *musique concrète* in Paris spliced tape by hand, and the Mellotron of the 1960s played a strip of tape for each key, the strings and flutes on records by the Beatles and the Moody Blues. The first digital samplers, the Fairlight CMI of 1979 and the E-mu Emulator of 1981, cost as much as a house; Peter Gabriel, Kate Bush and Stevie Wonder used them to put real-world sounds into songs.

Then samplers got cheap. Ensoniq's Mirage (1984) and Akai's S900 (1986) brought them into bedrooms and small studios, and in 1987 and 1988 two machines changed music for good: E-mu's SP-1200 and the Akai MPC60, which Roger Linn designed. Both put sampling and a sequencer in one box with pads, and both had a gritty sound of their own: the SP-1200 recorded at 12 bits and a low rate, which made drums crunchy and bass heavy. Producers such as Marley Marl, Pete Rock and later J Dilla built hip hop from them, sampling drum breaks from old funk records, chopping them into hits and playing them back in new orders. A few seconds of drumming from The Winstons' "Amen, Brother" (1969), sampled, chopped and sped up, became the backbone of jungle and drum and bass.

Since then, samplers have played everything: whole orchestras, drum kits recorded one hit at a time, and vocal chops in pop. Taste isn't any of those machines. It's a sampler of the late-1980s kind, with that kind's character (CHARACTER, section 8.10), and it works the way they do, so what you learn here will serve you on any sampler you ever meet, or in any music software, which all sample the same way.

8.1 Find a sound you like

Taste has one factory bank, FAC A, with 32 sounds, all made from the samples built into Taste (the **starter set**, section 8.4): drum kits, chopped breaks, instruments, and many of them the same sample turned into something quite different.

> **Try this**
>
> ■ SOUNDS › ❷ SOUND and go through them. A kit (BOOM KIT, STUDIO KIT, BREAK CHOP) wants every key played, top to bottom; an instrument (KEYS, PAD, STRINGS) wants chords and tunes. Look at the KIT/INSTR button at the top of the middle column: lit, the sound's a kit; out, an instrument.
>
> Then listen to the families: KEYS, KEYS BACK, KEYS PLUCK and KEYS DRONE are all one sample of a piano-like sound. TWO SPEEDS plays the same drum loop on eight keys at four speeds, two ways (section 8.7 says why they sound so different).

8.2 How Taste is laid out

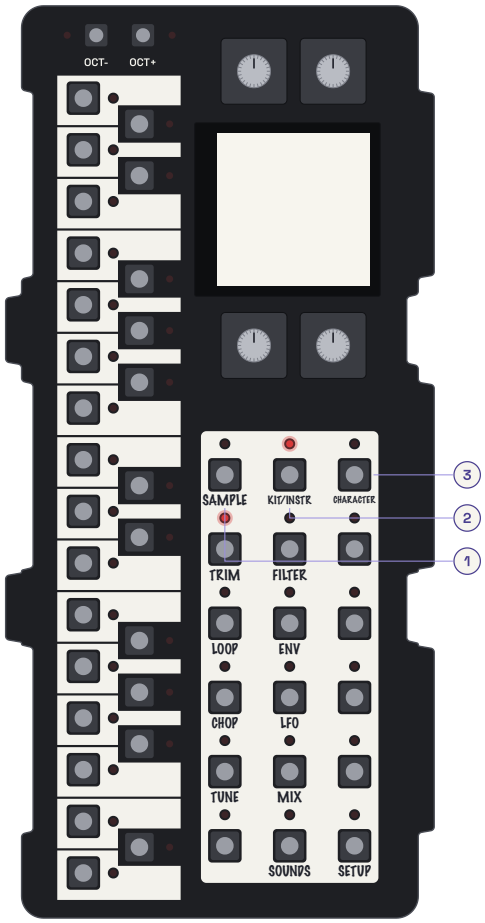


Figure 8.1 Taste's buttons. 1: the sample itself, SAMPLE to TUNE. 2: KIT/INSTR (lit in a kit), then how it sounds, FILTER to MIX, then SOUNDS. 3: CHARACTER, then SETUP.

1. **The left column** is the sample itself: SAMPLE (getting one), TRIM (which part plays), LOOP, CHOP and TUNE.
2. **The middle column** is KIT/INSTR at the top, then how it sounds: FILTER, ENV, LFO and MIX, then SOUNDS.
3. **The right column** is CHARACTER at the top and SETUP at the bottom.

In a kit, **the pages work on one key at a time**: the last one you played. Its name is in the top bar (`TRIM C3`), and its light stays dim so you can see which it is. Play another key and the pages show that one instead. In an instrument, the top bar says INST: there's only one sample to work on.

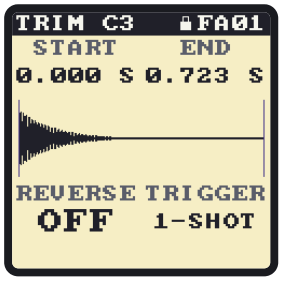


Figure 8.2 TRIM on the factory's BOOM KIT, key C3 (the top bar says which key): START and END above the waveform, REVERSE and TRIGGER below it.

The pages with a waveform (TRIM, LOOP, SAMPLE and CHOP) draw the sample across the middle of the screen, the part that plays dark, the rest pale, so you can see what you're doing.

8.3 The sounds you'll build

Your own slots start as INIT SOUND, and on Taste that's truly empty: a kit with nothing on any key. This chapter fills seven of them:

Slot	Sound	Sections	What it shows
UA01	MY KIT	8.4 and 8.5	putting samples on keys, and making them a kit
UA02	MY BREAK	8.6 and 8.7	chopping a drum loop, and fitting it to the tempo
UA03	MY KEYS	8.8	an instrument from one note
UA04	VOX PAD	8.8	a loop that holds a note for ever
UA05	SLOW CHOIR	8.9	STRETCH: pitch and length apart
UA06	DUSTY BREAK	8.10	CHARACTER: the old machines' grit
UA07	HELLO	8.11	a sample you record yourself

Every one starts from an empty slot, so there's no copying to do: just go to the slot. And there's nothing to save: Taste keeps everything as you go.

8.4 The starter set

Taste comes with 37 samples built in, ready to use on any key: drum hits (from Strike's kits, and a real acoustic kit, the ST… ones), a drum loop (BREAK), notes and chords from the other units (KEYS C4, PAD C4, BASS C3, STRINGS, BELLS C4, VOX C4, EXPANSE, SWEEP), and some real-world sounds (RAIN, CHOIR, and a spoken line, WORDS). Press ■ SAMPLE twice for the SAMPLES page, the list of every sample Taste has:



Figure 8.3 The SAMPLES page (SAMPLE pressed again): every sample, its length, and a star for the starter set's.

> **SAMPLE** (encoder 1) which sample: the list below shows it, with its length in seconds. A star (*) means one of the starter set; your own come first, without one.

> **NAME** (encoder 2) your own samples only: turn to change the letter, push for the next one.

> **USE** (hold encoder 3) puts the sample on the key being edited (or makes it the instrument): the whole of it, with every setting back to the start.

> **SAVE** (hold encoder 4) writes it onto the Root's card as a WAV file (section 8.12).

MY KIT, part 1: samples on keys

Go to ■ SOUNDS › ❶ BANK › USER A › ❷ SOUND › UA01 and name it MY KIT. It's a kit with nothing on it. Now put a drum on each of five keys, the same keys Strike uses, so a beat for one plays on the other:

1. Press the top key, C3, to choose it. Then ■ SAMPLE, ■ SAMPLE *again* for SAMPLES, ❶ SAMPLE until it shows ST KICK, and hold ❸ USE. The screen says ON THIS KEY NOW. Play C3: a real kick drum.
2. Press E3 (the third white key), find ST SNARE, hold USE.
3. A3 (the sixth white key): CLAP, Strike's handclap.
4. B3 (the seventh): ST CL HAT. And C4 (the eighth): ST OP HAT.

Play a beat: kick, hats, snare. Half a real drum kit and a drum machine's clap, on one unit. That's already more than most drum machines could do.

8.5 Trimming, tuning and mixing a key

Every key has its own settings, on the pages from TRIM to MIX, for the key being edited.

TRIM START, END, REVERSE and TRIGGER

START, **END** (encoders 1 and 2, in seconds) where the key starts and stops playing in its sample. Each click moves a thousandth of the waveform shown; push the encoder to zoom in on its marker (×8, ×64, back) for finer moves.

REVERSE (encoder 3, ON or OFF) plays it backwards, from END to START.

TRIGGER (encoder 4, 1-SHOT or GATE) 1-SHOT: one press plays it all, as a drum should. GATE: it plays while the key is held and stops when you let go.

TUNE COARSE, FINE, BEATS and ROOT

COARSE (encoder 1, -24 to +24) semitones up or down. On a sampler, up also means faster and shorter, down slower and longer, as with a tape or a record played at the wrong speed (unless TIME is STRETCH, section 8.7).

FINE (encoder 2, -50 to +50) cents, for tuning by ear.

BEATS (encoder 3, OFF or 1-64) section 8.7.

ROOT (encoder 4) instruments only: section 8.8.

MIX LEVEL, PAN, CHOKE and NORMAL

LEVEL (encoder 1, 0-10) how loud this key is.

PAN (encoder 2, L50 to R50) where it sits, left to right.

CHOKE (encoder 3, OFF or GROUP 1-4) kits only: keys in the same group cut each other off, as a closed hi-hat stops an open one.

NORMAL (hold encoder 4) shows how loud the sample is at its loudest; hold to make it as loud as it can go without distorting. Your own recordings only: the starter set's are full already.

MY KIT, part 2: making it a kit

1. The hats choke each other: play B3, then ■ MIX › ❸ CHOKE › **GROUP 1**; play C4, the same. Now play the open hat and the closed one straight after: the open one is cut short.
2. Spread them, as a drummer's kit is spread: on B3, ❷ PAN › **L15**; on C4, ❷ PAN › **L25**.
3. A deeper kick: play C3, ■ TUNE › ❶ COARSE › **−3**. Lower and a little longer, because it plays slower.
4. The clap under the snare: play A3, ■ MIX › ❶ LEVEL › **7.0**.

MY KIT

An empty kit, with:

C3	ST KICK	COARSE −3	
E3	ST SNARE		
A3	CLAP	LEVEL 7.0	
B3	ST CL HAT	CHOKE GROUP 1	PAN L15
C4	ST OP HAT	CHOKE GROUP 1	PAN L25

Then the trick drum machines can't do: play E3, ■ TRIM › ❸ REVERSE › **ON**, and play it. A backwards snare, swelling up into nothing. Put it back OFF, or keep it on a key of its own (CHOP's COPY, section 8.6, puts a copy on another key).

8.6 Chopping

Here's the trick hip hop was built on. Take a drum loop, cut it into pieces, one hit each, and put each piece on a key. Now you can play the drummer's hits in any order you like.

■ CHOP SLICES, MODE, SENS and CHOP



Figure 8.4 CHOP on the whole break (BREAK SYNC's top key, C3): sixteen cuts shown dashed before you hold CHOP.

> **SLICES** (encoder 1, 2-27) how many pieces (EVENLY only).

> **MODE** (encoder 2) EVENLY: equal pieces, START to END, which suits a loop with a steady beat (16 for a bar of sixteenths). AT HITS: a new piece wherever the sound jumps up, a drum hit or a word. COPY: not a chop at all, but the key's sample and settings copied onto another key (encoder 1 becomes TO).

> **SENS** (encoder 3, 0-10) AT HITS only: higher finds quieter hits.

> **CHOP** (hold encoder 4) does it: the pieces go on the key being edited and the keys above it, one each. The waveform shows the cuts first, dashed, so you can see before you hold.

MY BREAK, part 1: a loop, kept whole

Go to UA02 and name it `MY BREAK`. First, the whole loop on the top key, to keep:

1. Press the bottom key, D5 (the last white key). ■ SAMPLE, ■ SAMPLE *again*, find BREAK and hold ❸ USE. Play it: one bar of drums, from Strike.
2. Make it a loop that plays while you hold it: ■ TRIM › ❹ TRIGGER › `GATE` and ■ LOOP › ❶ LOOP › `ON`. Hold D5: round and round.
3. Copy it to C3, to chop: ■ CHOP › ❷ MODE › `COPY`, ❶ TO › `C3`, and hold ❹ COPY. The screen says `COPIED TO C3`, and C3 is now the key being edited.

MY BREAK, part 2: chopped

1. On C3: ■ CHOP › ❷ MODE › **EVENLY** and ❶ SLICES › **16**. The dashed lines show sixteen cuts.
2. Hold ❹ CHOP. **16 SLICES**: the bar is now on the sixteen keys from C3 up, one sixteenth each, every key a 1-SHOT.
3. Play the keys from C3 up, one by one, in time: the break, as it was. Now play them in another order. Kick, kick, snare. Hats on their own. A snare twice. That's a new drum part, played by the same drummer.
4. Make one slice different: play the slice with the loudest snare, then ■ TRIM › ❸ REVERSE › **ON**; or ■ TUNE › ❶ COARSE › **−5** on a kick slice for a deeper one.

MODE AT HITS works the same way, but cuts wherever the sound jumps up instead of evenly, with ❸ SENS deciding how small a jump counts. It suits a loose loop, or a spoken line: the factory's WORDS CHOP is made that way, and section 8.11 uses it on your own voice. (A chop puts its pieces on the key being edited and the keys above it, so start low on the keyboard: on D5, the last key, there'd be room for only one.)

8.7 In time with the music

A loop was recorded at one tempo, and your song is probably at another. Taste has two ways to fit it, and they sound completely different.

> **BEATS** (TUNE, encoder 3, OFF or 1-64) how many beats this key's sample should last. Taste plays it faster or slower so that it does, at the tempo: Step's, when Step is on the chain, otherwise 120 BPM (the screen says which).

■ TUNE TIME (TUNE pressed again) TIME, SPEED and GRAIN



Figure 8.5 TIME (TUNE pressed again) on KEYS DRONE: STRETCH at 25%, so each note lasts four times as long at its own pitch.

TIME (encoder 1, CLASSIC or STRETCH) CLASSIC: speed and pitch go together, as on the old machines: faster is higher. STRETCH: pitch and speed come apart, so a note keeps its length whatever its pitch, and a loop keeps its pitch whatever its speed.

SPEED (encoder 2, STRETCH only, 25% to 400%) how fast it moves through the sample, without changing the pitch.

GRAIN (encoder 3, STRETCH only, 10-100 MS) how STRETCH does it: it plays the sample as many short overlapping pieces, grains, and this is their length. Longer suits smooth sounds, shorter suits drums.

MY BREAK, part 3: to the tempo

1 Play D5, the whole loop. ■ TUNE › 3 BEATS › `4` : one bar of four beats. Hold D5. It's faster now, a bar at 120 BPM, and higher, like a record played too fast: the classic sampler sound, and in jungle it's the whole point.

2 Now ■ TUNE *again* for TIME, and 1 TIME › `STRETCH` . Hold D5: the same tempo, but back at its own pitch. 3 GRAIN › `30 MS` , short, suits drums.

MY BREAK

An empty kit, with:

D5	`BREAK` `TRIGGER GATE` `LOOP ON` `BEATS 4` `TIME STRETCH` `GRAIN 30 MS`
C3 up	`BREAK, CHOP EVENLY 16`

Play the factory's TWO SPEEDS now: the same loop at four speeds CLASSIC on the first four white keys, and at the same four speeds STRETCH on the next four. Slower CLASSIC is lower and darker; slower STRETCH keeps its pitch and gets a little grainy. Neither is better. They're two different instruments.

8.8 Instruments

Now the other half: one sample, played across all the keys. Press ■ KIT/INSTR on one of your own sounds and it switches to its instrument (the light goes out, and the screen says `INSTRUMENT`); press it again for the kit. The two are kept separately, so switching never loses either.

ROOT (TUNE, encoder 4) the note at which the sample plays as it was recorded. If you sampled a C4, ROOT is C4, and every other key plays it that far higher or lower, in tune. The starter set's notes say theirs in their names: KEYS C4, BASS C3.

■ FILTER CUTOFF, RES and ENV ■ ENV ATTACK, DECAY, SUSTAIN and RELEASE ■ LFO RATE, PITCH and FILTER

CUTOFF, **RES** (FILTER, 0-10) a low-pass filter on the sample, as chapter 2 described; 10 is wide open. **ENV** (-50 to +50) how far the envelope moves it on each note.

ATTACK, **DECAY**, **SUSTAIN**, **RELEASE** (ENV, 0-10) the shape of each note, on top of the sample's own shape.

RATE, **PITCH**, **FILTER** (LFO, 0-10) one LFO for the whole sound: how fast, and how much it moves the pitch (vibrato) and the filter.

Build MY KEYS

Go to UA03, name it `MY KEYS`, and press ■ KIT/INSTR: `INSTRUMENT`. Then ■ SAMPLE, ■ SAMPLE *again*, find KEYS C4, and hold ❸ USE. ROOT is already C4.

Play a tune and a chord: one recorded note from Glass, playing all 27 keys. Listen to the very high and very low keys: the higher you go, the shorter and brighter (and more chipmunk-like) it gets; the lower, the longer and duller. That's how a sampler plays one note across a keyboard.

1. It stops dead when you let go, because an instrument starts on GATE: ■ ENV › ❹ RELEASE › `3.5`, and it fades instead.
2. Softer and warmer: ■ FILTER › ❶ CUTOFF › `7.0`, and ❸ ENV › `+20` so each note starts a little brighter.
3. Three notes at once are loud: ■ MIX › ❶ LEVEL › `8.0`.

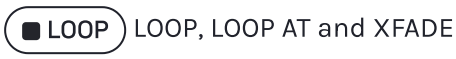
MY KEYS

An empty instrument, with:

SAMPLE	`KEYS C4`	`ROOT C4`
FILTER	`CUTOFF 7.0`	`ENV +20`
ENV	`RELEASE 3.5`	
MIX	`LEVEL 8.0`	

Loops: a note that lasts for ever

A sample is only as long as it is. KEYS C4 dies away in under two seconds, however long you hold the key. To hold a note for ever, a sampler **loops** part of it: it plays to the end, jumps back, and plays that part again, over and over, for as long as the key is down.



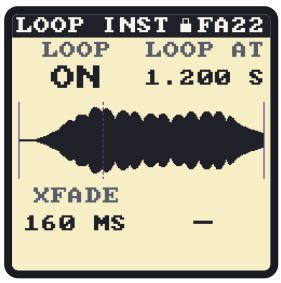


Figure 8.6 LOOP on VOX: LOOP AT drawn as a dashed line, and the crossfade that smooths the join.

> **LOOP** (encoder 1, ON or OFF) whether it loops.

> **LOOP AT** (encoder 2, in seconds) where it jumps back to, drawn as a dashed line on the waveform; it loops from there to END. Push to zoom in on it.

> **XFADE** (encoder 3, 0-1000 MS) a crossfade: over its last stretch, the loop fades into the sound just before LOOP AT, so the jump back doesn't click. Most loops need some.

Build VOX PAD

Go to UA04, name it **VOX PAD**, press ■ KIT/INSTR for the instrument, and USE VOX C4 (Glass's voice-like sound). ROOT stays C4.

1. Hold a chord: it fades out after three and a half seconds, as the sample does.
2. ■ LOOP › ❶ LOOP › `ON` and ❷ LOOP AT to about 1.200 S: the steady middle of the sample. Hold a chord: it goes on for ever. Listen for a tiny bump each time round, at the join.
3. ❸ XFADE › `160 MS`. The bump smooths away.
4. Make it a pad: ■ ENV › ❶ ATTACK › `5.0` and ❹ RELEASE › `6.0`; a slow, gentle vibrato with ■ LFO › ❶ RATE › `5.2` and ❷ PITCH › `1.2`; and ■ MIX › ❶ LEVEL › `8.0`.

VOX PAD

An empty instrument, with:

SAMPLE	`VOX C4`	`ROOT C4`	
LOOP	`LOOP ON`	`LOOP AT ABOUT 1.200 S`	`XFADE 160 MS`
ENV	`ATTACK 5.0`	`RELEASE 6.0`	
LFO	`RATE 5.2`	`PITCH 1.2`	
MIX	`LEVEL 8.0`		

LOOP AT moves a thousandth of the sample a click, so 1.200 S may show as 1.197 or 1.204: near enough. Try moving it while you hold a chord: some places loop smoothly, some wobble. Finding a good loop point was a craft on the old samplers, and still is.

8.9 STRETCH on an instrument

MY KEYS showed the sampler's oldest habit: notes up high come out short and squeaky, notes down low long and groaning, because the sample plays faster or slower. For drums and keys that's often lovely. For a voice it's a problem: a choir singing a low note shouldn't sound like a recording played too slowly. STRETCH fixes it: every note plays at its own pitch, at the sample's own speed.

Build SLOW CHOIR

Go to UA05, name it **SLOW CHOIR**, press ■ KIT/INSTR, and USE CHOIR: a real choir singing "aah".

1. The choir sings a B4, slightly sharp. Tell Taste: ■ TUNE › ❹ ROOT › **B4** and ❷ FINE › **−21**. Now B4 on the keys plays it as recorded, and every other key is in tune with Taste's other sounds.

2. Play a low chord, around G3. Deep and slow, a choir of giants. Now ■ TUNE *again* for TIME, ❶ TIME › **STRETCH**, and play the same chord: a real choir's voices again, singing lower.

3. Slower than they sang it: ❷ SPEED › **50%**. The "aah" now lasts twice as long, at the same pitch. ❸ GRAIN › **60 MS**, long, suits voices.

4. ■ ENV › ❹ RELEASE › **6.5** and ■ MIX › ❶ LEVEL › **8.0**.

SLOW CHOIR

An empty instrument, with:

SAMPLE	CHOIR
TUNE	FINE −21 · ROOT B4
TIME	TIME STRETCH · SPEED 50% · GRAIN 60 MS
ENV	RELEASE 6.5
MIX	LEVEL 8.0

Switch TIME back to CLASSIC for a moment and play the same chord: you'll hear exactly what STRETCH is doing. Then try SPEED at 25%, as slow as it goes: a held chord that takes ten seconds to sing its "aah".

Grains

STRETCH works by playing the sample as a stream of short overlapping pieces, grains, each one read at the note's pitch, while the place they're read from moves at SPEED. Done carefully, as Taste does, you hear the sound, not the grains. Make the grains the point instead, scatter them, freeze them, pile hundreds up at once, and you've got a different instrument altogether: that's Grit, chapter 9.

8.10 CHARACTER

The samplers of the late 1980s had a sound of their own. They stored their samples with less detail than a CD (12 bits instead of 16) and played them back at a lower rate, without smoothing between one sample and the next. That made drums crunchy and gritty, and bass heavier, and producers loved it: it's a big part of the sound of late-1980s and 1990s hip hop. CHARACTER is that, for the whole sound.



Figure 8.7 CHARACTER: CLEAN, 12-BIT and 12-BIT 26K, the chosen one marked, and what it's like underneath.

CLEAN as recorded, read smoothly: the sample as it is.

12-BIT 12 bits at 40 kHz: gritty, a little bite.

12-BIT 26K 12 bits at 26 kHz: grittier and darker, the crunchiest.

Build DUSTY BREAK

Copy MY BREAK into UA06: ■ SOUNDS › ❷ SOUND › UA02 › ❸ COPY TO › UA06 › *hold* ❸, and name it DUSTY BREAK. Then press ■ CHARACTER and turn encoder 1 to 12-BIT 26K. Play the slices: the same drummer, on an old record.

Give the whole loop on D5 a darker filter, as a worn record would: play D5, then ■ FILTER › ❶ CUTOFF › 6.0. Now play MY BREAK and DUSTY BREAK one after the other. Two different decades from one sample.

8.11 Recording your own

The starter set is a start. The real fun of a sampler is your own sounds: your voice, a guitar, a chord from another unit, the kitchen. Taste records from the Root's inputs (its built-in microphone, or a microphone or line-level sound plugged in), or from any unit on the chain.

■ SAMPLE SOURCE, THRESH, LENGTH and REC

> **SOURCE** (encoder 1) what to record: the Root's LINE IN, MIC IN or BUILT-IN MIC (its own microphone); ALL, everything on the chain mixed together; any other unit by name (GLASS 2…) to sample just that one; or CARD, a WAV file from the Root's card (section 8.12). A meter under the waveform shows how loud the source is.

> **THRESH** (encoder 2, OFF or -60 to -6 DB) OFF: recording starts the moment you push REC. Otherwise it waits (**WAITING**) until the sound gets louder than this, so nothing's wasted on silence before you start; a marker on the meter shows where.

> **LENGTH** (encoder 3, 1-60 S) the longest it will record. Push the encoder on a Root input to see **GAIN** instead, how loud the input comes in (the same as SETUP's ROOT GAIN).

> **REC** (push encoder 4) start (or wait, with THRESH); push again to stop early.

A recording goes on the key being edited (or becomes the instrument), and replaces what was there. Your recordings are kept in Taste's own memory, about 83 seconds of them, and the SAMPLE page shows how much is **FREE**.

Build HELLO

Your own voice, on every key. Go to UA07, name it **HELLO**, and press ■ KIT/INSTR for the instrument.

1. ■ SAMPLE › ❶ SOURCE › **BUILT-IN MIC** : the Root's own microphone, the little hole in its top. Talk, and watch the meter move.
2. ❷ THRESH › **-30 DB**, so it waits for you, and ❸ LENGTH › **3 S**.
3. Push ❹ REC. It says WAITING. Lean towards the Root and say "hello" clearly. It starts as you speak, and stops after three seconds.
4. Trim off the silence after it: ■ TRIM › ❷ END back until the dark part ends just after your word. Then make it as loud as it can be: ■ MIX › ■ *hold* 4 NORMAL.
5. Your voice plays at its own pitch on C4 (ROOT). Play it up and down the keys: from a giant to a mouse.

Then try it with ■ TUNE *again* and TIME STRETCH: "hello" at every pitch, always the same length, a choir of you. Or press KIT/INSTR, record a few words onto keys of the kit, and CHOP AT HITS one of them into syllables.

Sampling another unit

With another unit on the chain, SOURCE can be that unit by name: play Glass or Volt and Taste records exactly what it plays, nothing else. This is how most of the starter set was made. Sample a chord from Ember and play it as an instrument, and every key plays the whole chord, the trick behind house music's chord stabs.

8.12 The card

The Root's memory card is where samples come in from a computer and go out to one. Taste uses a folder on it called TASTE.

Loading ■ SAMPLE › ❶ SOURCE › **CARD** : ❷ FILE chooses a WAV file from the folder, and pushing ❹ LOAD puts it on the key being edited. Any WAV will do: mono or stereo, any rate; Taste turns it into its own kind.

Saving On the SAMPLES page, hold ❹ SAVE: the sample shown is written to the folder as a WAV, named after it, to keep or to use on a computer.

8.13 Effects

Taste has no effects of its own, beyond its filter and CHARACTER. With Haze joined on, Taste's sounds get echo and reverb like any synth's, and Haze's DRIVE and DJ FILTER work beautifully on chopped loops.

8.14 Taste's SETUP

> **BEND** (THIS UNIT, encoder 2, 0-12) how far a MIDI keyboard's pitch bend wheel bends an instrument's notes.

> **WHEEL** (encoder 3, 0-10) how much vibrato the mod wheel adds.

> **VELO** (encoder 4, OFF or ON) ON: harder is louder, from a MIDI keyboard or Step.

From MIDI, a kit plays from C2 (note 36) upwards: C2 plays the top key, C#2 the next, and so on. An instrument plays the notes as they come.

> ### What you have at the end of this chapter
>
> Seven sounds of your own: MY KIT, MY BREAK, MY KEYS, VOX PAD, SLOW CHOIR, DUSTY BREAK and HELLO. And the sampler's whole bag of tricks: a sound on a key, chopped, reversed, tuned, looped, stretched and roughed up. Now go and record something nobody else has.

Chapter 9 - Grit

A sound broken into thousands of tiny pieces, and played back as a cloud.

Take a sound, any sound, and cut out a tiny piece of it: a twentieth of a second, say, faded in and out so it doesn't click. That's a **grain**. On its own it's a little blip. Now play another grain, from a slightly different place, and another, and another, dozens a second, overlapping, some higher, some lower, some backwards, scattered left and right. You hear a **cloud**: the colour of the sound you started with, but frozen, smeared, shimmering or broken into glitter. That's **granular synthesis**, and Grit is built for it.

Grit's grains come from one of two places:

- **A SAMPLE**: one of the samples built in (the same starter set as Taste's), one you've captured, or one from the card. Every factory sound from FA01 to FA27 is one of these, so they all play on Grit on its own.
- **LIVE**: whatever is happening right now, from another unit on the chain, or from the Root's microphone, kept in a rolling recording a few seconds long. Grit plays grains from the last moments, and FREEZE stops time.

Granular synthesis: where it came from, and who uses it

The idea came from physics. In 1946 the physicist Dennis Gabor suggested that any sound could be thought of as a stream of tiny "acoustic quanta", grains, each too short to hear as a note on its own. In 1958 the composer Iannis Xenakis made music from it the hard way, cutting magnetic tape into tiny pieces and splicing them back together by hand. Computers made it practical: Curtis Roads wrote granular programs in the 1970s, and in 1986 Barry Truax ran the first granular synthesis in real time, so it could be played rather than calculated overnight.

For a long time it lived in universities and electronic music studios. Then computers got fast enough to do it on a laptop, and now it's in music software and synthesisers of every size: the frozen pads under film scores, the shimmering textures of ambient music, vocal smears and glitches in electronic music and pop, and stretched-out sounds that last far longer than the recording they came from. Granular synthesis is a method rather than a machine, so there's no one famous instrument to compare Grit with. Learn it here and you'll recognise it everywhere.

9.1 Find a sound you like

Grit has one factory bank, FAC A, with 32 sounds. The first 27 play from samples, so you can hear them all now. The last five, whose names start LIVE, work on whatever's playing: another unit on the chain, or the Root's microphone (section 9.8).

> **Try this**
>
> ■ SOUNDS › ❷ SOUND and go through FA01 to FA27. Hold notes and chords for a while: a lot of them change as they go. Some play on their own as soon as you reach them, without a key (SLOW TIME, RAIN CLOUD, DEEP DRONE, TOM DRONE, CLOCKWORK…): section 9.6 says why. On STUTTER, SCATTER and BROKEN KIT, play the white keys and then the black ones.

9.2 How Grit is laid out

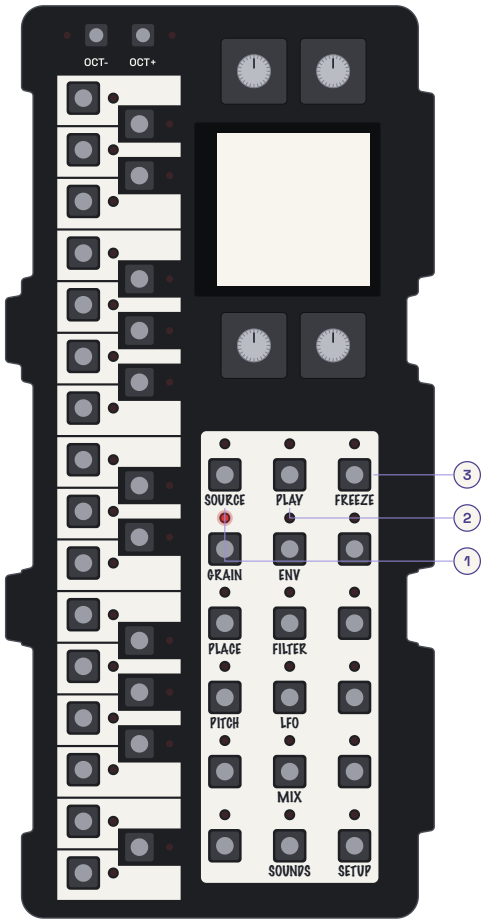


Figure 9.1 Grit's buttons. 1: where the grains come from and what they are, SOURCE to PITCH. 2: how they play, PLAY to MIX, then SOUNDS. 3: FREEZE (lit while frozen), then SETUP.

1 **The left column** is where the grains come from and what they are: SOURCE, GRAIN, PLACE and PITCH.

2 **The middle column** is how they play: PLAY, ENV, FILTER, LFO and MIX, then SOUNDS.

3 **The right column** is FREEZE at the top (lit while frozen) and SETUP at the bottom.

The pages in the left column draw the sound the grains come from across the middle of the screen, and every grain on it as a little fade shape, where it's reading and how loud: the cloud you're hearing, as a picture. Turn SIZE or SPRAY and you'll see the shapes change as you hear it.



Figure 9.2 GRAIN on CATHEDRAL, a note held: the sound across the middle, every grain drawn where it's reading as a little fade shape, and how many are playing.

9.3 The sounds you'll build

Your own slots start as INIT SOUND: Taste's PAD sample, played from the keys, every setting plain, so it makes a sound as soon as you play. This chapter builds seven:

Slot	Sound	Section	What it shows
UA01	MIST	9.4	the grain itself: size, how many, their shape
UA02	SLOW BREAK	9.5	moving through a sound slowly, and lining the grains up
UA03	SHIMMER BELLS	9.5 and 9.6	scattering grains, and pitching them up
UA04	BREAK PLACES	9.7	places on the white keys, stutters on the black
UA05	HOVER	9.7	a sound that plays itself
UA06	MIST MOVING	9.8	the filter and LFO on MIST
UA07	MY FREEZE	9.9	LIVE, FREEZE and CAPTURE, from the Root's microphone

Each one starts from an empty slot of USER A, so just go to the slot. To choose a sample for it, ■ SOURCE : with FROM on SAMPLE, ❷ SAMPLE turns through every sample Grit has.

9.4 The grain

GRAIN SIZE, DENSITY, SHAPE and ALIGN

SIZE (encoder 1, 5-500 MS) how long each grain is. Short grains, under about 20 MS, are too short to hear the sound's pitch in them: buzzy and gritty. Long ones, over about 100 MS, are smooth and you hear the sound itself.

DENSITY (encoder 2, 1-200 a second) how many grains start each second. Few, and you hear them as separate blips; many, and they melt into a cloud.

SHAPE (encoder 3, 0-10) how each grain fades in and out: 0 a smooth hump, all fade; 10 sharp, nearly square, so every grain clicks in. The screen draws it.

ALIGN (encoder 4, ON or OFF) ON: each new grain is lined up with the one playing, so their waves join smoothly (the trick Taste's STRETCH uses). OFF: raw, with a little roughness where they overlap.

Try this first

On INIT SOUND, in UA01, hold one note and turn SIZE right down to 5 MS: a buzz, the pad gone. Bring it up slowly and the pad comes back, first gritty, then smooth. Now SIZE at about 50 MS and DENSITY right down to 1/S: one grain a second, a little "blip" of pad each time. Turn DENSITY up and the blips crowd together, then join into a sound. Last, SHAPE at 10.0 with DENSITY around 10/S: every grain starts with a tick.

That's everything granular synthesis is made of: how long, how many, and what shape.

Build MIST

Still in UA01, name it **MIST**. A soft, wide cloud of the pad:

1. Long, smooth grains, plenty of them: GRAIN › 1 SIZE › 315 MS, 2 DENSITY › 41/S, 3 SHAPE › 0.0.
2. A little scatter, so they don't all read the same place: PLACE › 3 SPRAY › 3.0 (section 9.5).
3. Spread wide, left and right: MIX › 2 SPREAD › 10.0, and 1 LEVEL › 6.0, as a cloud this thick is loud.
4. Swelling in and fading out: ENV › 1 ATTACK › 5.0, 4 RELEASE › 7.0.

MIST

INIT SOUND (the PAD sample), with:

GRAIN	SIZE 315 MS	DENSITY 41/S	SHAPE 0.0
PLACE	SPRAY 3.0		
ENV	ATTACK 5.0	RELEASE 7.0	
MIX	LEVEL 6.0	SPREAD 10.0	

Hold a chord. The pad you know, but hovering: it doesn't move on through the sample, it stays where it is, breathing. That's section 9.5's subject.

9.5 Where the grains come from

■ PLACE WHERE, SCAN, SPRAY and REVERSE

WHERE (encoder 1, 0-100%) the place in the sound the grains read from: 0% the start, 100% the end. (LIVE, 100% is now, and lower is further back.) The screen marks it on the waveform.

SCAN (encoder 2, -200% to +200%) how fast WHERE moves on while a note plays. 0% stands still, on one moment for ever; +100% moves at the speed the sound was recorded, so you hear it play through; +10% takes ten times as long; negative goes backwards.

SPRAY (encoder 3, 0-10) how scattered the grains are around WHERE: 0, every grain from the same place; 10, from all over.

REVERSE (encoder 4, 0-100%) the chance a grain plays backwards: 50%, half of them.

Build SLOW BREAK

A drum loop slowed right down, without slowing its pitch. Go to UA02 and name it **SLOW BREAK**.

1 ■ SOURCE › ❷ SAMPLE › **BREAK** : Taste's drum loop.

2 Start at the beginning and move through it at a tenth of its speed:
■ PLACE › ❶ WHERE › **0%** , ❷ SCAN › **+10%** , ❸ SPRAY › **0.0** .

3 Grains short enough for drums, lined up so they join cleanly: ■ GRAIN › ❶ SIZE › **50 MS** , ❷ DENSITY › **41/S** , ❸ SHAPE › **2.0** , ❹ ALIGN › **ON** .

SLOW BREAK

INIT SOUND, with:

SOURCE	SAMPLE BREAK
PLACE	WHERE 0% · SCAN +10% · SPRAY 0.0
GRAIN	SIZE 50 MS · DENSITY 41/S · SHAPE 2.0 · ALIGN ON

Hold C4. The break plays at a tenth of its speed: each hit of the drums stretched into a long, ringing smear, but at the pitch it was. Hold it and watch WHERE creep along the waveform. Then turn ALIGN OFF and listen to the roughness come in where the grains overlap; and back ON.

Now play with SCAN while you hold the note: +100% and it's the break at its own speed, played by grains; -50% and it goes backwards at half speed; 0% and it hangs on one moment, a drum hit frozen into a tone.

SHIMMER BELLS, part 1: scattered

Go to UA03 and name it **SHIMMER BELLS**. ■ SOURCE › ❷ SAMPLE › **BELLS C4** : Glass's bell.

1 Read from near the strike, where the bell is loudest, and scatter the grains around it: ■ PLACE › ❶ WHERE › **5%** , ❸ SPRAY › **4.0** .

2 Medium grains, plenty of them: ■ GRAIN › ❶ SIZE › **199 MS** , ❷ DENSITY › **53/S** .

Hold a note: not a bell ringing out, but the bell's sound hovering, glittering, as if many bells were being struck very softly. The pitch comes next.

9.6 The pitch of the grains

■ PITCH PITCH, FINE, RANDOM and JUMPS

PITCH (encoder 1, -24 to +24) semitones, for every grain. Grit's pitch and speed are apart, as Taste's STRETCH: an octave up doesn't play any faster.

FINE (encoder 2, -50 to +50 CT) fine tuning.

RANDOM (encoder 3, OFF up to 12 ST) each grain detuned at random, up to this much either way: a little, a chorus; a lot, a swarm.

JUMPS (encoder 4, OFF, OCTAVE or OCT+5TH) some of the grains an octave up (or an octave and a fifth): a shimmer above the sound.

SHIMMER BELLS, part 2: the shimmer

On SHIMMER BELLS: ■ PITCH › 4 JUMPS › `OCT+5TH` and 3 RANDOM › `48 CT`. Then ■ MIX › 2 SPREAD › `10.0`, 1 LEVEL › `10.0`, and ■ ENV › 4 RELEASE › `7.0`.

SHIMMER BELLS

INIT SOUND, with:

SOURCE	`SAMPLE BELLS C4`
PLACE	`WHERE 5%` `SPRAY 4.0`
GRAIN	`SIZE 199 MS` `DENSITY 53/S`
PITCH	`RANDOM 48 CT` `JUMPS OCT+5TH`
ENV	`RELEASE 7.0`
MIX	`LEVEL 10.0` `SPREAD 10.0`

Play slow chords. The bells now sparkle an octave and a fifth above themselves, each grain a fraction out of tune with the next. Push RANDOM up to 3.0 ST and the shimmer becomes a swarm; take JUMPS off and it settles back down.

9.7 Three ways to play

■ PLAY PLAY, SYNC, GLIDE and ROOT

PLAY (encoder 1) **NOTES**: the keys play the cloud at their pitch, chords and all, as an instrument. **PLACES**: the 16 white keys are 16 places along the sound, from the start (the top key) to the end, and the black keys **stutter**: they repeat what was played last, at note lengths in time with the tempo. **ALWAYS**: the cloud plays all the time, without a key; a key retunes it while you hold it, and lets go back to the sound's own pitch.

SYNC (encoder 2, OFF, 1/4 … 1/32) grains on the beat: a new grain every quarter note, eighth, triplet, sixteenth and so on, at Step's tempo (120 without it), instead of DENSITY's steady stream. The stutters always keep time.

GLIDE (encoder 3, OFF or a time) how long the pitch slides from one key to the next.

ROOT (encoder 4, C2 to C6) the key that plays the sound at its own pitch: C4 to start. The screen says so at the bottom (`C4 PLAYS IT AS IT WAS`), and while you hold another key, by how much it's moved.

In PLACES, the black keys stutter at these lengths, lowest key first: 1/4, 1/8, 1/8 triplet, 1/16, 1/16 triplet, 1/32; then 1/4, 1/8, 1/16 and 1/32 backwards; and last, 1/64. A stutter repeats whatever was played last, so play a place, then hold a black key to make it stutter, then let go.

Build BREAK PLACES

Go to UA04 and name it **BREAK PLACES**. ■ SOURCE › ❷ SAMPLE › **BREAK** .

1 ■ PLAY › ❶ PLAY › **PLACES** . Each place plays on from where it starts, so: ■ PLACE › ❷ SCAN › **+100%** , at the break's own speed, and ❸ SPRAY › **0.0** .

2 Short, sharp grains to keep the drum hits crisp: ■ GRAIN › ❶ SIZE › **40 MS** , ❸ SHAPE › **8.5** .

3 Each place starts and stops exactly with your finger: ■ ENV › ❶ ATTACK › **0.0** , ❹ RELEASE › **1.0** .

BREAK PLACES

INIT SOUND, with:

SOURCE	SAMPLE BREAK	
PLAY	PLAY PLACES	
PLACE	SCAN +100%	SPRAY 0.0
GRAIN	SIZE 40 MS	SHAPE 8.5
ENV	ATTACK 0.0	RELEASE 1.0

Play the white keys, top to bottom, one at a time: the break, from sixteen places along it. Tap them in a new order for a new beat, like Taste's chopped break, but with no chopping. Now play a place and hold a black key under it: the hit stutters, a quarter note at a time, then faster and faster as you go down the black keys. A drum roll from one hit.

Build HOVER

A sound that plays itself. Go to UA05 and name it **HOVER**. ■ SOURCE › ❷ SAMPLE › **CHOIR**.

1 ■ PLAY › ❶ PLAY › **ALWAYS**. It starts at once, with no key.

2 Hold one moment of the choir: ■ PLACE › ❶ WHERE › **40%**, ❷ SCAN › **0%**, ❸ SPRAY › **1.5**.

3 Long, smooth, lined-up grains, and an octave above: ■ GRAIN › ❶ SIZE › **251 MS**, ❷ DENSITY › **41/S**, ❸ SHAPE › **0.0**, ❹ ALIGN › **ON**; ■ PITCH › ❹ JUMPS › **OCTAVE**; ■ MIX › ❷ SPREAD › **10.0**.

4 The choir sings a B4, so tell Grit: ■ PLAY › ❹ ROOT › **B4**.

HOVER

INIT SOUND, with:

SOURCE	SAMPLE CHOIR
PLAY	PLAY ALWAYS · ROOT B4
PLACE	WHERE 40% · SCAN 0% · SPRAY 1.5
GRAIN	SIZE 251 MS · DENSITY 41/S · SHAPE 0.0 · ALIGN ON
PITCH	JUMPS OCTAVE
MIX	SPREAD 10.0

A choir's "aah", held for ever, with an octave floating over it. Now hold a key: the whole cloud moves to that note, and the screen says by how much (G4, say: **G4: 3RD DOWN**). Let go, and it goes back to B4. Hold a key and turn WHERE: a different moment of the choir, a different vowel.

9.8 Shaping the cloud

■ ENV ATTACK, DECAY, SUSTAIN and RELEASE ■ FILTER CUTOFF, RES and TYPE ■ LFO RATE, WHERE, SIZE and FILTER ■ MIX LEVEL, SPREAD, DRY and FEEDBK

ATTACK, **DECAY**, **SUSTAIN**, **RELEASE** (ENV, 0-10) the shape of each note's cloud: how it swells in and fades away.

CUTOFF, **RES** (FILTER, 0-10) and **TYPE** (LOW, HIGH or BAND): LOW keeps the bottom, HIGH the top, BAND a slice in the middle.

RATE (LFO, in Hz) one LFO, and how much it moves **WHERE** (the place, so the cloud wanders through the sound), **SIZE** (the grains grow and shrink) and **FILTER**.

LEVEL and **SPREAD** (MIX, 0-10) how loud, and how widely the grains are scattered left and right. **DRY** and **FEEDBK** are for LIVE: section 9.9.

Build MIST MOVING

Copy MIST into UA06: ■ SOUNDS › ❷ SOUND › UA01 › ❸ COPY TO › UA06 › *hold* ❸. Then make the cloud wander:

1. A band of the pad, slightly ringing: ■ FILTER › ❸ TYPE › BAND, ❶ CUTOFF › 6.0, ❷ RES › 4.0.
2. The LFO, slowly: ■ LFO › ❶ RATE › 0.09 HZ, once every eleven seconds. Then ❷ WHERE › 3.0, so the cloud drifts back and forth through the sample, and ❹ FILTER › 5.0, so the band sweeps up and down.

Name it MIST MOVING, hold a chord and wait. It never sounds the same twice. Watch WHERE's marker drift on the PLACE page while it plays.

9.9 LIVE, FREEZE and CAPTURE

Now the other half of Grit. Set FROM to anything other than SAMPLE, and Grit listens instead: it keeps the last few seconds of what it hears in a rolling recording, the **buffer**, and plays its grains from that.

■ SOURCE FROM, BUFFER, GAIN and CAPTURE



Figure 9.3 SOURCE on CATHEDRAL: FROM, and the sample it plays. On a LIVE sound the second box is BUFFER, and a level meter shows what Grit hears.

FROM (encoder 1) what to listen to: the Root's LINE IN, MIC IN or BUILT-IN MIC (its own microphone); ALL, every other unit on the chain; one unit by name (GLASS 2…); SAMPLE; or CARD, a WAV file from the Root's card (the GRIT folder).

BUFFER (encoder 2, 1-20 S) how much of the past the grains can reach. With WHERE at 100%, they play what's happening now; at 50%, from halfway back.

GAIN (encoder 3, the Root's inputs only) how loud the input comes in.

> **CAPTURE** (hold encoder 4) keeps what's in the buffer now as a sample (CAP 1, CAP 2…), to play later as a SAMPLE, here or on Taste via the card.

And then the button: **FREEZE**, top right. Press it and the buffer stops recording: whatever was just heard is held, and the grains play from it for as long as you like. Press it again and Grit listens once more. When you freeze, the frozen moment's ROOT becomes the lowest key that was held at that instant, on any unit on the chain, so the keys then play the frozen chord in tune.

Build MY FREEZE

This one uses the Root's own microphone, so you need nothing else; wear headphones, or Grit will hear itself through the speaker. Go to UA07 and name it `MY FREEZE`.

1. ■ SOURCE › ❶ FROM › `BUILT-IN MIC`, and ❷ BUFFER › `4 S`. Talk or hum, and watch the level meter and the scrolling waveform.
2. Read from just behind now, standing still once frozen: ■ PLACE › ❶ WHERE › `95%`, ❷ SCAN › `0%`, ❸ SPRAY › `1.5`.
3. Smooth, long grains: ■ GRAIN › ❶ SIZE › `158 MS`, ❷ DENSITY › `31/S`, ❸ SHAPE › `0.0`. And a slow swell: ■ ENV › ❶ ATTACK › `5.5`, ❹ RELEASE › `6.5`.
4. Now sing one long note, "aaah", and while you're still singing, press ■ FREEZE. Stop singing. Play a key: your voice, frozen, held as long as you like. Play chords: a choir of you.

MY FREEZE

INIT SOUND, with:

SOURCE	`FROM BUILT-IN MIC`	`BUFFER 4 S`	
PLACE	`WHERE 95%`	`SCAN 0%`	`SPRAY 1.5`
GRAIN	`SIZE 158 MS`	`DENSITY 31/S`	`SHAPE 0.0`
ENV	`ATTACK 5.5`	`RELEASE 6.5`	

Keep it: hold ■ SOURCE › ❹ CAPTURE while it's frozen, and it becomes a sample, CAP 1. Then unfreeze and try again with a different sound: a clap, a whistle, a laugh. A frozen clap is a strange, metallic drone.

The factory's LIVE ICE is a cousin of this one. Try it, and the other LIVE sounds, on the microphone too.

Grit on another unit

With another unit on the chain, FROM can be that unit, and Grit becomes something between an instrument and an effect: it plays clouds of whatever the other unit is playing, as it plays. Two settings on the MIX page are for that:

> **DRY** (MIX, encoder 3, 0-100%) how much of the other unit's own sound the Root still plays. 100%: you hear it and Grit's cloud together. 0%: only the cloud (the factory's LIVE GATE uses that).

> **FEEDBK** (MIX, encoder 4, 0-10) feedback: some of Grit's own output goes back into its buffer, so grains of grains build up into a shimmering trail. LIVE SHINE uses a little; a lot runs away with itself.

Put Haze on Grit's left to give the cloud echo and reverb as well.

9.10 Samples, the card, and Grit's SETUP

Press ■ SOURCE again for the SAMPLES page, as on Taste: every sample Grit has, your own first (captures, and samples loaded from the card), then the starter set. ❶ SAMPLE chooses, ❷ NAME renames your own, hold ❸ USE to play it, and hold ❹ SAVE to write it to the card's GRIT folder as a WAV. A sample of yours that no sound uses can be deleted: turn encoder 4 to DELETE and hold it.

> **BEND** (SETUP, THIS UNIT, 0-12) how far a MIDI keyboard's pitch bend wheel bends the cloud, in semitones.

> **WHEEL** (WHERE or SPRAY) what a MIDI keyboard's mod wheel moves: WHERE, further back through the sound, or SPRAY, more scatter.

> **VELO** (OFF or ON) ON: harder is louder, from a MIDI keyboard or Step.

> **PEDAL** (SUSTAIN or FREEZE) what a sustain pedal does: holds notes, or works the FREEZE button with your foot, so both hands stay on the keys.

What you have at the end of this chapter

Seven sounds of your own: MIST, SLOW BREAK, SHIMMER BELLS, BREAK PLACES, HOVER, MIST MOVING and MY FREEZE. And the whole of granular synthesis: how long, how many, what shape, where from, what pitch, and how to play it. Any sound you can record, Grit can turn into a cloud.

Chapter 10 - Haze

Echo, reverb, chorus and drive: the room, the pedals and the tape, for every synth on the chain.

Every other unit in this book makes sounds. Haze doesn't make any: it takes the sounds the others make and does something to them. It's an **effects unit**: the echo off a canyon wall, the reverb of a concert hall, the shimmer of a chorus, the growl of a guitar pedal, all in one box, shared by every synth on the chain.

Chapter 2, section 2.5, said what the four main effects are. This chapter is about using them: setting them up, shaping them, and building effects of your own that make a plain sound come alive.

> ### You need two units for this chapter
>
> Haze makes no sound of its own: it needs another unit to work on. So this chapter needs two units joined to the Root, Haze and one other, of any kind (section 10.1 says what to do with it).
>
> If you've only got one unit, you can't use Haze yet, but that's no reason to stop: you can design really good sounds with the one you have, and the chapter on it is where to do it. Skip on to chapter 11.

Effects: where they came from, and who uses them

The first effect was a room. Recordings in the 1930s and 40s got their reverb from where they were made, or from a special "echo chamber": a bare concrete room with a loudspeaker at one end and a microphone at the other. Studios that couldn't build one used a **plate**, a big sheet of steel hung in a box, shaken by the sound, from the late 1950s. Organ and guitar amplifiers used a **spring**, which is why surf guitar sounds the way it does.

Echo came from tape. A loop of tape running past a record head and then a playback head plays the sound back a moment later; feed it back round and it repeats, a little darker and more worn each time. Tape echo machines of the 1960s and 70s became instruments in their own right: in Jamaica, King Tubby and Lee "Scratch" Perry built a whole kind of music, dub, out of echoes and reverb, turning a song into space.

Flanging was discovered by pressing a thumb on the flange of a spinning tape reel, to slow one copy of a recording against another (the Small Faces' "Itchycoo Park", 1967, is the famous one). By the 1970s pedals did it, and phasers and choruses with it. Guitarists stood on them; keyboard players ran synthesisers through them. A plain electric piano through a chorus is the sound of half the ballads of the 1980s.

Distortion was a fault before it was a sound: an overloaded amplifier or a damaged speaker. Then it was a pedal: fuzz made the riff of the Rolling Stones' "(I Can't Get No) Satisfaction" in 1965, and overdrive and distortion pedals have been on every rock record since. Electronic music borrowed it too: an acid bass line run into a distortion pedal is a sound of its own.

Dance music added two of its own. DJ mixers grew a single filter knob, turned to make a record go muffled and then bright again, the build before a drop. And producers found that if the kick drum turned everything else down for a moment each time it hit, the whole track breathed in and out with it: the **pump**, all over French house records such as Stardust's "Music Sounds Better With You" (1998).

Haze has a version of each, and what you learn here will work on any effects unit, pedal or piece of music software you meet.

10.1 What you need for this chapter

Haze needs something to work on, and this chapter uses Glass's factory sounds for every example, so you can hear exactly what the book describes. If your other unit is already a Glass, you're ready. If it's something else, turn it into one for the length of this chapter. Reflashing keeps every kind's sounds (chapter 1, section 1.9), so when you flash it back, your own sounds are just where you left them.

On the other unit (not Haze), press ■ SETUP until you reach the RESET page, then:

❶ RESET › **REFLASH** › ❸ BECOME › **GLASS** › *hold* ❹

It starts again as Glass. Glass's factory sounds are in FAC A, and you'll use five of them, chosen on Glass's own SOUNDS page:

Glass's sound	Code	Used for
E.PIANO 1	FA11	reverb, chorus and the pump
STRINGS 1	FA04	a big hall
MARIMBA	FA22	echo
BASS 1	FA15	drive
BRASS 1	FA01	the DJ filter

■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **FA11** (on Glass)

> ### Which unit is which
>
> From here on, a route on its own is on Haze. When a step is on Glass, the book says so.

10.2 Find a preset you like

Haze's keys don't play notes. The **white keys** are sixteen presets, one press each: press one and Haze switches to it and sends every synth on the chain into it, so you hear it straight away on whatever you play. They start as the first sixteen factory presets: PLATE, BIG HALL, CAVE, SPRING, TAPE ECHO, DUB ECHO, SLAPBACK, ECHO HALL, CHORUS, FLANGER, PHASER, PUMP, ACID DRIVE, FUZZ BASS, WARM DRIVE and DJ SWEEP, lowest key first.

> **Try this**
>
> On Glass, choose E.PIANO 1 (FA11). Play a few notes and short chords on Glass, and with your other hand press Haze's white keys one at a time, from the bottom up. Some are obvious at once (BIG HALL, TAPE ECHO); some need Glass to play a sound they suit (the drives want a bass). Note down the ones you like.

The **black keys** are tap tempo: tap two or more in time and the echoes that follow the tempo (section 10.8) follow you. The TAP button does the same. Their lights flash the beat.

Haze has 32 factory presets in all, FAC A, on the SOUNDS page: ■ SOUNDS › ❶ BANK › FAC A › ❷ PRESET. The white keys are your own copies of the first sixteen, USER A 01 to 16, so you can change them; the factory ones stay as they are.

10.3 How Haze is laid out

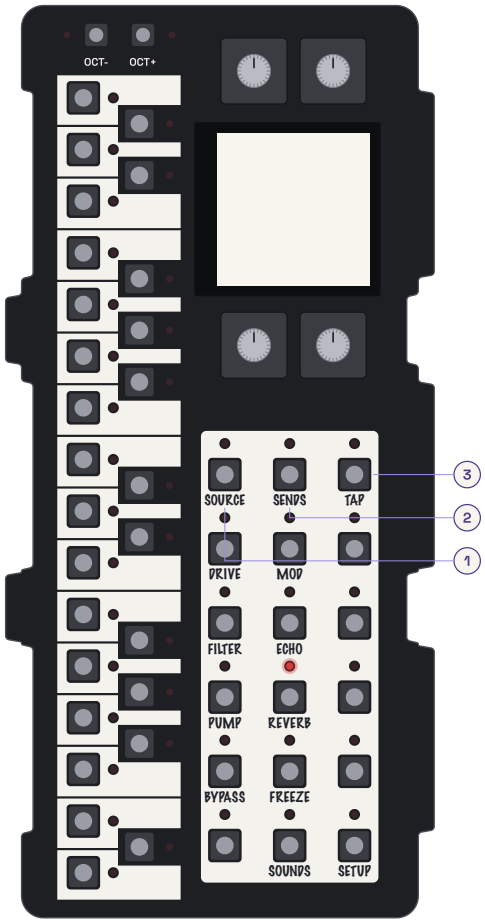


Figure 10.1 Haze's buttons. 1: COLOUR, for one synth: SOURCE, DRIVE, FILTER, PUMP, then BYPASS. 2: SPACE, for every synth: SENDS, MOD, ECHO, REVERB, then FREEZE, and SOUNDS at the bottom. 3: TAP, and SETUP at the bottom.

1 **The left column, COLOUR**, works on one synth you choose: SOURCE (which synth), DRIVE (distortion), FILTER (the DJ filter) and PUMP. Under them, BYPASS switches COLOUR off for a moment, to compare.

2 **The middle column, SPACE**, is shared by every synth: SENDS (how much of each goes in), MOD (chorus, flanger, phaser), ECHO and REVERB. Under them, FREEZE, and SOUNDS at the bottom.

3 **The right column** has TAP at the top and SETUP at the bottom.

Every setting reads 0 to 99, or a time, or a name; turn its encoder. Two pages have a second page: press ECHO or REVERB again.

10.4 Two halves: SPACE and COLOUR

First, two words you'll meet all through this chapter, and on any effects unit. A sound with no effect on it, just as the synth makes it, is **dry**. The effect's part of the sound, the echoes or the reverb's tail or the chorus's moving copy, is **wet**. Most effects are a mix of the two: the dry sound you play, with some wet around it.

On Haze, **WET** is also the name of how much of a synth goes into its effects: turn a synth's WET up and more of it goes in, so more echo or reverb comes out around it. Its dry sound doesn't change.

Haze has two halves, and they work in two different ways, the two ways a mixing desk uses effects.

- **SPACE** is for the effects that add something around a sound: chorus, echo, reverb. Each synth sends Haze a share of itself, its **WET**, and Haze adds only the effect. The synth's own sound still plays at the Root, as it always does. So turning a synth's WET up adds more echo or reverb without taking anything away, and every synth can share the same room.
- **COLOUR** is for the effects that change a sound itself: drive, the DJ filter, the pump. They only make sense on one sound at a time, so COLOUR works on one synth you choose, and plays it in place of the synth's own sound. MIX says how much: at 100% you hear only Haze's version.

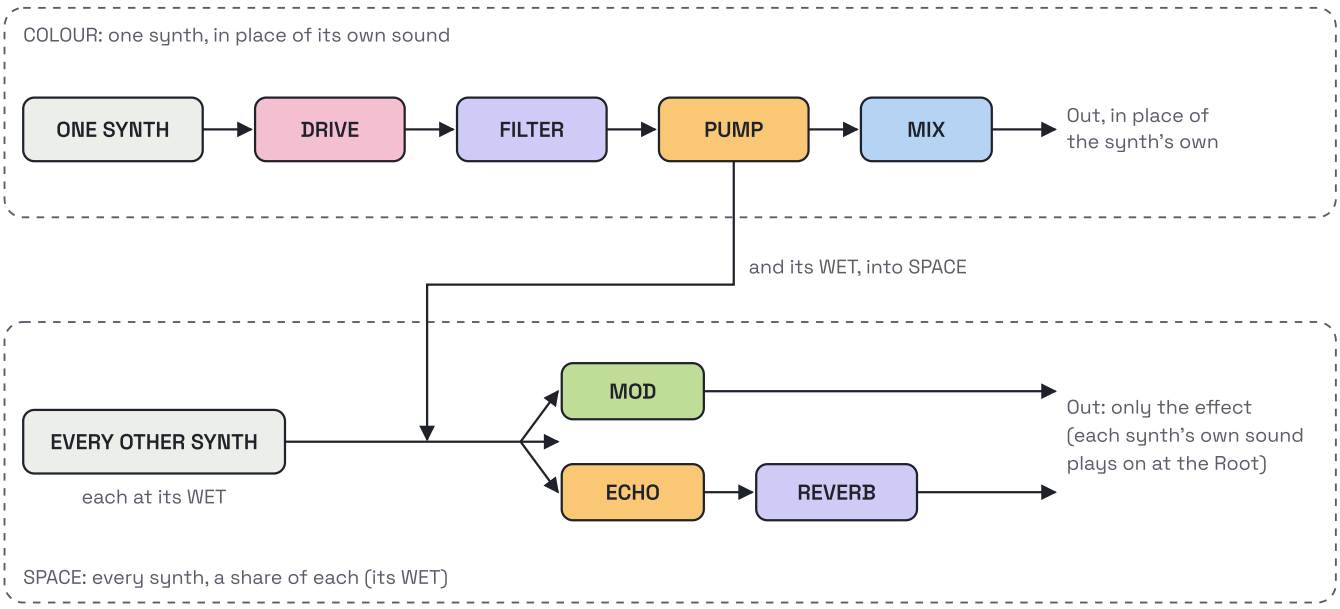


Figure 10.2 Haze's two halves. COLOUR works on one synth and plays it in place of its own sound; SPACE takes a share of every synth (its WET) and adds only the effect, while each synth's own (dry) sound plays on at the Root.

The two connect: the synth that COLOUR works on can also be sent into SPACE at its WET, after its drive and filter, as a guitar's pedals come before the amplifier's reverb.

10.5 The presets you'll build

This chapter builds eight presets of your own, in USER A 17 to 24. Those slots start as INIT FX: a plate reverb at a moderate level and everything else off, a plain place to start. (USER A 01 to 16 are the white keys; you can copy your favourites there at the end, section 10.14.)

■ SOUNDS › ❶ BANK › USER A › ❷ PRESET › UA17

Name each one as you build it, on the SOUNDS page with ❹ NAME (turn for the letter, push for the next one), as on every unit. By the end, USER A will hold:

Slot	Preset	Section	On Glass's	What it shows
UA17	MY PLATE	10.7	E.PIANO 1	a plate reverb, and pre-delay
UA18	MY HALL	10.7	STRINGS 1	a big, dark hall
UA19	MY ECHO	10.8	MARIMBA	a tape echo in time with the tempo
UA20	MY DUB	10.8	MARIMBA	runaway repeats, ping-pong, a spring, and FREEZE
UA21	MY CHORUS	10.9	E.PIANO 1	chorus, flanger and phaser
UA22	MY DRIVE	10.10, 10.11	BASS 1	choosing a synth for COLOUR, and distortion
UA23	MY SWEEP	10.12	BRASS 1	the DJ filter
UA24	MY DUCK	10.13	E.PIANO 1	the pump, ducking the echo

10.6 Sending a sound into Haze

Choose UA17 on Haze, and on Glass choose E.PIANO 1 (FA11). Play: you hear the electric piano, dry, just as before. Nothing goes into Haze until you send it.

■ SENDS › ❶ GLASS › 30

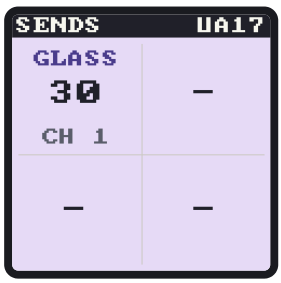


Figure 10.3 SENDS, with Glass on the chain: how much of Glass goes into Haze's SPACE, its WET.

SENDS has one encoder for every other unit on the chain that makes a sound (Step, which only plays the others, has none), four to a page (press SENDS again for the next four), each named and with its MIDI channel underneath. Turn Glass's to 30 and play again: now there's a room around the piano. That's INIT FX's plate reverb. Turn WET up and down while you play: 0 is none, 99 is drenched. WET belongs to the synth and the chain, not to the preset, so it stays where you put it whichever preset you choose.

WET and LEVEL

Two things set how loud an effect is: how much of each synth goes in (its WET, on SENDS), and how much of the effect comes out (each effect's LEVEL or VOLUME). WET is how a synth sits in the room; LEVEL is the room itself. Set LEVEL for the preset, and WET for each sound.

10.7 Reverb

A reverb is thousands of echoes so close together that they blur into one tail. Haze's REVERB page has four settings:

TYPE (REVERB, encoder 1) PLATE: bright and dense, the classic studio sound, flattering on almost anything. HALL: bigger and slower, a large room. SPRING: twangy and bouncy, with a chirp on sharp sounds.

DECAY (REVERB, encoder 2) how long the tail lasts, as a time: 0.3S to 8.0S for a plate or hall, 0.4S to 4.0S for a spring.

TONE (REVERB, encoder 3, 0-99) how bright the tail stays. Low: a tail that darkens as it fades, like a big soft room. High: bright and splashy.

LEVEL (REVERB, encoder 4, 0-99) how loud the reverb is.

MY PLATE

On UA17, with Glass's E.PIANO 1 sent at 30, turn ■ REVERB › ❷ DECAY from 1.0S to 1.1S, and ❸ TONE down from 60 to 55. Play short chords and listen to the tail: the piano now sounds as if it's in a small, bright studio.



Figure 10.4 The REVERB page, partway through building MY PLATE.

Now press REVERB again for its second page:

■ REVERB REVERB 2 › ❶ PRE-DLY › 20MS



Figure 10.5 REVERB pressed again: REVERB 2, with PRE-DLY.

PRE-DLY is a short wait before the reverb starts, 0 to 198 ms. In a real room the first reflections take a moment to come back from the walls, and a bigger room means a longer wait. It also keeps the start of each note clean: the piano's attack is heard on its own, and the reverb blooms just after. Try 100MS and play short notes: you can hear the gap. Then go back to 20MS, just enough to keep the piano clear. Turn ■ REVERB › 4 LEVEL down to 50 if it's grown, and name the preset MY PLATE.

MY PLATE (UA17)

A small, bright studio around an electric piano.

REVERB	TYPE PLATE · DECAY 1.1S · TONE 55 · LEVEL 50
REVERB 2	PRE-DLY 20MS
SOURCE	KEY WET 30
SENDS	GLASS 30

MY HALL

Choose UA18 on Haze and STRINGS 1 (FA04) on Glass. A hall wants a long, slow sound to fill it. Set:

■ REVERB › 1 TYPE › HALL › 2 DECAY › 4.3S › 3 TONE › 40 › 4 LEVEL › 60

Hold a chord, then let go and listen to it hang in the air for seconds. TONE at 40 makes the tail darker than the strings, so it sits behind them rather than on top. On REVERB 2, set PRE-DLY to 60MS: a big room, a long wait. Then on ■ SOURCE › 4 KEY WET › 35 (section 10.14 explains it), and name it MY HALL.

MY HALL (UA18)

A big, dark hall behind slow strings.

REVERB	TYPE HALL · DECAY 4.3S · TONE 40 · LEVEL 60
REVERB 2	PRE-DLY 60MS
SOURCE	KEY WET 35

> **Less than you think**
>
> Reverb is the easiest effect to overdo. A sound with too much sounds far away and washed out, and on a busy track every part blurs into the next. Turn LEVEL up until you can hear it clearly, then back until you only miss it when it's gone.

10.8 Echo

Haze's echo works the way a tape echo does: the sound is recorded onto a loop and played back a moment later, and each repeat goes round again, a little darker and more worn. Its page:

RATE (ECHO, encoder 1) how long until each repeat: 1000MS down to 60MS (FREE), or pushed, a note value that follows the tempo (below).

REPEATS (ECHO, encoder 2, 0-99) how much of each repeat goes round again: low, one or two echoes; high, a long trail; near the top, more than ever went in (below).

VOLUME (ECHO, encoder 3, 0-99) how loud the echoes are. INIT FX starts at 0: no echo.

TONE (ECHO, encoder 4, 0-99) how bright the echoes are. Lower is darker and more like old tape.

MY ECHO

Choose UA19 on Haze and MARIMBA (FA22) on Glass, and make sure Glass is still sent (SENDS, 30). Short, sharp notes show off an echo best. Turn ■ ECHO › 3 VOLUME › `60` and play a single note: it comes back every 321 milliseconds, quieter each time.

Echoes sound best in time with the music. Push RATE:

■ ECHO › *push* 1 › 1 RATE › `1/8 D`

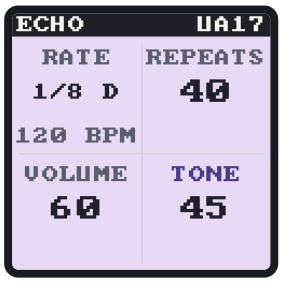


Figure 10.6 The ECHO page with RATE pushed: in note values, following the tempo, here 1/8 D at 120 BPM.

Now RATE is a note value, 1/32 up to 1/2 (T for triplets, D for dotted), at the tempo shown under it. With Step on the chain it's Step's tempo; without it, tap one in on TAP or the black keys, two taps or more. Play a simple tune at 120 beats a minute (tap it in), and the dotted eighth's echo makes its own rhythm against yours: the trick behind a thousand guitar riffs.

Set ❷ REPEATS › 40 and ❹ TONE › 45, turn the reverb down on ■ REVERB › ❹ LEVEL › 15 so the echoes stay clear, and name it MY ECHO.

MY ECHO (UA19)

Echoes in time with the tune, a dotted eighth apart.

ECHO	RATE 1/8 D	REPEATS 40	VOLUME 60	TONE 45
REVERB	LEVEL 15			
SOURCE	KEY WET 30			

MY DUB

Dub engineers played the echo like an instrument. Choose UA20, keep MARIMBA, and set an echo a quarter note apart, dark, with lots of repeats:

■ ECHO › *push* ❶ › ❶ RATE › 1/4 › ❷ REPEATS › 75 › ❸ VOLUME › 65 › ❹ TONE › 25

Play one note and listen: the repeats go on and on, getting darker. Now turn REPEATS up slowly, past 90. Somewhere near the top more goes round than came in: the echo stops dying away and grows, howling and distorting on its own, the way a tape echo does when it's pushed. Turn it back to 75 before it gets out of hand. That edge, turned up and down by hand, is a dub sound.

Press ECHO again for ECHO 2:

■ ECHO ECHO 2 › ❶ PINGPONG › ON

Now the repeats bounce from left to right and back, which is best heard on headphones. (Ping-pong reads the tape twice, so each side's gap can't be longer than half a second: at slow tempos the longest values are shortened.)

Last, the room:
■ REVERB › ❶ TYPE › SPRING › ❷ DECAY › 1.6S › ❸ TONE › 40 › ❹ LEVEL › 45. The echoes go into the reverb, so each repeat lands in its own twangy space. Name it MY DUB.

Now play a phrase and press **FREEZE**, the button under REVERB. Its light comes on, and the echo and reverb stop taking anything new in and hold what they have, round and round, for as long as you like. Play over the top: your new notes are dry against the frozen wash. Press FREEZE again and the wash fades away as normal. It's a performance button: it isn't kept with the preset.

MY DUB (UA20)

Dark echoes bouncing side to side into a spring; ride REPEATS and FREEZE by hand.

ECHO	`RATE 1/4` `REPEATS 75` `VOLUME 65` `TONE 25`
ECHO 2	`PINGPONG ON`
REVERB	`TYPE SPRING` `DECAY 1.6S` `TONE 40` `LEVEL 45`
SOURCE	`KEY WET 30`

10.9 Chorus, flanger and phaser

The MOD page holds three close relatives, all made by mixing a sound with a copy of itself that moves:

> **MODE** (MOD, encoder 1) OFF; CHORUS, two copies swaying slightly in pitch, one left and one right, for width and warmth; FLANGER, a very short echo fed back into itself, the jet-plane whoosh; PHASER, harmonics cancelled out by a sweeping filter, a softer, rounder swirl.

> **RATE** (MOD, encoder 2) how fast it sways: 0.08HZ to 4.00HZ, or pushed, once per note value, from 1/16 to 16/1 (sixteen whole notes), following the tempo.

> **DEPTH** (MOD, encoder 3, 0-99) how far.

> **LEVEL** (MOD, encoder 4, 0-99) how loud the moving copies are against the synth's own sound.

MY CHORUS

Choose UA21 on Haze and E.PIANO 1 on Glass again. Set a slow, gentle chorus:

■ MOD › ❶ MODE › `CHORUS` › ❷ RATE › `0.21HZ` › ❸ DEPTH › `55` › ❹ LEVEL › `60`

Play chords and hold them. The piano spreads out across both speakers and gains a slow, liquid wobble. Turn LEVEL down to 0 and back up to hear the difference: dry, it's a plain piano. With chorus it's the sound of a thousand records. Turn the reverb down to ■ REVERB › ❹ LEVEL › `20`, set ■ SOURCE › ❹ KEY WET › `45` (chorus likes more), and name it MY CHORUS.

MY CHORUS (UA21)

A wide, warm, swaying electric piano.

MOD	`MODE CHORUS` `RATE 0.21HZ` `DEPTH 55` `LEVEL 60`
REVERB	`LEVEL 20`
SOURCE	`KEY WET 45`

Try this

On MY CHORUS, turn ■ MOD › ❶ MODE to FLANGER, then PHASER, holding a chord. Then push ❷ RATE and turn it to 2/1: the sweep now takes two whole notes, a slow rise and fall that keeps time with the music. Turn MODE back to CHORUS and RATE back (push it again for 0.21HZ) before you move on, or keep the version you like best: it's your preset.

10.10 COLOUR: choosing a synth

The left column works on one synth. First you choose it:

■ SOURCE › ❶ SYNTH › GLASS



Figure 10.7 SOURCE, with Glass chosen: Glass now plays through COLOUR, at MIX 100%.

SYNTH (SOURCE, encoder 1) NONE, or any other unit on the chain.

MIX (SOURCE, encoder 2, 0-100%) how much of Haze's version you hear. The Root plays the rest of the synth's own sound, so at 100% you hear only Haze's, and at 50% half and half, the way a pedal's blend knob works.

KEY WET (SOURCE, encoder 4, 0-99) the preset's send for its white key (section 10.14).

Like WET, SYNTH and MIX belong to the chain, not the preset: they stay as you set them whichever preset you choose.

BYPASS, under PUMP, switches COLOUR off and on. While it's lit, the chosen synth plays clean, all of it from the Root (and still sent into SPACE at its WET). Use it to compare: does the drive make the sound better, or just louder?

10.11 Drive

Drive is what happens when a sound is pushed harder than it can go: the tops of the wave are squashed flat, and the squashing adds new harmonics. A little sounds warm and thick; a lot sounds hard and aggressive. DRIVE has three kinds, after the three classic kinds of pedal:

> **MODE** (DRIVE, encoder 1) OFF; OVERDRIVE, the gentlest: mostly the middle of the sound breaks up, the bass stays clean; DISTORTION, harder and buzzier; FUZZ, the most extreme, a thick, square buzz.

> **DRIVE** (DRIVE, encoder 2, 0-99) how hard it's pushed.

> **TONE** (DRIVE, encoder 3, 0-99) darker or brighter.

> **LEVEL** (DRIVE, encoder 4, 0-99) how loud it comes out.

MY DRIVE

Choose UA22 on Haze and BASS 1 (FA15) on Glass, and make sure SOURCE's SYNTH is GLASS and MIX is 100%. Play a few bass notes. Now:

■ DRIVE › 1 MODE › **DISTORTION** › 2 DRIVE › **55**



Figure 10.8 DRIVE on MY DRIVE: DISTORTION, and LEVEL set so it's no louder than the clean sound.

The clean, round FM bass turns gritty and hard. Press BYPASS on and off while you play. The driven version is louder, and louder always sounds better at first (chapter 2, section 2.6), so it isn't a fair test. Turn ■ DRIVE › 4 LEVEL down until switching BYPASS on and off changes the sound but not how loud it is: about 31. Now you can tell whether you like it. Try OVERDRIVE and FUZZ too, and TONE: each needs LEVEL set again.

Finish with a touch of room, ■ REVERB › 4 LEVEL › **15** and ■ SOURCE › 4 KEY WET › **15**, and name it MY DRIVE.

MY DRIVE (UA22)

A gritty, distorted bass, no louder than the clean one.

DRIVE	`MODE DISTORTION` `DRIVE 55` `TONE 50` `LEVEL 31`
REVERB	`LEVEL 15`
SOURCE	`SYNTH GLASS` `MIX 100%` `KEY WET 15`

10.12 The DJ filter

The FILTER page is a DJ mixer's filter: one knob that goes both ways.

> **FILTER** (FILTER, encoder 1) OFF in the middle. Turn it down and a low-pass filter closes, from LP 17.8K down to LP 60, taking the top off until only a muffled thud is left. Turn it up and a high-pass filter opens, from HP 17 up to HP 6.0K, taking the bass away until only a thin fizz is left.

> **RESO** (FILTER, encoder 2, 0-99) resonance: a peak at the filter's edge, which makes a sweep sing.

MY SWEEP

Choose UA23 on Haze, BRASS 1 (FA01) on Glass, with Glass still chosen on SOURCE. Hold a note, and turn ■ FILTER › ❶ FILTER slowly down from OFF, then all the way back up past OFF and on into the high-pass, and back to OFF. That's the DJ's move: down to build tension, back up to release it, or up through the high-pass to thin a sound out before the drop.

Turn ❷ RESO up to 20 and sweep again: the edge of the filter whistles as it moves. More RESO whistles more, but it also makes the sound louder at the edge: a big, loud chord through a lot of RESO can go over the top, and Haze rounds it off rather than letting it crack. If it starts to sound harsh, turn RESO down.

Leave FILTER at LP 2.0K, a warm, muffled brass to sweep up from; add ■ REVERB › ❹ LEVEL › `20` and ■ SOURCE › ❹ KEY WET › `20`, and name it MY SWEEP. When you call it up, sweep the filter to taste: it's meant to be moved.

MY SWEEP (UA23)

A muffled brass, ready to be swept open by hand.

FILTER	`FILTER LP 2.0K` `RESO 20`
REVERB	`LEVEL 20`
SOURCE	`SYNTH GLASS` `KEY WET 20`

10.13 The pump

The pump turns something down each time another sound plays, and lets it back up after. Its page:

> **KEY** (PUMP, encoder 1) the synth whose sound does the ducking, NONE or any unit on the chain. Like SYNTH, it belongs to the chain, not the preset.

> **DEPTH** (PUMP, encoder 2, 0-99) how far it ducks.

> **RELEASE** (PUMP, encoder 3) how long it takes to come back up, 40MS to 800MS.

> **TARGET** (PUMP, encoder 4) what gets ducked: COLOUR (the chosen synth), SPACE (the echo and reverb), or BOTH.

In a dance track the KEY is the kick drum, and everything else pumps in time with it. That needs a drum unit as well as a synth, three units in all, and the Music Production manual does it. With one synth, the pump has a cleverer use: the synth can duck its own echoes.

MY DUCK

Choose UA24 on Haze and E.PIANO 1 on Glass, sent on SENDS at about 35. First an echo and a hall:

■ ECHO › *push* ❶ › ❶ RATE › 1/4 › ❷ REPEATS › 55 › ❸ VOLUME › 70

■ REVERB › ❶ TYPE › HALL › ❷ DECAY › 2.2S › ❹ LEVEL › 40

Play a busy phrase: it's a mess, echoes piling on top of the notes. Now the pump:

■ PUMP › ❶ KEY › GLASS › ❷ DEPTH › 80 › ❸ RELEASE › 182MS › ❹ TARGET › SPACE

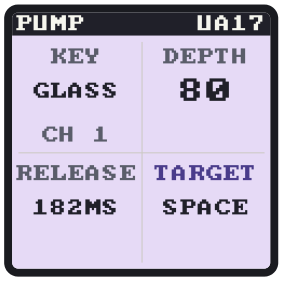


Figure 10.9 PUMP on MY DUCK: Glass is its own KEY, and the pump ducks only the SPACE section, so the echoes get out of the way of the notes.

Play the same phrase. While you play, the echoes and reverb duck out of the way; the moment you stop, they bloom up into the gap. Your notes stay clear, and the space fills in around them. Engineers do this to vocals all the time. RELEASE sets how quickly the space comes back: shorter for fast playing, longer for slow. Name it MY DUCK.

MY DUCK (UA24)

Echoes and a hall that get out of the way while you play.

PUMP	`KEY GLASS`	`DEPTH 80`	`RELEASE 182MS`	`TARGET SPACE`
ECHO	`RATE 1/4`	`REPEATS 55`	`VOLUME 70`	`TONE 50`
REVERB	`TYPE HALL`	`DECAY 2.2S`	`LEVEL 40`	
SOURCE	`KEY WET 35`			

10.14 The white keys, KEY WET and Haze's SETUP

A white key does two things: it calls up its preset, USER A 01 to 16, and it sets every synth's send to that preset's **KEY WET** (SOURCE, encoder 4). That's why each recipe in this chapter has a KEY WET: it's how much of everything goes in when you call the preset up with a key, so a reverb comes in gently and a chorus strongly. Choosing a preset on the SOUNDS page doesn't touch the sends.

To put one of your presets on a white key, copy it over one of the sixteen: on the SOUNDS page, choose it, then 3 COPY TO the slot of the key you want (UA01 is the lowest white key, UA16 the highest) and *hold* 3. The preset that was there is replaced; the factory ones are always on FAC A to copy back.

Haze's SETUP has one setting of its own, on THIS UNIT beside the MIDI channel:

> **KEYS** (encoder 2) + SENDS: a white key calls up its preset and sets every synth's send to its KEY WET, as above. PRESET: a white key only calls up the preset, and every synth's send stays where you set it.

> **What you have at the end of this chapter**
>
> Eight presets of your own: MY PLATE, MY HALL, MY ECHO, MY DUB, MY CHORUS, MY DRIVE, MY SWEEP and MY DUCK. Between them they use every page on Haze. They work on any synth, not only on Glass: flash your other unit back to what it was (its own sounds are all still there), send it into Haze, and try each one on its sounds. A good effect changes how you play, so play.

Chapter 11 - Where next

You know how sound is made now. The rest is up to you.

11.1 What you've learned

Think back to chapter 2. Pitch, tone, shape, movement, effects, loudness: six things that every sound is made of. Since then you've built sounds out of each of them, from scratch, by hand.

You've built tones from sine waves, one operator modulating another, the way the bell-like sounds of the 1980s were made. You've started from bright, buzzy waves and carved them with a filter, the way the classic analogue synthesisers did, for basses, leads, pads and brass. You've made drums from nothing but oscillators and noise. You've moved a sound between four waves at once, played back recordings across the keyboard, and broken sounds into thousands of grains and scattered them. And you've put all of it into a room, an echo, a chorus and a distortion pedal.

If you've worked through every chapter, you now know more about how sound is made than most musicians ever learn. Plenty of people play synthesisers for years without knowing why a filter whistles, what an operator does or where a reverb's tail comes from. You do. And you know it the best way there is: not from reading about it, but from turning the encoders and listening to what happened.

Two things to take away from all of it, the same two the book started with.

The first: **a good sound is one you like.** Not the one that's most like a famous record, not the one with the most going on, not the one somebody else told you was right. Every recipe in this book is a starting point. If you turned something further than the recipe said and liked it better, you were right and the recipe was wrong.

The second: **the best sounds are found, not planned.** Remember the beginning of the book: some of the most famous sounds in music were accidents, a machine used the wrong way, a setting turned too far, a mistake somebody liked. Every time you turned something past where it was meant to go, just to hear what happened, you were doing what those people did. Keep doing it. Turn things too far. Send a sound somewhere it was never meant to go. Try the setting that can't possibly work. Most of the time it won't. The times it does are the sounds that will be yours.

11.2 Your unit, from now on

Start with the biggest thing you've got out of this book, which isn't in the box at all. **What you've learned works on every synthesiser there is.** Operators, filters, envelopes, LFOs, wavetables, samples, grains, sends and inserts: these are the same ideas, with the same names, on every synthesiser, every effects unit and every piece of music software you'll ever sit down at. The knobs will be in different places and the screens will look different, but you'll know what they do, and you'll know where to start. Most people faced with a new synthesiser scroll through its presets. You can make your own.

And the unit in front of you isn't finished with you just because the book is. It was never only a way to learn: it's an instrument, and it's yours.

- **Play it.** A unit and a Root are a complete synthesiser: headphones or speakers on the Root, a power bank or a phone charger for power, and you can play anywhere, on its own keys or from a MIDI keyboard. Turned sideways (chapter 1), it's a little keyboard instrument for a desk, a sofa or a train.
- **Play it from a computer.** Plug the Root's USB-C into a computer and the computer sees it as a MIDI device: music software can play your unit, from a recorded part or from the keyboard plugged into the computer, and an editor or librarian can send it sounds (Glass's, chapter 3). The sound itself doesn't travel over USB: take it from the Root's LINE OUT into the computer's audio interface, or into a mixer, and record it like any other instrument.
- **Play it with other instruments.** The Root's MIDI IN takes notes from any keyboard, sequencer or drum machine with a MIDI OUT, each unit on its own channel (chapter 1, section 1.10), so your unit can be one voice in a bigger setup. MIDI OUT passes a copy of what comes in on to the next instrument (THRU), and once you have Step it sends Step's notes and its clock out too, so your other instruments can play along and keep time with it.
- **Keep building.** Every one of your banks has room for sounds you haven't made yet. Back them up to a card (chapter 1, section 1.9), swap them with friends, and when you're ready, flash the unit as something else and start on its sounds too. Your sounds are never lost: every kind keeps its own.

11.3 The whole Tessel

This book has been about making sounds, one unit at a time. But a Tessel was never meant to be limited to just one unit. Units snap together side by side, as many as eight on a Root, and the moment you join a second one, everything changes.

Join a Volt and a Strike and you have a bass line and a beat. Add Ember or Glass for chords, Haze for the room, Taste for a vocal chop, Grit for a texture drifting underneath. Each unit is a complete instrument with its own screen and its own keys, so there are no menus to dig through: everything you've learned in

this book, on every unit, at your fingertips at once. And because every unit can be flashed as any other, the set you own can change shape with the music you're making: today a techno rig, tomorrow an ambient one.

There are two modules this book hasn't covered. The first is the one that turns a row of synths into music: **Step**, the sequencer. Step plays every unit on the chain at once, sixteen tracks of notes on a timeline of bars and beats, with accents, slides, chords, an arpeggiator, swing and a mixer, and it remembers which sound every unit was on, so a whole song comes back at the press of a button.

The second is **Pilot**, the controller. It makes no sound of its own: it gathers the knobs you reach for most, from any units on the chain, onto pages of four, so a filter on one unit, an echo on another and the tempo can all sit under your fingers together. Its keys call up scenes, every unit switching sound at once. Pilot's changes are always temporary, so you can twist a sound as far as you like in the middle of a set and it's still waiting, untouched, for next time.

Step has a book of its own: **Making Music**, the companion to this one. Where this book taught you how sounds are made, that one teaches you what to do with them: how Step works, and then how real tracks are built, one style of electronic music at a time, from techno and acid to house and beyond, starting with just three units and growing as your setup grows. Each chapter says which modules it needs, so you can always start with what you've got.

Put it together and a Tessel is whatever you need it to be:

- **A synthesiser**, on its own, in your bag, ready wherever you are.
- **A groovebox**: Step, a drum unit and a bass, everything in one row, no computer anywhere.
- **A live instrument**: songs that come back at a press, sounds you can grab and twist with your hands, Haze's FREEZE and DJ filter, Pilot's pages and scenes, every part of the music in reach on stage.
- **A whole studio**: write, arrange, mix and play out a finished track from first idea to last note, without ever opening a computer. Sample your own voice through the Root's microphone, record a friend's guitar into Taste, and it's all in the same row.
- **A part of a bigger setup**: MIDI in, MIDI out, a computer over USB, a mixer on the LINE OUT. A Tessel plays well with others.

Start where you are, with what you've got, and add a module when the music asks for one. Every one you add makes every other one do more.

Now go and make something

That's the end of the book, and the start of the good bit. Make sounds nobody's heard before. Make music with them. Get things wrong on purpose and keep the mistakes you like.

And when you've made something, we'd love to hear it. Tell us what you've done with your Tessel: the sounds, the tracks, the strange setups and the happy accidents. Write to us at Utopian Instruments, **[address to come]**.

Enjoy what you make.

Quick reference

Every button, every page, every encoder, on one page per unit.

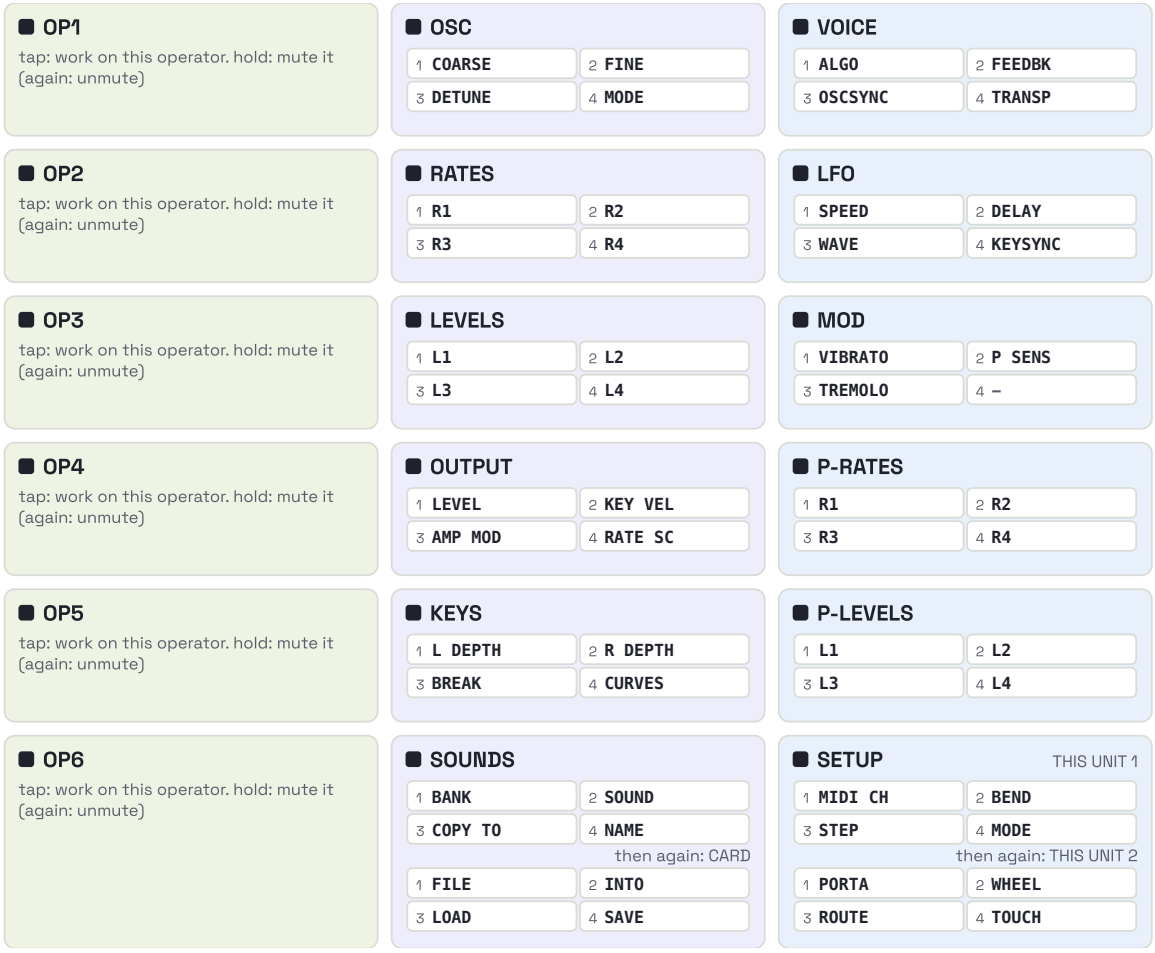
For when you know what you want and just need to find it. Each unit's map shows its buttons as they sit on the panel, with the four boxes of each page in the corners where they'll appear on the screen: encoder 1 top left, 2 top right, 3 bottom left, 4 bottom right. A button with a second page shows it underneath (press the button again to get there). Under each map, a line on every control.

Every unit

SOUNDS the SOUNDS button	
BANK	which bank: FAC A… (the factory's) or USER A… (your own)
SOUND	which sound in it; it plays as soon as you reach it (on Strike: KIT)
COPY TO	turn to choose one of your slots; hold the encoder to copy the sound there
NAME	your own sounds: turn to change the letter, push for the next letter
THIS UNIT SETUP (the first pages)	
MIDI CH	the unit's MIDI channel: AUTO, or one you choose (never changed after)
2–4	the unit's own settings, listed under each unit below
RESET SETUP, the last THIS UNIT page	
RESET	encoder 1: SETTINGS, EVERYTHING, REFLASH (become another kind), another type kept in the unit's flash, EVERY TYPE, TOTAL RESET (wipe everything: a blank unit) or TEST (the unit's own test, then back as it was)
FIRMWARE	the firmware version; on REFLASH, BECOME: which kind, from the ones the Root carries
RESET	encoder 4: hold for three seconds to do it
ROOT SETUP, after THIS UNIT (the Root's own settings)	
INPUT	which input units record from: LINE, MIC (with mic power) or BUILT-IN; also on Taste's and Grit's recording pages
GAIN	how loud the input comes in, in dB; also on Taste's and Grit's recording pages
MIDI OUT	OFF, NOTES (Step's), CLOCK (Step's tempo), BOTH, THRU (a copy of MIDI IN) or ALL
CARD	turn to choose, hold to run: BACKUP (every unit and the Root, into a new BACKUP-n), RESTORE (the newest backup back into the units here), FORMAT (empties the card)
ROOT2 SETUP, the last page	
MIDI IN	CHANNELS: each unit plays its own channel; FOLLOW: notes go to the unit last touched
FROM CH	in FOLLOW, the channel the Root listens on
SOUNDS	UNITS' (each kind's sounds from its leftmost unit) or ROOT'S: for a Root that already has sounds on it

KEEP	encoder 4: hold for three seconds to do SOUNDS' choice
KEYBOARD	while OCT- or OCT+ is held
OCTAVE	moves the keys by octaves (also: tap OCT- or OCT+)
SEMI	moves the keys by semitones
LOWEST	the lowest note the keys now play
HIGHEST	the highest; press OCT- and OCT+ together to put them back

Glass



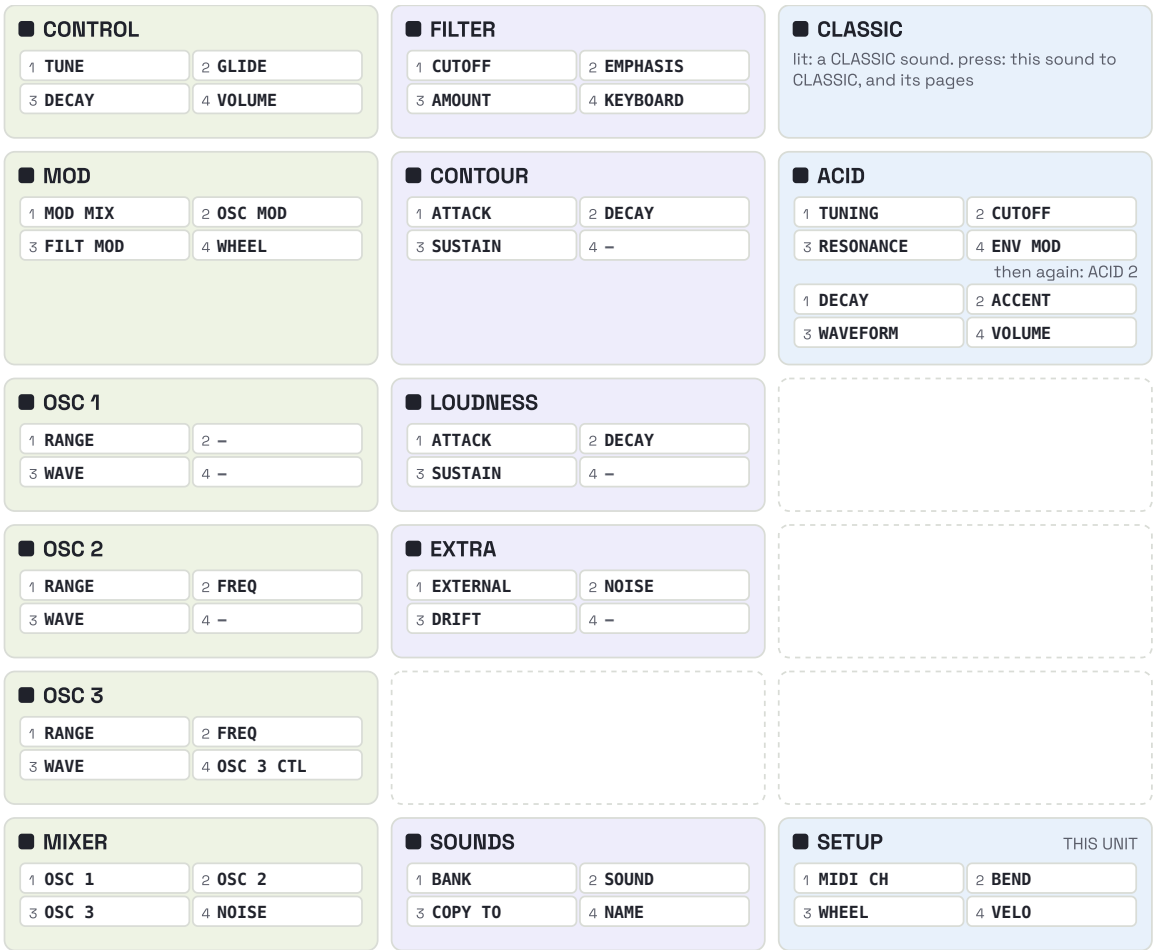
The five pages in the middle column belong to whichever operator is chosen (the top bar says which: OP1 OSC). The right-hand column's belong to the whole sound.

OSC the chosen operator's pitch	
COARSE	its pitch as a multiple of the note: 0.5, 1-31 (FIXED: 1, 10, 100 or 1000 Hz)
FINE	in between, 0-99; the result shows under both (X2.50, or Hz)
DETUNE	a hair either side of in tune, -7 to +7
MODE	RATIO (follows the keys) or FIXED (the same pitch on every key)

RATES the chosen operator's envelope: speeds (99 fastest)	
R1–R4	how fast it travels to L1, L2, L3, and (after you let go) back to L4
LEVELS the chosen operator's envelope: corners, 0-99	
L1–L3	the corners it travels through; it rests on L3 while you hold the key
L4	where it starts and ends (above 0: a drone)
OUTPUT the chosen operator's level and how it follows playing	
LEVEL	how much it puts out, 0-99: a carrier's loudness, a modulator's brightness
KEY VEL	how much playing harder raises it, 0-7
AMP MOD	how much tremolo reaches it, 0-3
RATE SC	how much faster its envelope runs higher up, 0-7
KEYS the chosen operator's level across the keyboard	
L DEPTH	how much it changes below BREAK, 0-99
R DEPTH	how much it changes above BREAK, 0-99
BREAK	the note it changes either side of, A-1 to C8
CURVES	the shape each side, left then right: -L, -E, +E, +L
VOICE the whole sound	
ALGO	which of the 32 algorithms; shows the diagram while you turn (push to close)
FEEDBK	how hard the looped operator modulates itself, 0-7
OSCSYNC	ON: operators start their waves afresh on every note
TRANSP	moves the whole sound, -24 to +24 semitones
LFO the slow wobble	
SPEED	how fast, 0-99
DELAY	how long it waits after a key goes down, then fades in, 0-99
WAVE	TRI, SAW-, SAW+, SQR, SINE or S&H
KEYSYNC	ON: the wobble starts afresh on every note
MOD how far the LFO moves things	
VIBRATO	how far it moves the pitch, 0-99
P SENS	how much of the vibrato this sound lets through, 0-7
TREMOLO	how far it moves the level, 0-99 (only on operators with AMP MOD)
P-RATES, P-LEVELS the pitch envelope: the same as RATES and LEVELS, moving the pitch	
L1–L4	50 is no change; above bends up, below bends down
CARD SOUNDS pressed again: whole banks on the Root's card (GLASS folder)	
FILE	which bank file
INTO	which of your banks, USER A-D

LOAD	hold: the file's 32 sounds replace that bank's
SAVE	hold: that bank becomes a new file on the card
THIS UNIT	Glass's own settings (SETUP)
BEND	how far pitch bend goes, 0-12 semitones
STEP	ON: bends in whole semitones
MODE	POLY (16 notes) or MONO (one, legato)
PORTA	glide time, OFF or 1-99
WHEEL	the mod wheel's range, 0-99
ROUTE	what the mod wheel moves: OFF, PITCH, AMP or BOTH
TOUCH	what aftertouch moves: the same choices

Volt

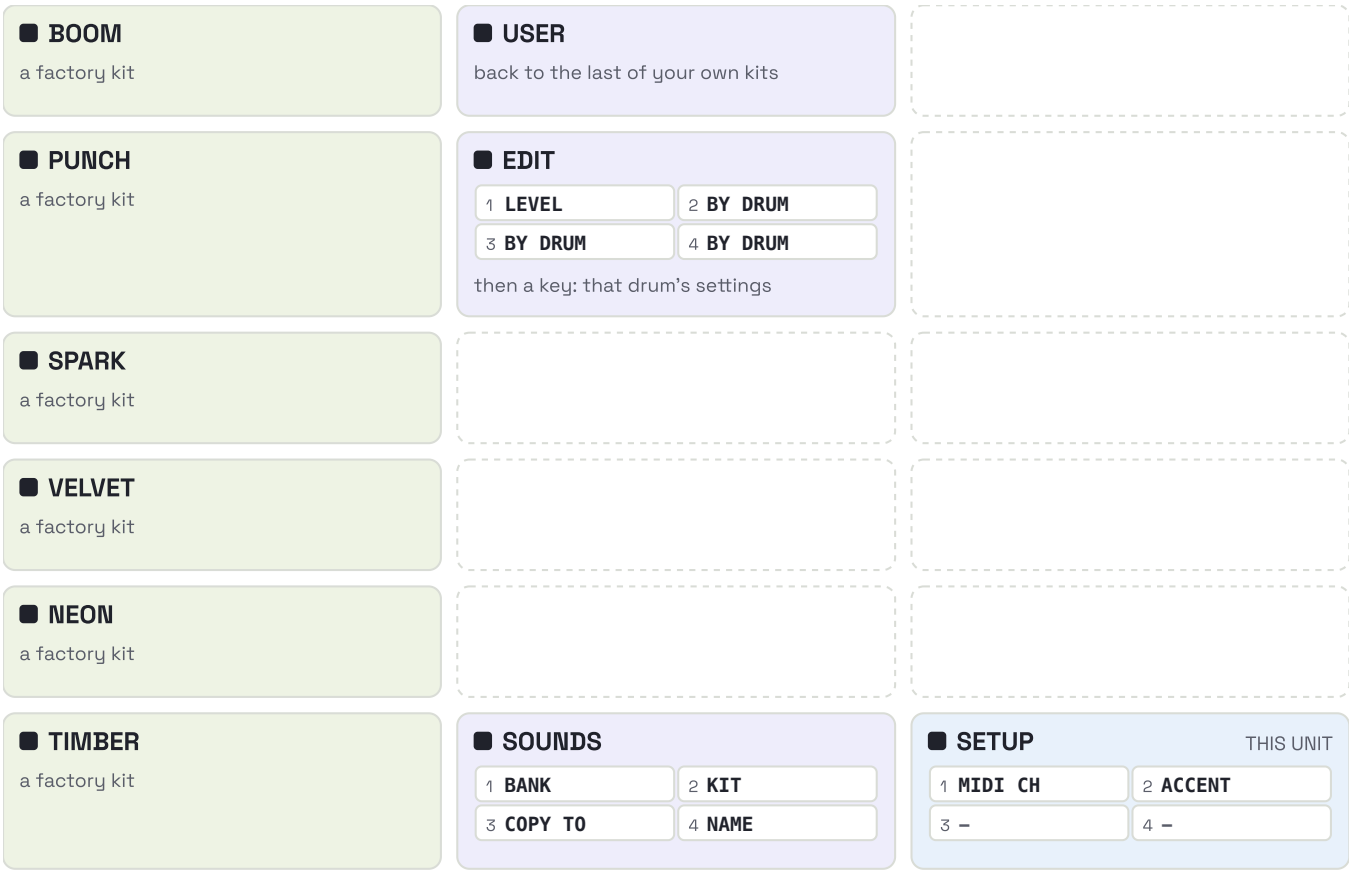


A CLASSIC page pressed while the sound is ACID (or the other way round) shows for as long as you hold the button, and changes nothing. Pushing an encoder works the switch beside that control (ON/OFF under the value).

CONTROL the whole instrument	
TUNE	fine tuning, -2.5 to +2.5 semitones
GLIDE	glide time, 0-10; push: glide on or off
DECAY	the release switch: ON, notes die away at DECAY's speed; OFF, they stop
VOLUME	the sound's level, 0-10
MOD what oscillator 3 or noise moves	
MOD MIX	0: oscillator 3 does the moving; 10: noise; in between, both
OSC MOD	ON: it moves the oscillators' pitch (vibrato)
FILT MOD	ON: it moves the filter (a wah, a wobble)
WHEEL	how much, 0-10; a MIDI keyboard's mod wheel moves it too
OSC 1-3 the three oscillators	
RANGE	the octave: LO, 32', 16', 8', 4', 2' (bigger is lower)
FREQ	OSC 2 and 3: tuned against OSC 1, -7 to +7 semitones
WAVE	TRIANGLE, SHARKTOOTH (OSC 3: REVERSE SAW), SAWTOOTH, SQUARE, WIDE, NARROW
OSC 3 CTL	OSC 3: ON follows the keys; OFF stays at one pitch (a drone, or a modulator)
MIXER how much of each goes on	
OSC 1–3, NOISE	levels, 0-10 (past about 5, the filter growls); push: that one on or off
FILTER the low-pass filter	
CUTOFF	where it starts to cut, -5 to +5
EMPHASIS	resonance, 0-10: from about 7 it whistles
AMOUNT	how far the CONTOUR opens it on each note, 0-10
KEYBOARD	how much it follows the keys: OFF, 1 (a third), 2 (two thirds), 1+2 (fully)
CONTOUR, LOUDNESS the filter's contour and the loudness contour	
ATTACK	time to the top, 0-10 (shown in ms or s)
DECAY	time to fall to SUSTAIN (and, with the DECAY switch on, the release)
SUSTAIN	the level it holds while the key is down, 0-10
EXTRA extras	
EXTERNAL	Volt's output fed back into its mixer, 0-10; push: on or off
NOISE	WHITE or PINK
DRIFT	ON: the oscillators wander a few cents, as analogue ones do
ACID an ACID sound (press ACID again for ACID 2)	
TUNING	the pitch, fine
CUTOFF	where the filter sits, 0-10
RESONANCE	the squelch, 0-10

ENV MOD	how far each note sweeps the filter, 0-10
DECAY	ACID 2: how long the sweep takes to close, 0-10
ACCENT	ACID 2: how much harder accented notes hit, 0-10
WAVEFORM	ACID 2: SAW or SQUARE
VOLUME	ACID 2: the sound's level, 0-10
THIS UNIT Volt's own settings (SETUP)	
BEND	how far pitch bend goes, 0-12 semitones
WHEEL	what moves the WHEEL encoder: MOD WHEEL, BREATH, EXPRESSION, AFTERTOUCH or OFF
VELO	OFF: every CLASSIC note equally loud; ON: harder is louder

Strike

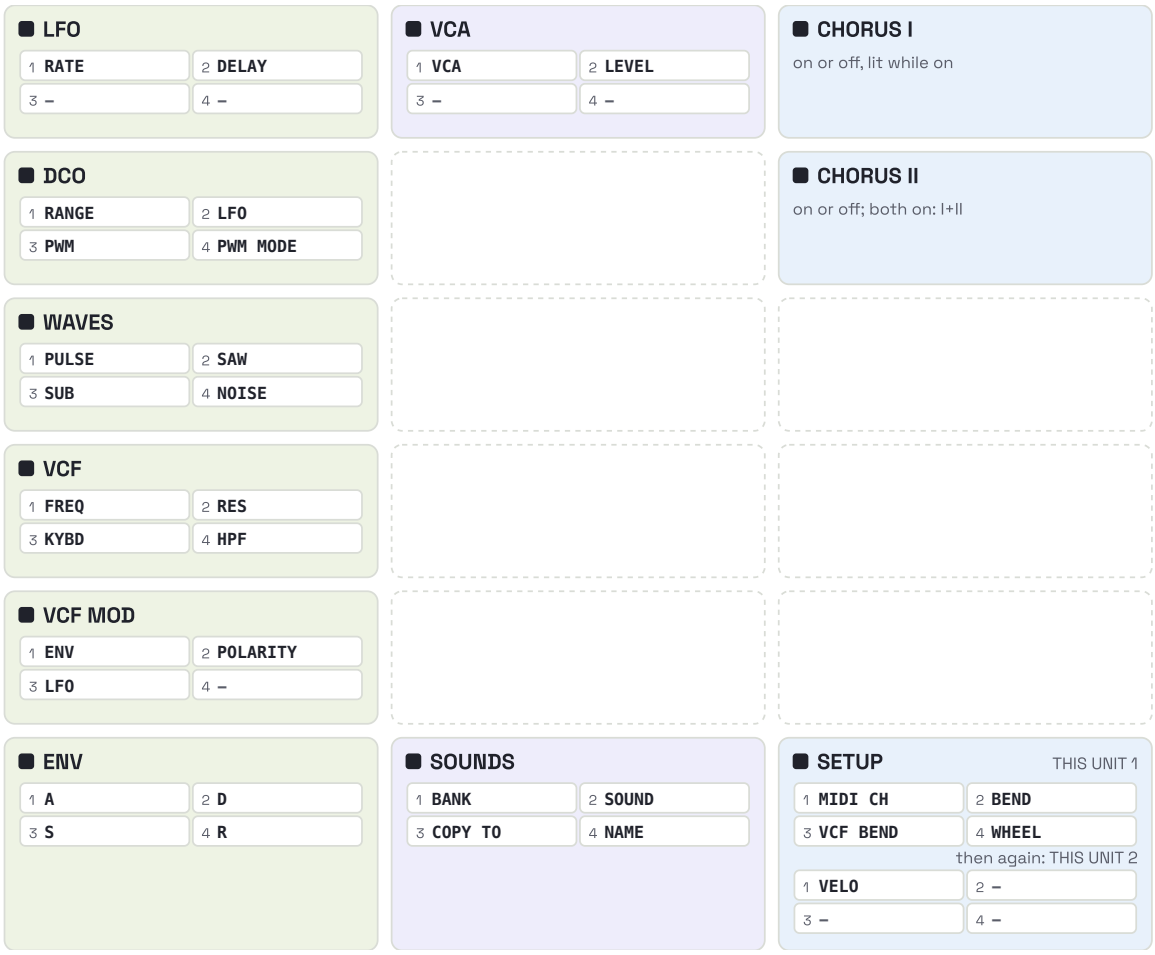


The white keys are the drums, the same on every kit: 1 KICK, 2 PERC, 3 SNARE, 4 TAMB, 5 RIM, 6 CLAP, 7 CL HAT, 8 OP HAT, 9 LO TOM, 10 MID TOM, 11 HI TOM, 12 CRASH, 13 RIDE, 14 COWBELL, 15 CLAVES, 16 MARACAS. In one of your own kits, hold a key: turn any encoder for that key's drum from another kit, or NONE.

EDIT the drum last played; encoder 1 is always LEVEL. Push: the factory's value	
BOOM	kick: TUNE, DECAY, TONE. snare: TUNE, TONE, SNAPPY. toms: TUNE, TYPE (TOM or CONGA). open hat: DECAY. cymbal: DECAY, TONE. cowbell: TUNE. the rest: LEVEL only

PUNCH	kick: TUNE, DECAY, ATTACK. snare: TUNE, TONE, SNAPPY. toms: TUNE, DECAY. hats: DECAY. crash and ride: TUNE. the rest: LEVEL only
SPARK, VELVET	LEVEL only, on every drum
NEON	every drum: TUNE, BEND (how fast the pitch falls), DECAY
TIMBER	kick: TUNE, DECAY, BEATER. snare: TUNE, DECAY, SNARES (0: off). hats, toms, crash, ride: TUNE, DECAY. rim, cowbell, claves: TUNE. clap, maracas, tambourine: DECAY
THIS UNIT Strike's own setting (SETUP)	
ACCENT	how much louder an accented step from Step is: OFF or +1 to +10 dB

Ember

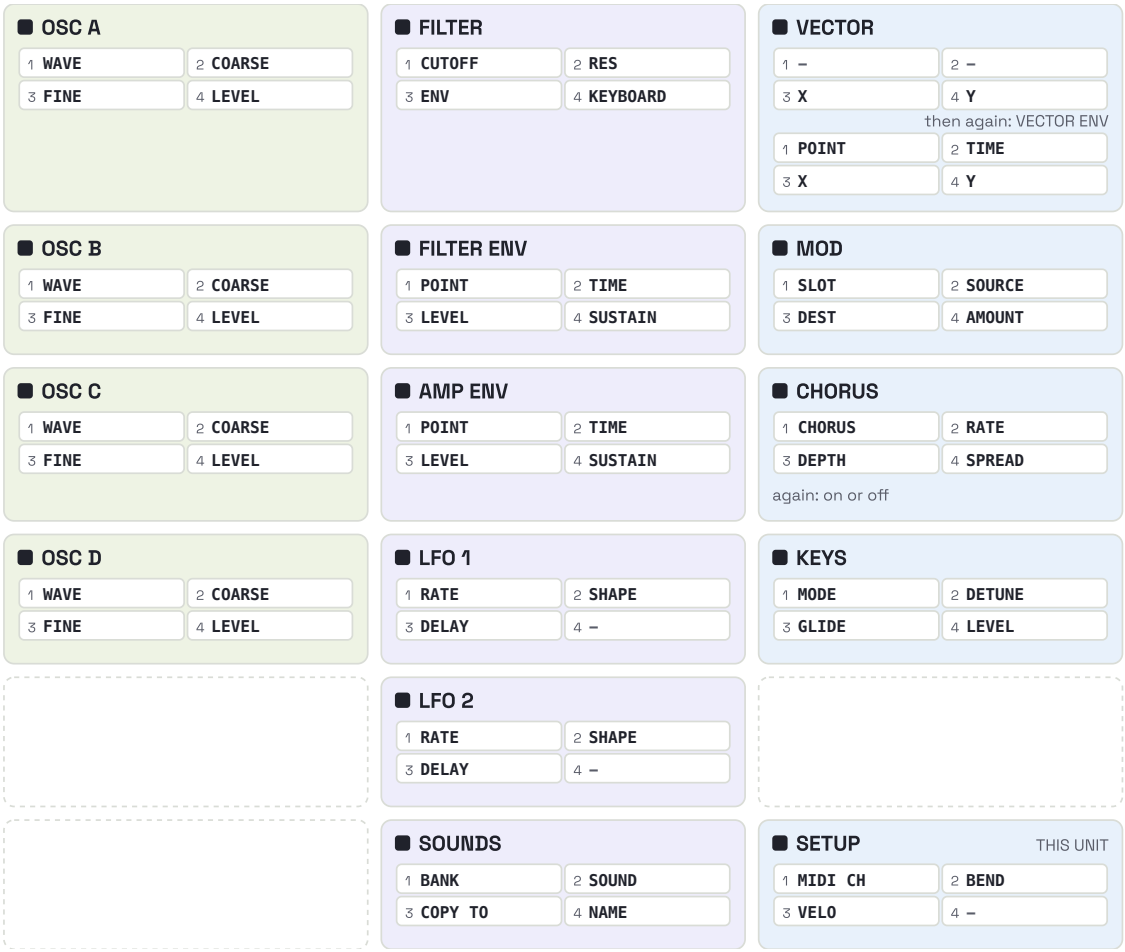


Every setting is for all six voices at once. Sliders read 0-10.

LFO the slow wobble, shared by every voice	
RATE	how fast, 0-10
DELAY	how long it waits before fading in, from the first key of each phrase, 0-10
DCO the oscillator	
RANGE	the octave: 16', 8', 4'

LFO	vibrato: how far the LFO moves the pitch, 0-10
PWM	the pulse's width (MANUAL), or how far the LFO moves it (LFO), 0-10
PWM MODE	LFO or MANUAL
WAVES what the oscillator makes	
PULSE, SAW	each on or off
SUB	a square an octave below, 0-10
NOISE	hiss, 0-10
VCF the low-pass filter, and the high-pass after all six voices	
FREQ	where it starts to cut, 0-10
RES	resonance, 0-10: near the top it whistles
KYBD	how much it follows the keys, 0-10
HPF	0 a little bass boost, 1 flat, 2 and 3 cut the bass
VCF MOD what moves the filter	
ENV	how far the envelope opens it, 0-10
POLARITY	NORMAL opens it; INVERT closes it
LFO	how far the LFO moves it, 0-10
ENV the one envelope: the filter (by VCF MOD's ENV) and the loudness (unless VCA is GATE)	
A, D, S, R	attack, decay, sustain level, release, 0-10
VCA the loudness	
VCA	ENV: shaped by the envelope; GATE: on while the key is down
LEVEL	the sound's volume, 0-10
CHORUS the two buttons	
I	a slow, gentle sway
II	faster
I+II	both on: a fast, shallow shimmer
THIS UNIT Ember's own settings (SETUP)	
BEND	how far pitch bend goes, 0-12 semitones
VCF BEND	how far the same wheel moves the filter, 0-10
WHEEL	how much vibrato the mod wheel adds, 0-10
VELO	OFF: every note equally loud; ON: harder is louder

Axis



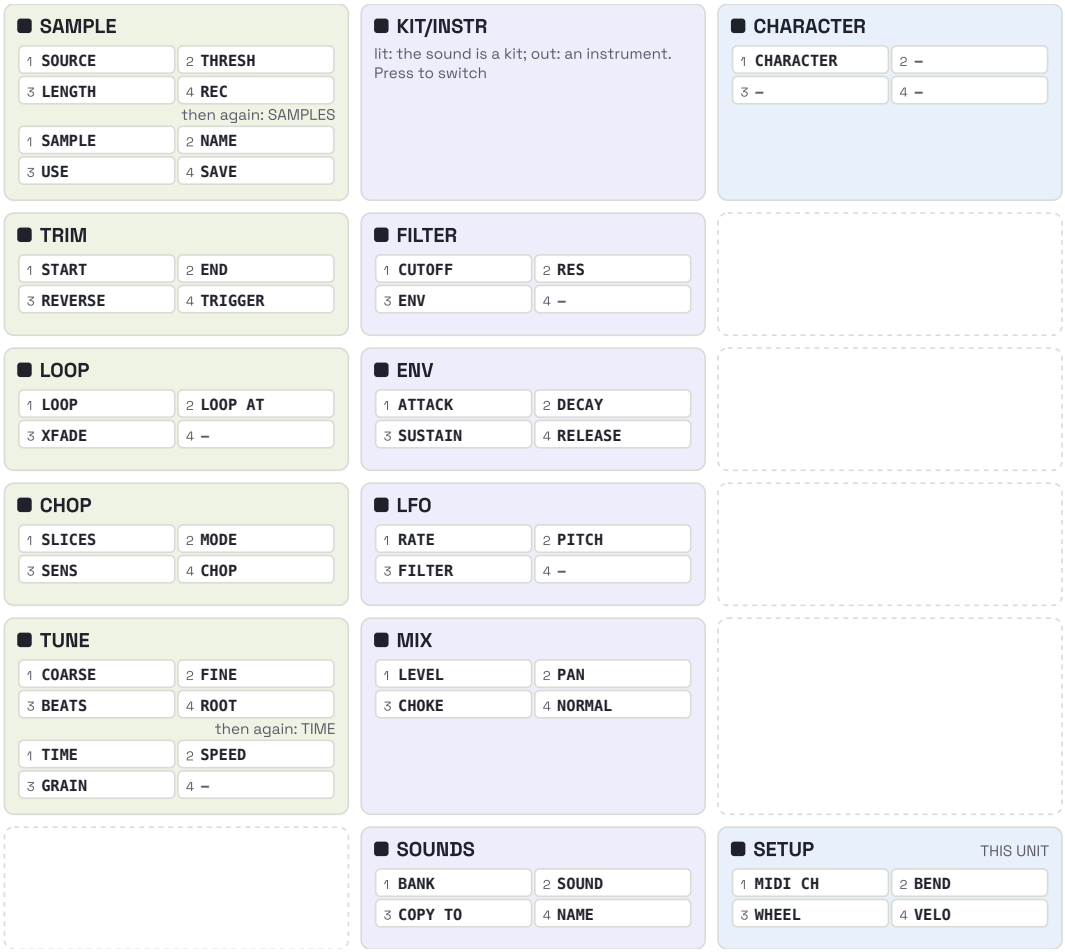
On every envelope page, push encoder 1 for its loop: SUSTAIN, LOOP AT, REPEATS, BOUNCE (push again for the points). The VECTOR page's top boxes are empty: the diamond is drawn there.

OSC A-D each corner's oscillator	
WAVE	which of the 128 waves, 0-127 (eight families of 16)
COARSE	pitch, -24 to +24 semitones
FINE	-100 to +100 cents
LEVEL	how loud this corner is, 0-10
VECTOR the diamond (press again for its envelope)	
X	towards A (-100) or C (+100)
Y	towards B (-100) or D (+100)
VECTOR ENV moves the point while the note plays	
POINT	which of the five points, 0-4
TIME	how long to get to it, 4 ms to 41 s
X, Y	where it is on the diamond

FILTER the resonant low-pass filter	
CUTOFF	where it cuts, in Hz
RES	resonance, 0-10
ENV	how far FILTER ENV moves it, -100 to +100
KEYBOARD	how much it follows the keys, 0-10
FILTER ENV, AMP ENV five-point envelopes	
POINT	which point, 0-4
TIME	how long to get to it, 4 ms to 41 s
LEVEL	its level, 0-10
SUSTAIN	the point it holds at while the key is down
LOOP AT	loop view: where a loop goes back to
REPEATS	loop view: OFF, 1-7 or FOR EVER
BOUNCE	loop view: ON, back and forth
LFO 1, LFO 2 two slow wobbles (what they move is set on MOD)	
RATE	how fast, in Hz
SHAPE	TRIANGLE, SAW UP, SAW DOWN, SQUARE or RANDOM
DELAY	how long before it fades in
MOD six connections: a source moving a destination	
SLOT	which connection, 1-6
SOURCE	LFO 1, LFO 2, FILT ENV, AMP ENV, VELOCITY, WHEEL, KEY, PRESSURE or NONE
DEST	PITCH, CUTOFF, RES, LEVEL, X, Y, PAN, WAVE A-D or WAVES (WAVE SCAN)
AMOUNT	how much, -100 to +100
CHORUS the stereo chorus (press CHORUS again: on or off)	
CHORUS	ON or OFF
RATE	how fast, in Hz
DEPTH	how deep, 0-10
SPREAD	how wide the voices sit, 0-10
KEYS how it plays	
MODE	POLY (8 notes), UNISON (all eight on one) or MONO
DETUNE	UNISON: how far apart, 0-10
GLIDE	OFF or a time
LEVEL	the sound's volume, 0-10
THIS UNIT Axis's own settings (SETUP)	
BEND	how far pitch bend goes, 0-12 semitones

VELO	ON: harder is louder; OFF: every note the same

Taste



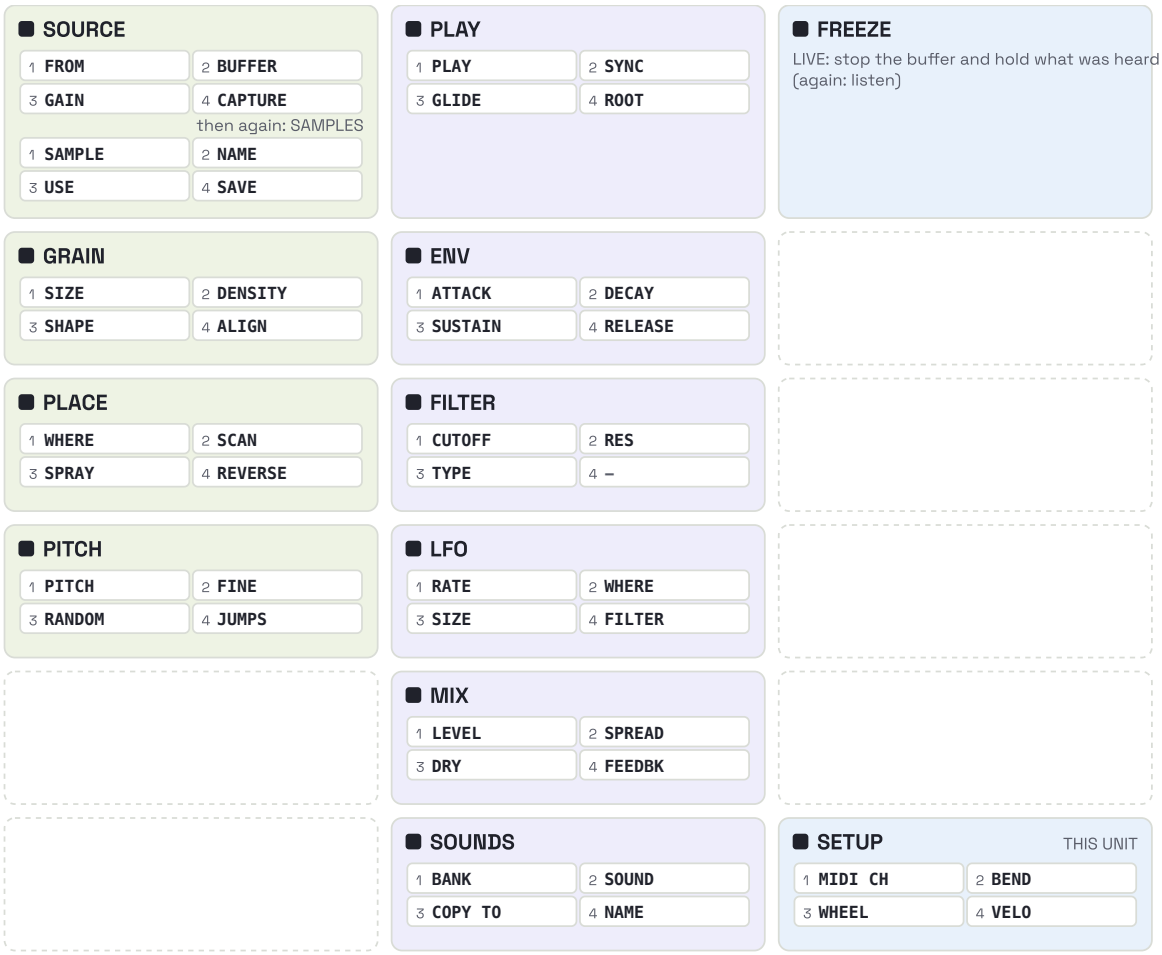
In a kit the pages work on the key last played (its name is in the top bar). On TRIM and LOOP, push a marker's encoder to zoom in on it. SAMPLE from CARD: encoder 2 is FILE and encoder 4 LOAD; from a Root input, push encoder 3 for GAIN.

SAMPLE recording onto the key being edited (or the instrument)	
SOURCE	LINE IN, MIC IN, BUILT-IN MIC, ALL, a unit by name, or CARD
THRESH	OFF (start at once) or -60 to -6 dB: wait until the sound is this loud
LENGTH	the longest take, 1-60 s (push on a Root input: GAIN)
REC	push to record, push again to stop
SAMPLES every sample (SAMPLE pressed again)	
SAMPLE	which one; your own first, then the starter set (*)
NAME	your own: turn for the letter, push for the next
USE	hold: on the key being edited (or the instrument)

SAVE	hold: onto the card's TASTE folder as a WAV
TRIM which part plays	
START, END	where it starts and stops, in seconds (push: zoom)
REVERSE	ON: backwards
TRIGGER	1-SHOT (plays to the end) or GATE (while held)
LOOP holding a note for ever	
LOOP	ON or OFF
LOOP AT	where it jumps back to (push: zoom)
XFADE	the crossfade over the join, 0-1000 ms
CHOP slices onto the keys from this one up	
SLICES	how many, 2-27 (COPY: TO, the key to copy onto)
MODE	EVENLY, AT HITS, or COPY
SENS	AT HITS: how small a jump counts, 0-10
CHOP	hold: do it (COPY: copy it)
TUNE pitch	
COARSE	-24 to +24 semitones
FINE	-50 to +50 cents
BEATS	OFF or 1-64: last this many beats at the tempo
ROOT	an instrument's own note
TIME TUNE pressed again	
TIME	CLASSIC (speed is pitch) or STRETCH (apart)
SPEED	STRETCH: 25% to 400%
GRAIN	STRETCH: the grains' length, 10-100 ms
FILTER the low-pass filter	
CUTOFF	where it cuts, 0-10
RES	resonance, 0-10
ENV	how far the envelope moves it, -50 to +50
ENV each note's shape	
ATTACK, DECAY, SUSTAIN, RELEASE	0-10 each
LFO one for the whole sound	
RATE	how fast, 0-10
PITCH	vibrato, 0-10
FILTER	how much it moves the filter, 0-10

MIX the key's level and place	
LEVEL	0-10
PAN	L50 to R50
CHOKE	kits: OFF or GROUP 1-4 (cut each other off)
NORMAL	hold: your own sample as loud as it goes
CHARACTER the whole sound	
CHARACTER	CLEAN, 12-BIT (40 kHz) or 12-BIT 26K
THIS UNIT Taste's own settings (SETUP)	
BEND	how far pitch bend goes, 0-12 semitones
WHEEL	how much vibrato the mod wheel adds, 0-10
VELO	ON: harder is louder

Grit

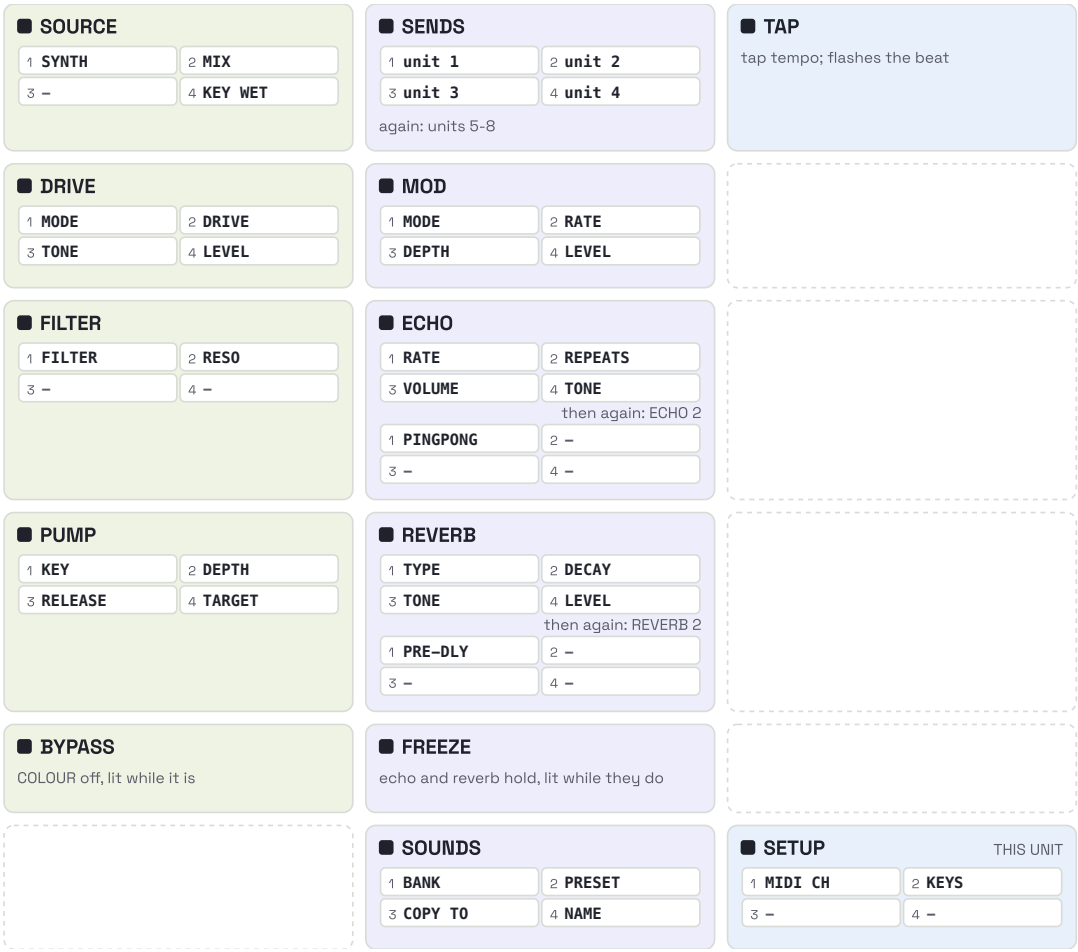


On a SAMPLE sound, SOURCE's second box is SAMPLE (turn to choose) and CAPTURE is empty. In PLACES, the white keys are 16 places and the black keys stutter (1/4 to 1/32, backwards, 1/64). SETUP's second page: PEDAL.

SOURCE where the grains come from (SOURCE again: SAMPLES)	
FROM	LINE IN, MIC IN, BUILT-IN MIC, ALL, a unit by name, SAMPLE or CARD
BUFFER	LIVE: how much of the past, 1-20 s (SAMPLE: which sample)
GAIN	a Root input: how loud it comes in
CAPTURE	LIVE: hold to keep the buffer as a sample
SAMPLES every sample	
SAMPLE	which one
NAME	your own: turn for the letter, push for the next
USE	hold: the sound plays it
SAVE	hold: onto the card's GRIT folder (turn clockwise for DELETE, an unused sample of yours)
GRAIN the grain	
SIZE	how long, 5-500 ms
DENSITY	how many a second, 1-200
SHAPE	the fade: 0 smooth to 10 sharp
ALIGN	ON: each grain lined up with the last
PLACE where in the sound	
WHERE	0-100% (LIVE: 100% is now)
SCAN	how fast WHERE moves, -200% to +200% (0 stands still)
SPRAY	how scattered, 0-10
REVERSE	the chance a grain plays backwards, 0-100%
PITCH the grains' pitch	
PITCH	-24 to +24 semitones
FINE	-50 to +50 cents
RANDOM	each grain detuned at random, OFF to 12 semitones
JUMPS	OFF, OCTAVE or OCT+5TH: some grains above
PLAY how the keys play it	
PLAY	NOTES, PLACES or ALWAYS
SYNC	OFF, or grains on the beat: 1/4 … 1/32
GLIDE	OFF or a time
ROOT	the key that plays it as it was, C2-C6
ENV each note's cloud	
ATTACK, DECAY, SUSTAIN, RELEASE	0-10 each

FILTER after the grains	
CUTOFF	0-10
RES	resonance, 0-10
TYPE	LOW, HIGH or BAND
LFO one slow wobble	
RATE	how fast, in Hz
WHERE, SIZE, FILTER	how much it moves each, 0-10
MIX	
LEVEL	0-10
SPREAD	grains scattered left and right, 0-10
DRY	LIVE from a unit: how much of its own sound still plays, 0-100%
FEEDBK	LIVE: Grit's output back into its buffer, 0-10
THIS UNIT Grit's own settings (SETUP)	
BEND	how far pitch bend goes, 0-12 semitones
WHEEL	what the mod wheel moves: WHERE or SPRAY
VELO	ON: harder is louder
PEDAL	second page: SUSTAIN (holds notes) or FREEZE

Haze



White keys: USER A 01-16, one press each, every synth sent at the preset's KEY WET (unless SETUP's KEYS is PRESET). Black keys: tap tempo, as TAP. SYNTH, MIX, KEY and the sends belong to the chain, not the preset.

SOURCE COLOUR's synth	
SYNTH	NONE, or the unit COLOUR works on
MIX	how much of Haze's version you hear, 0-100%; the Root plays the rest dry
KEY WET	the send every synth gets when the preset's white key is pressed, 0-99
SENDS SPACE's input	
each unit	its WET: how much of it goes into MOD, ECHO and REVERB, 0-99
DRIVE distortion on COLOUR's synth	
MODE	OFF, OVERDRIVE, DISTORTION or FUZZ
DRIVE, TONE, LEVEL	how hard, how bright, how loud, 0-99
FILTER the DJ filter on COLOUR's synth	
FILTER	OFF in the middle; down, a low-pass (LP 17.8K to LP 60); up, a high-pass (HP 17 to HP 6.0K)

RESO	a peak at the filter's edge, 0-99
PUMP ducking, in time with another sound	
KEY	NONE, or the unit whose sound does the ducking
DEPTH	how far, 0-99
RELEASE	how long it takes to come back, 40MS-800MS
TARGET	COLOUR, SPACE or BOTH
MOD chorus, flanger, phaser	
MODE	OFF, CHORUS, FLANGER or PHASER
RATE	0.08HZ-4.00HZ; push: a note value, 1/16 to 16/1, following the tempo
DEPTH, LEVEL	how far, how loud, 0-99
ECHO the tape echo	
RATE	1000MS-60MS; push: a note value, 1/32 to 1/2, following the tempo
REPEATS	how much goes round again, 0-99: near the top it runs away
VOLUME, TONE	how loud, how bright, 0-99
PINGPONG	ECHO 2: the repeats alternate left and right
REVERB the room	
TYPE	PLATE, HALL or SPRING
DECAY	how long: 0.3S-8.0S (a spring: 0.4S-4.0S)
TONE, LEVEL	how bright the tail, how loud, 0-99
PRE-DLY	REVERB 2: the wait before it starts, 0-198MS
THIS UNIT Haze's own setting (SETUP)	
KEYS	+ SENDS: a white key sets every synth's send to its KEY WET; PRESET: it only calls up the preset

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